Quartus II 軟體使用練習

練習三種輸入方法：

(1). Bottom-Up

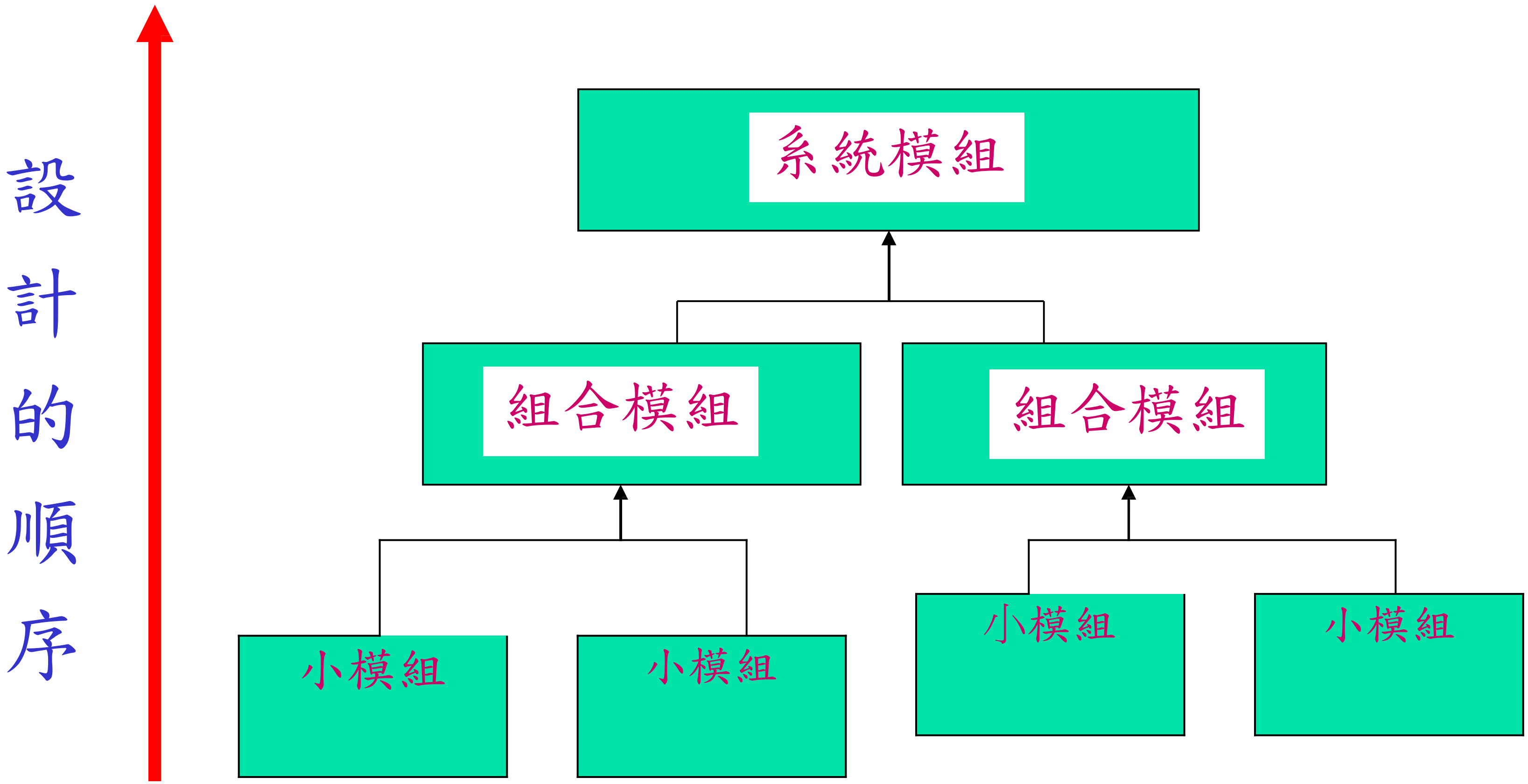
(2). Top-Down

(3). One-Chip

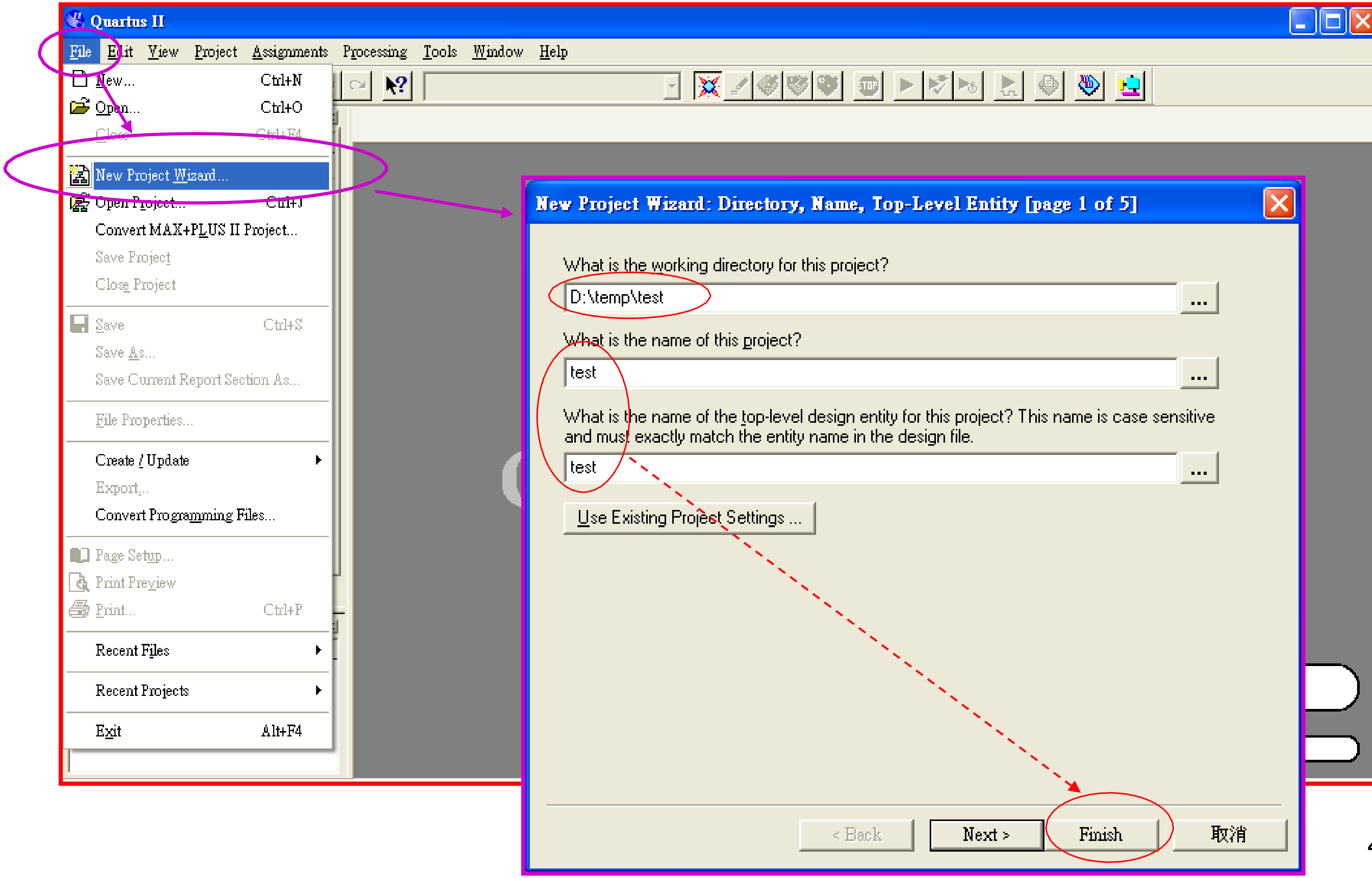
Bottom-Up Design

Bottom-Up Design Flow

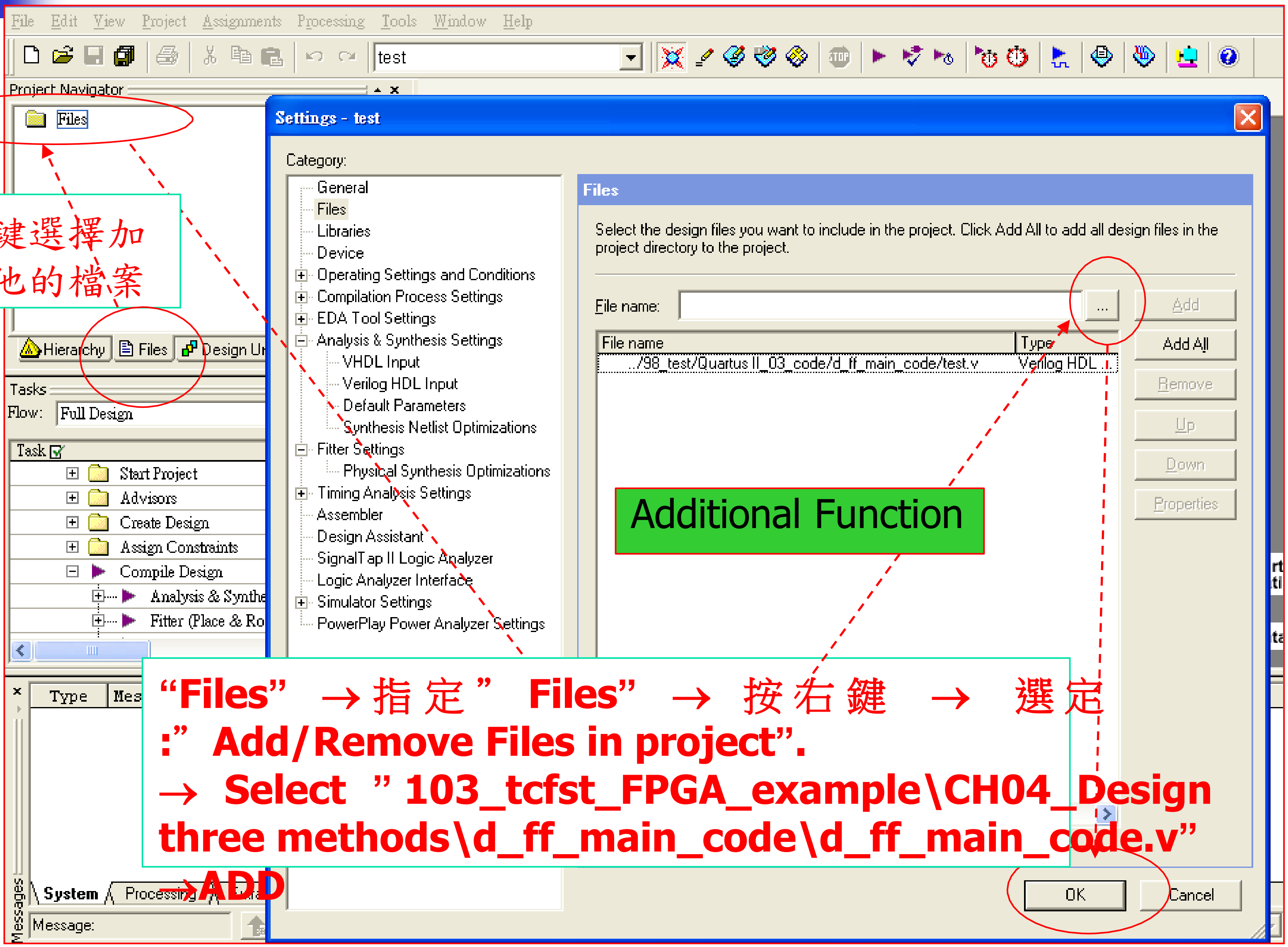
Bottom-Up Design



Bottom-Up Design (1)



Bottom-Up Design (2)

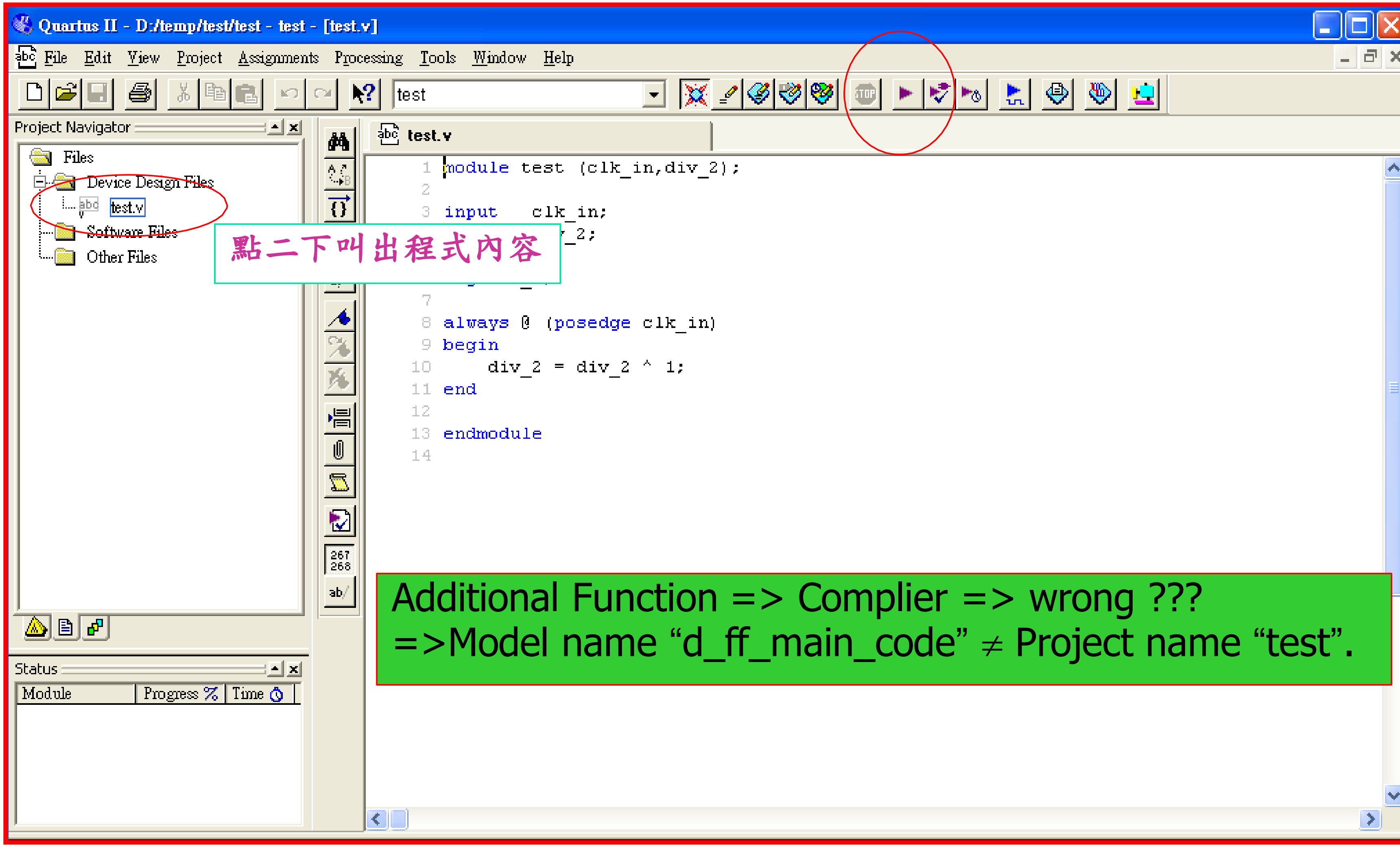


按右鍵選擇加入其他的檔案

Additional Function

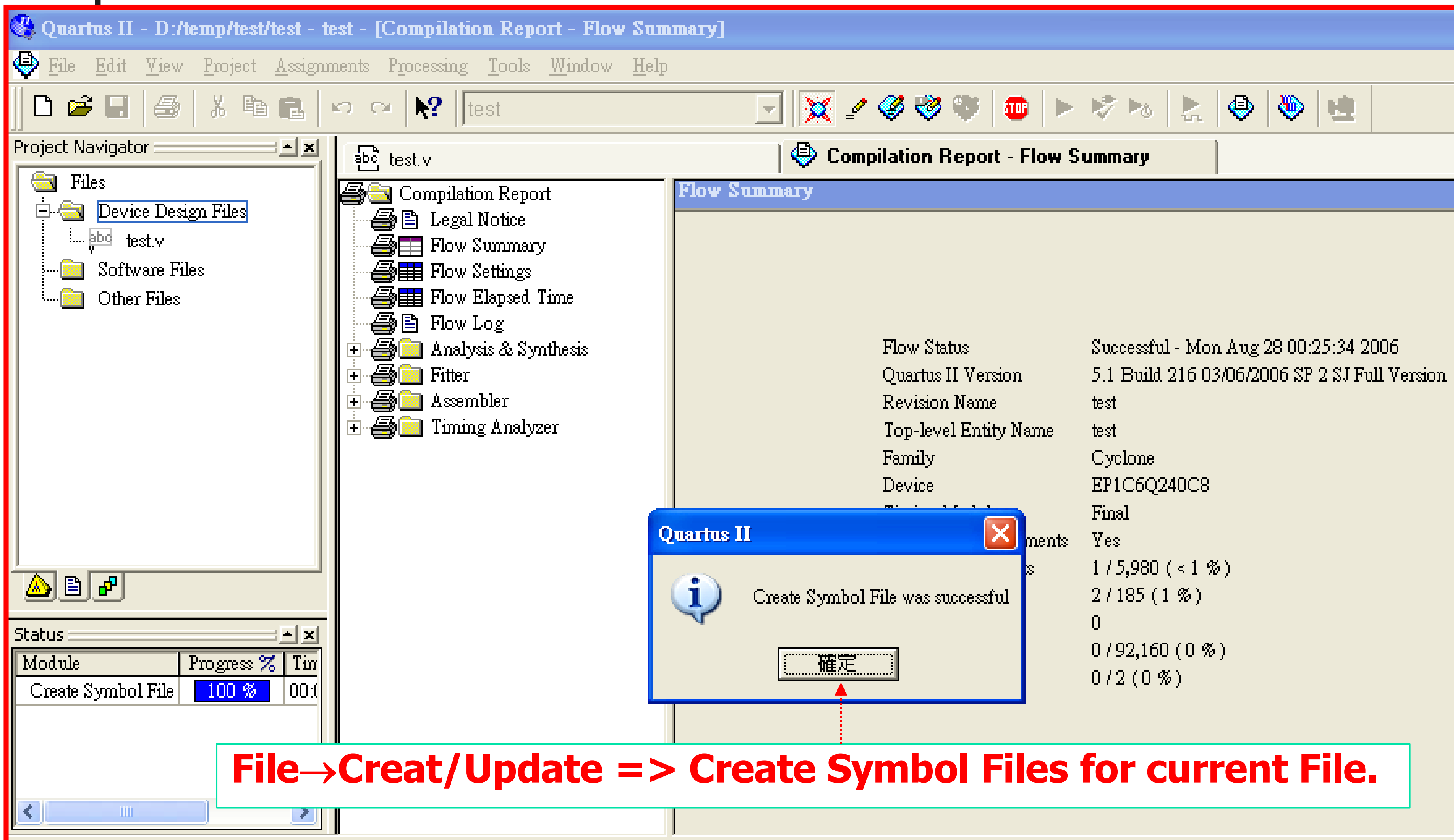
“Files” → 指定 ” Files” → 按右鍵 → 選定 :” Add/Remove Files in project”.
→ Select ” 103_tcfst_FPGA_example\CH04_Design three methods\d_ff_main_code\d_ff_main_code.v”
→ADD

Bottom-Up Design (3)

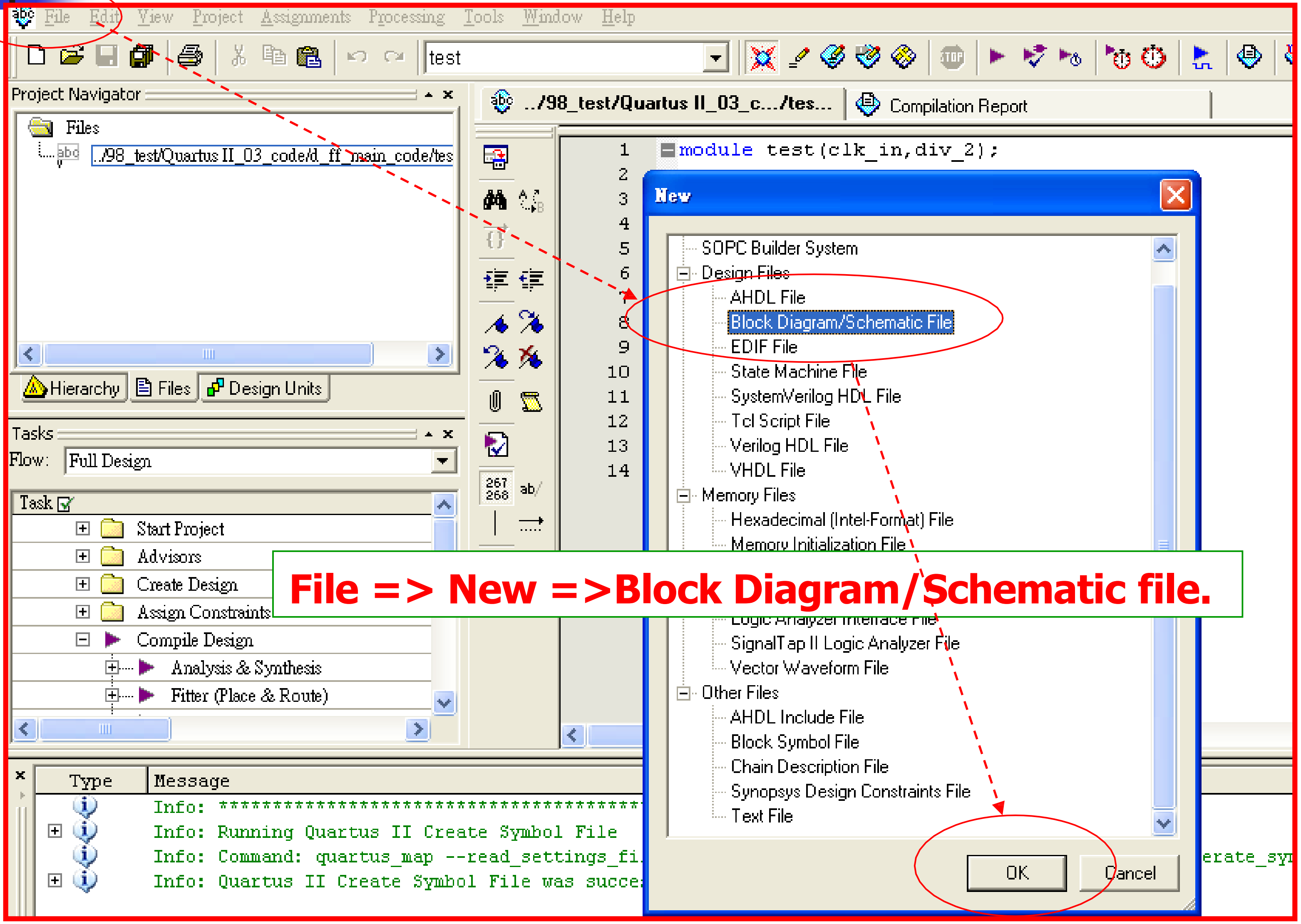


Additional Function => Complier => wrong ???
=>Model name “d_ff_main_code” ≠ Project name “test”.

Bottom-Up Design (4)

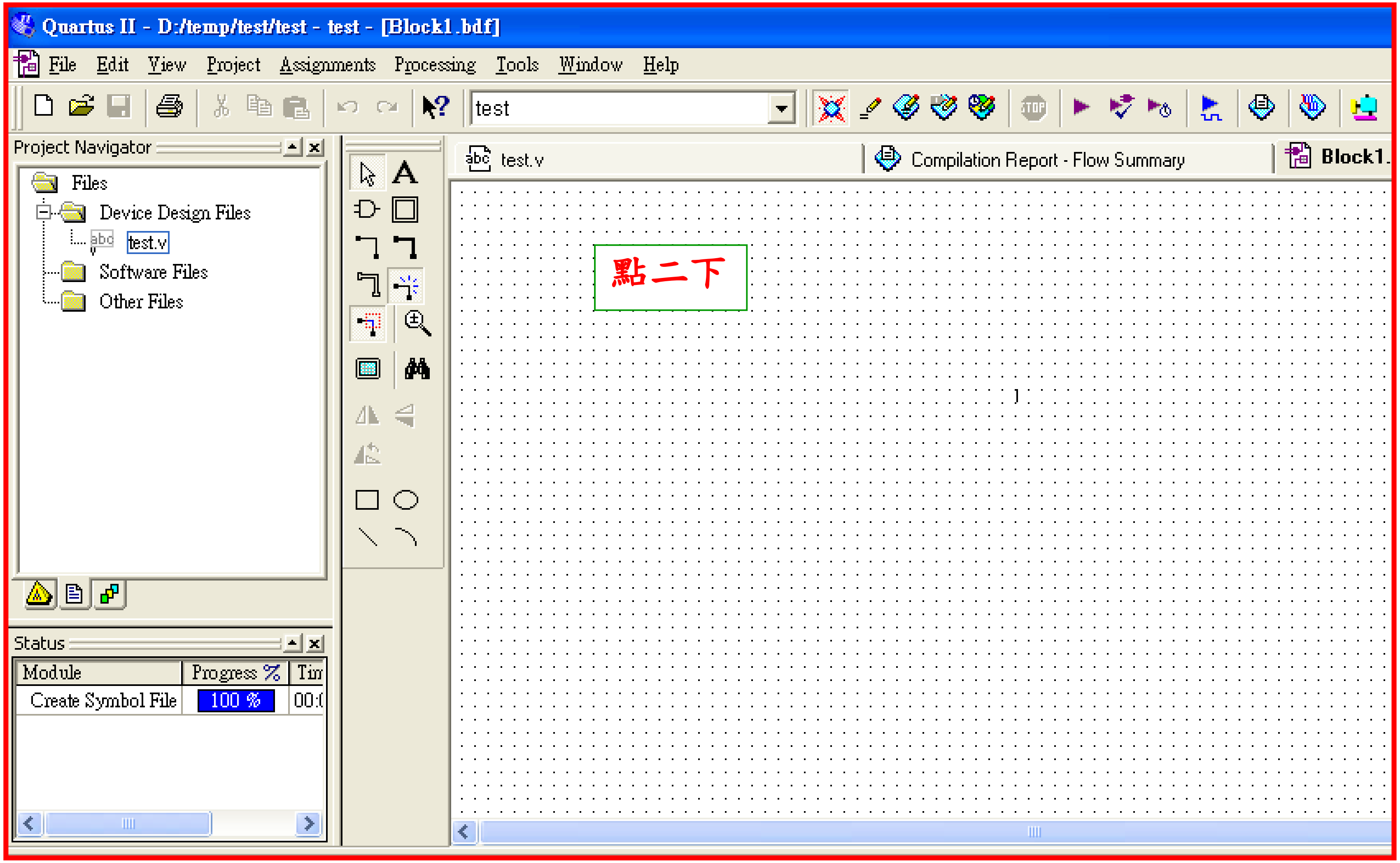


Bottom-Up Design (5)

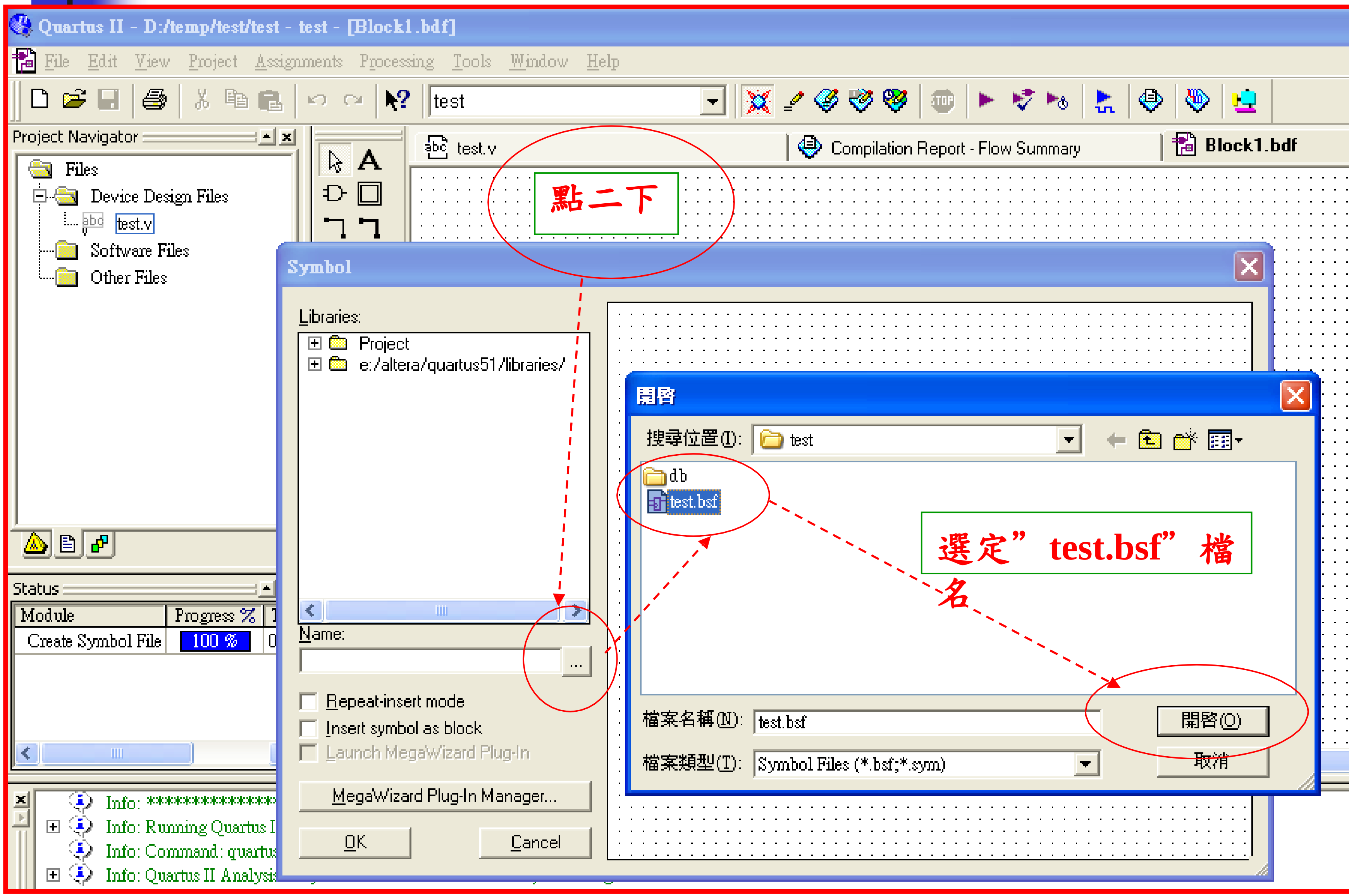


File => New =>Block Diagram/Schematic file.

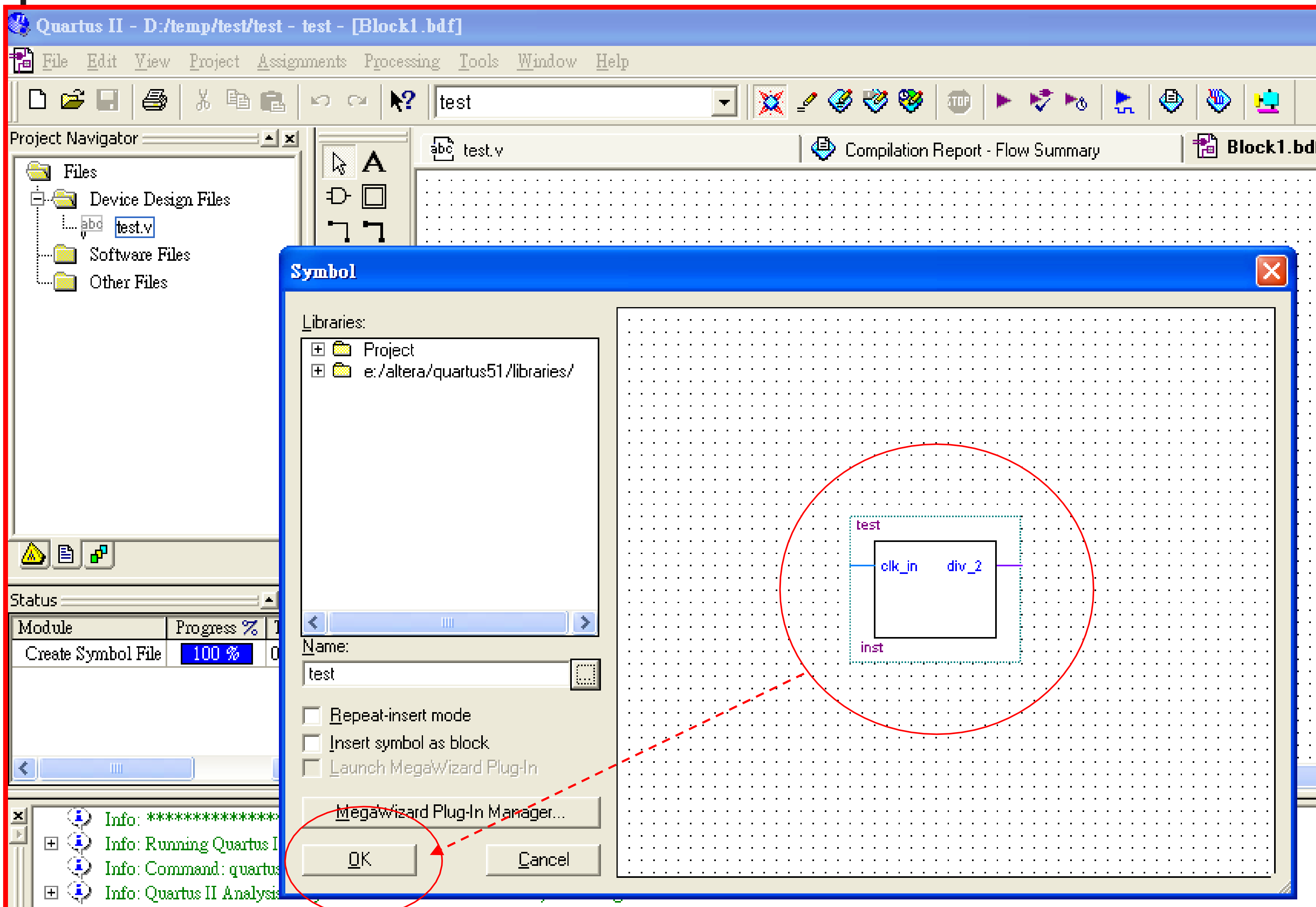
Bottom-Up Design (6)



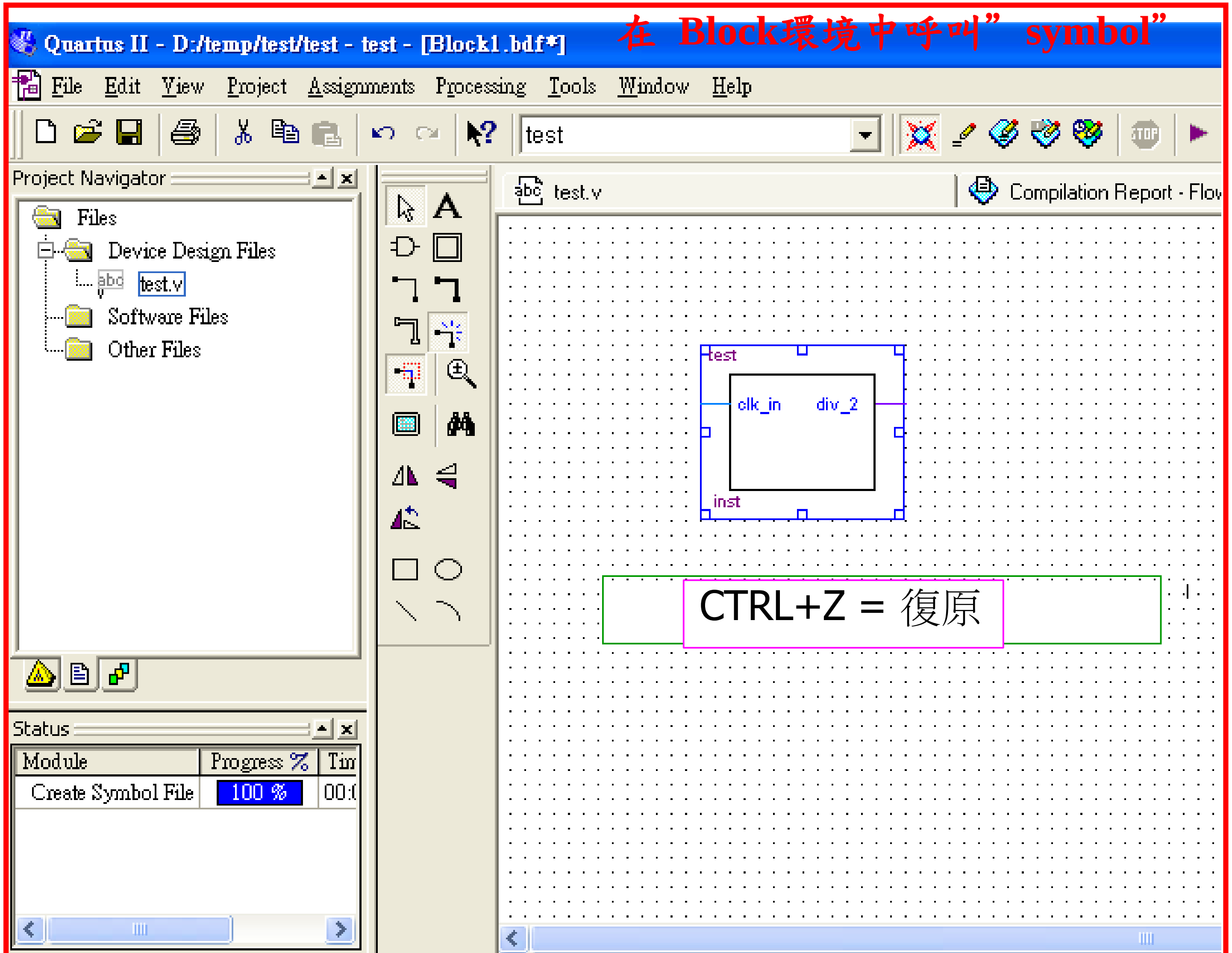
Bottom-Up Design (7)



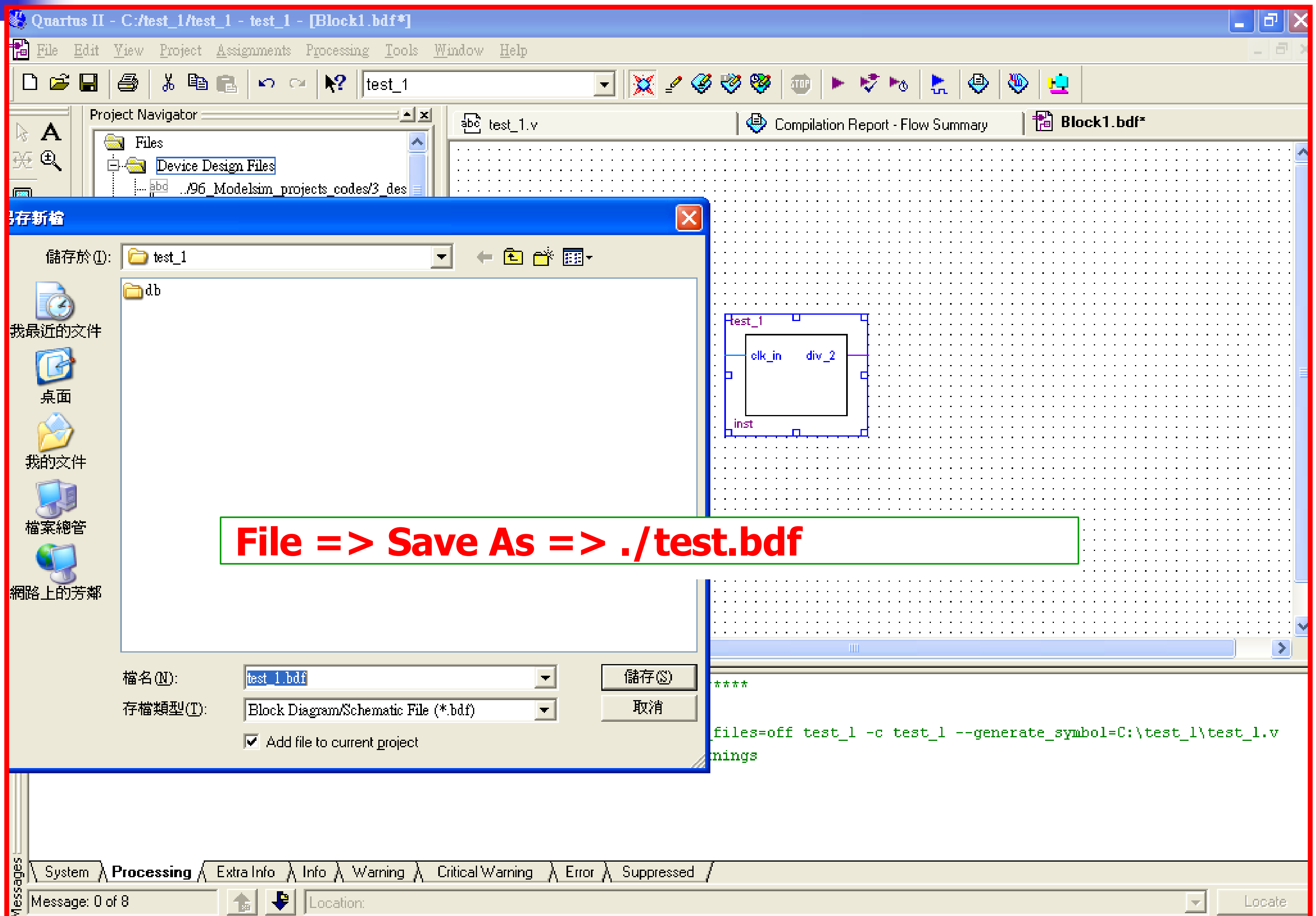
Bottom-Up Design (8)



Bottom-Up Design (9)

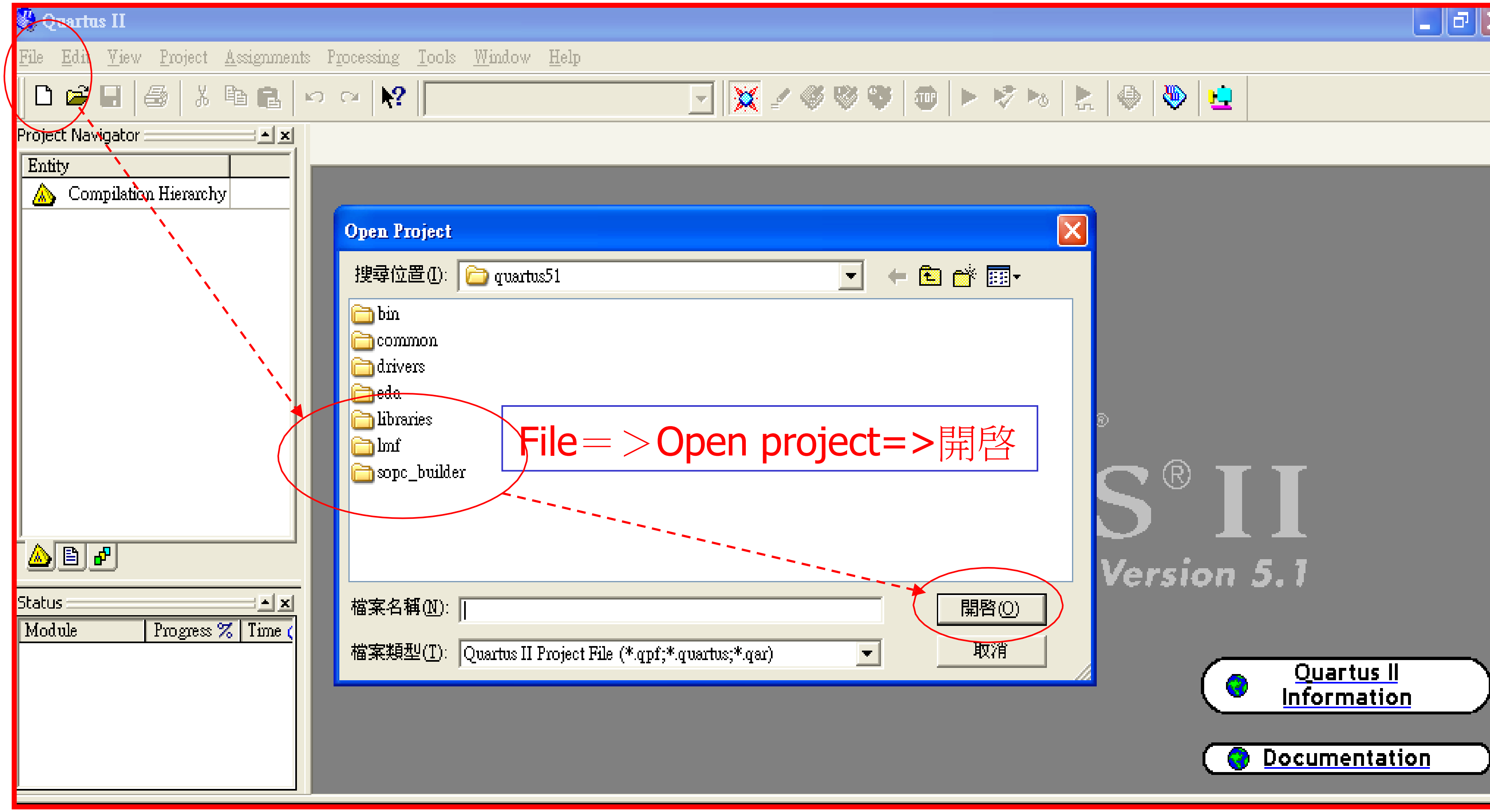


Bottom-Up Design (9)

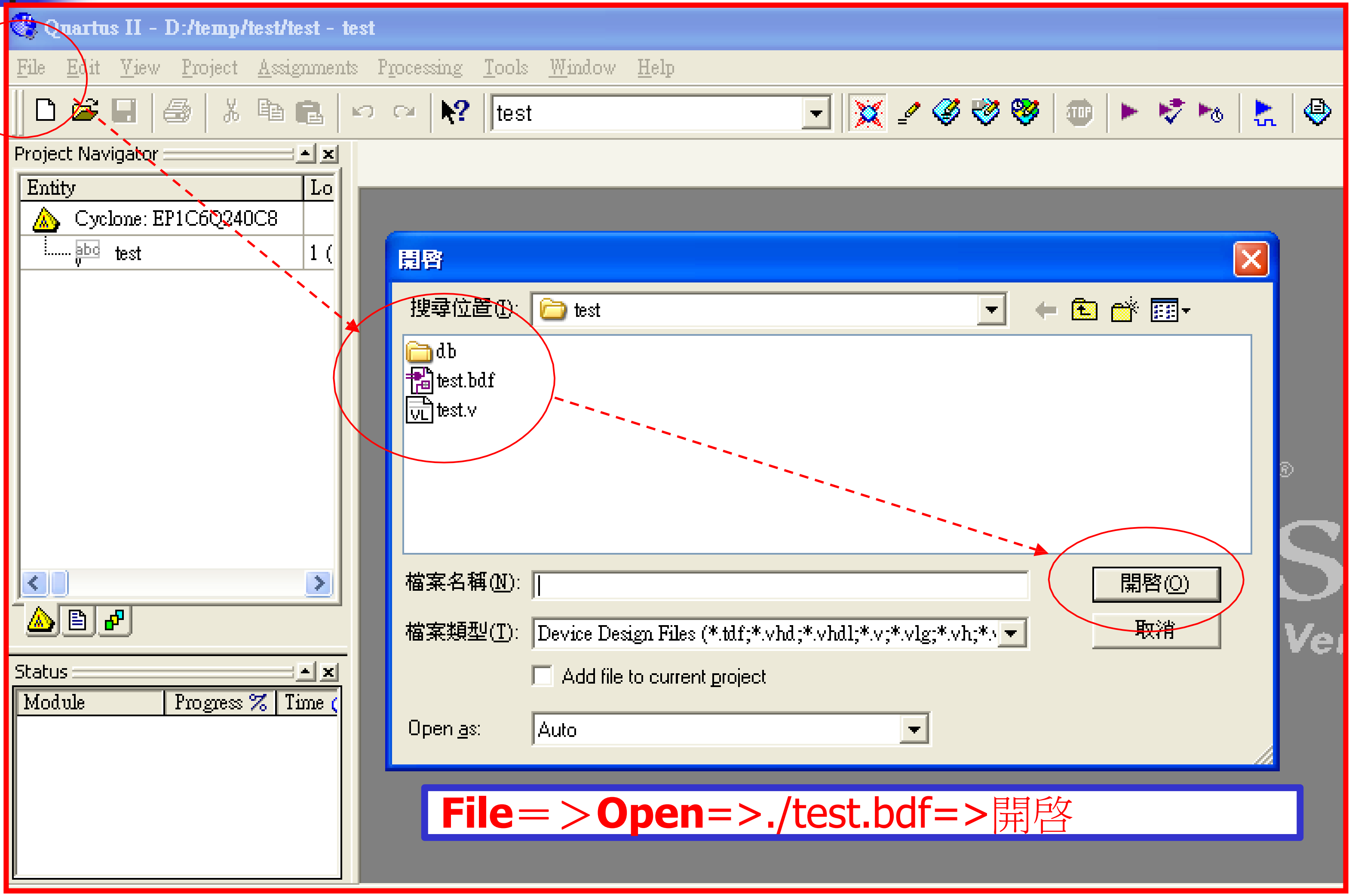


File => Save As => ./test.bdf

Bottom-Up Design (10)

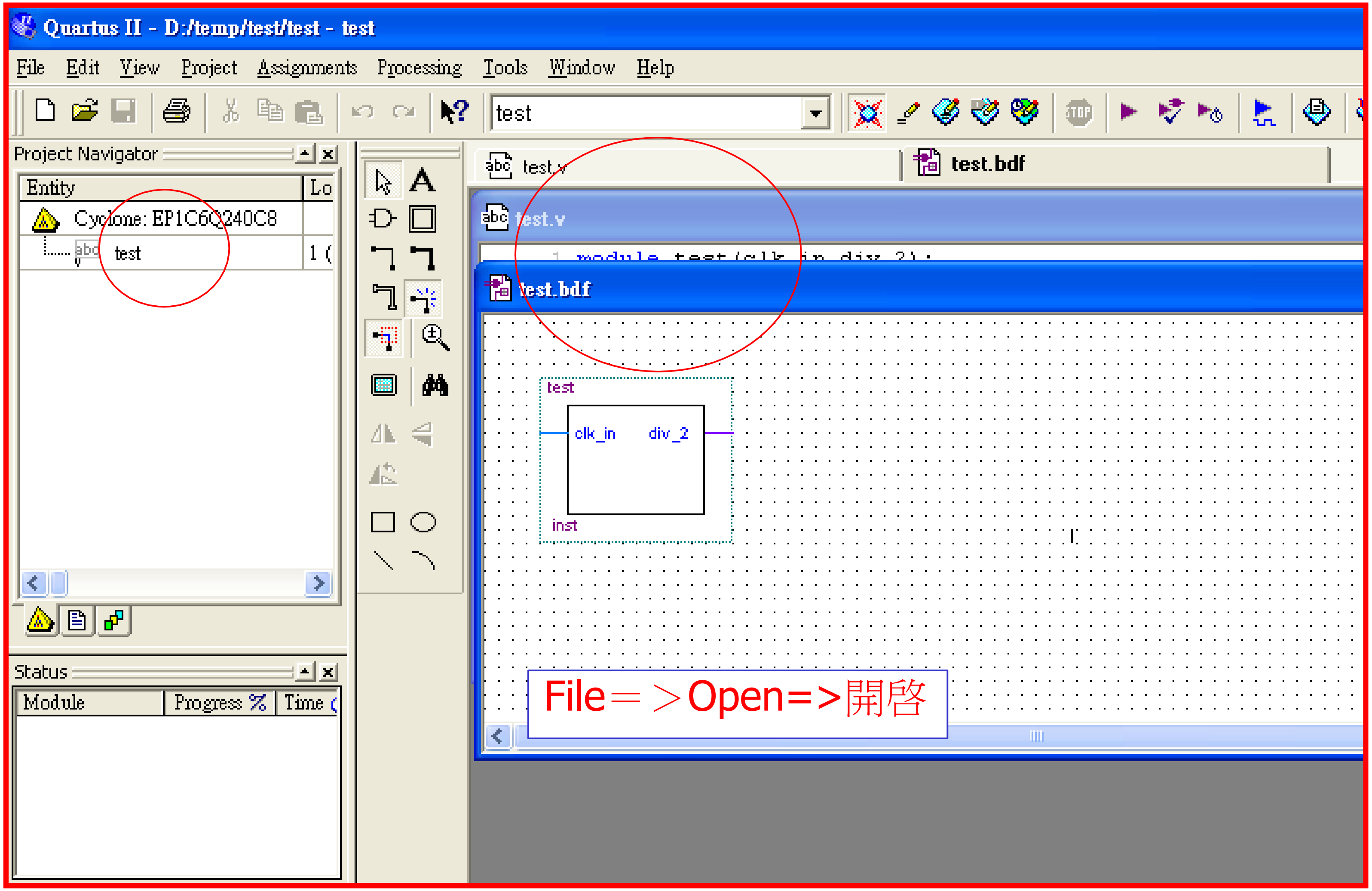


Bottom-Up Design (11)

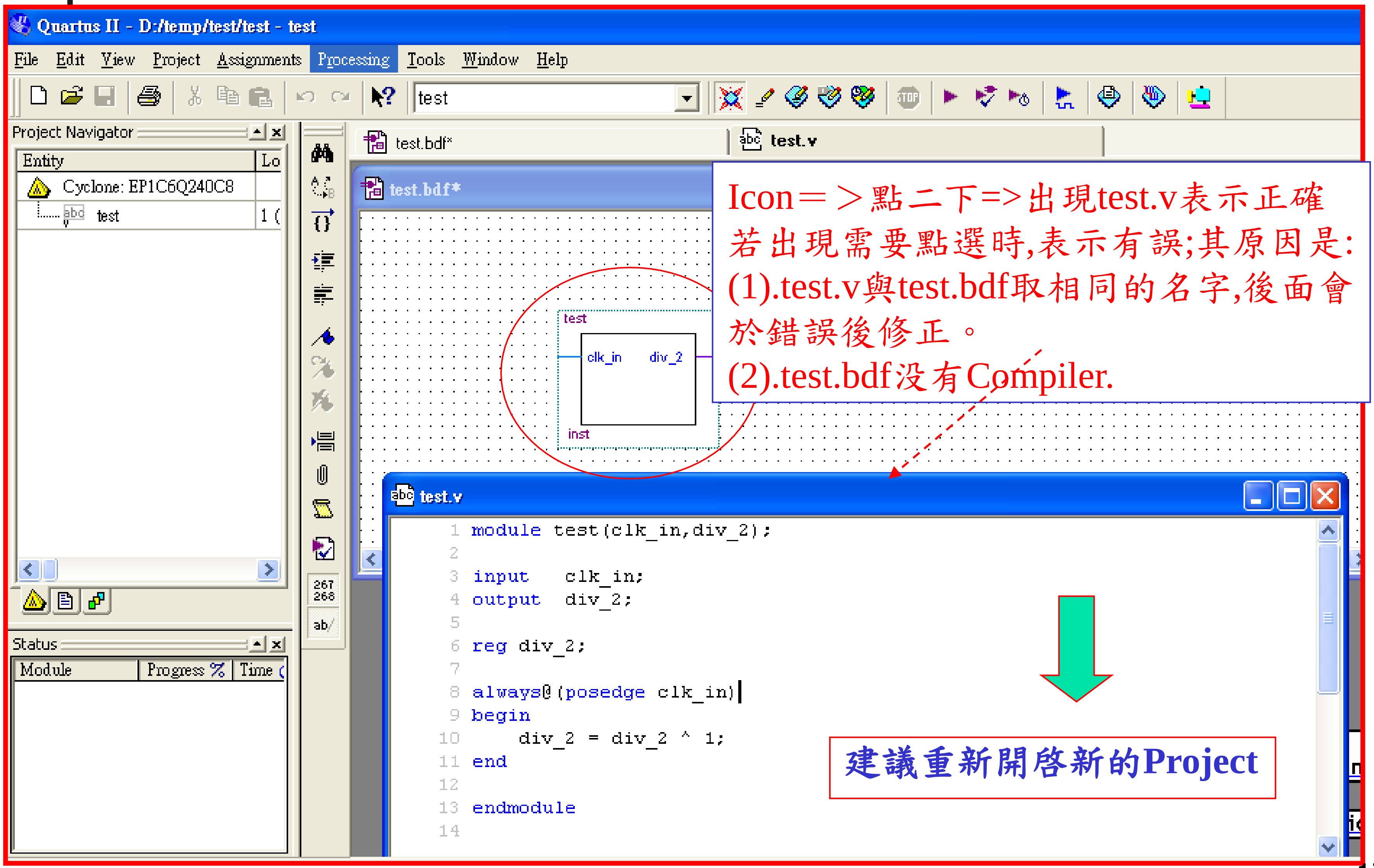


File＝＞Open=>./test.bdf=>開啓

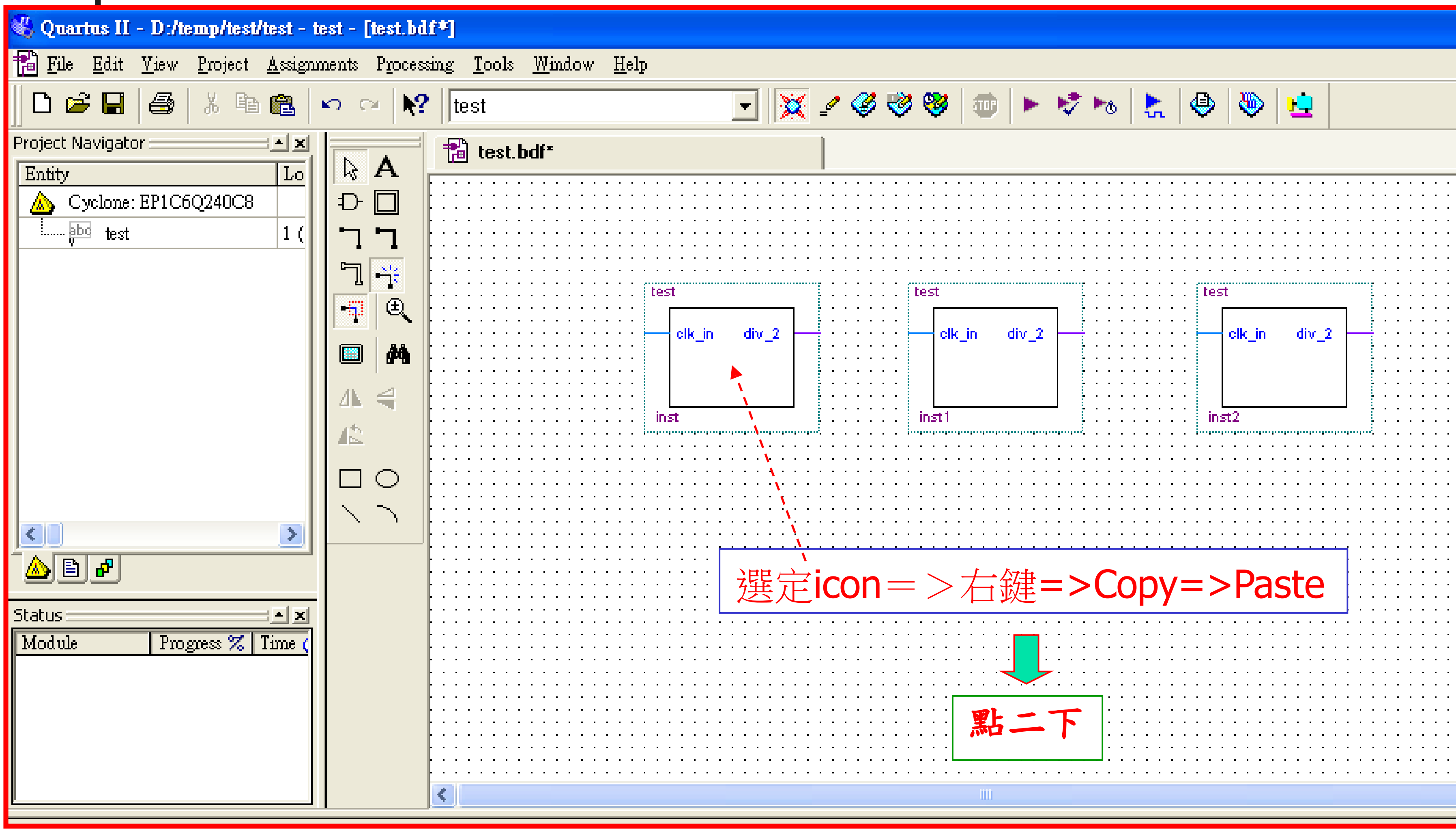
Bottom-Up Design (12)



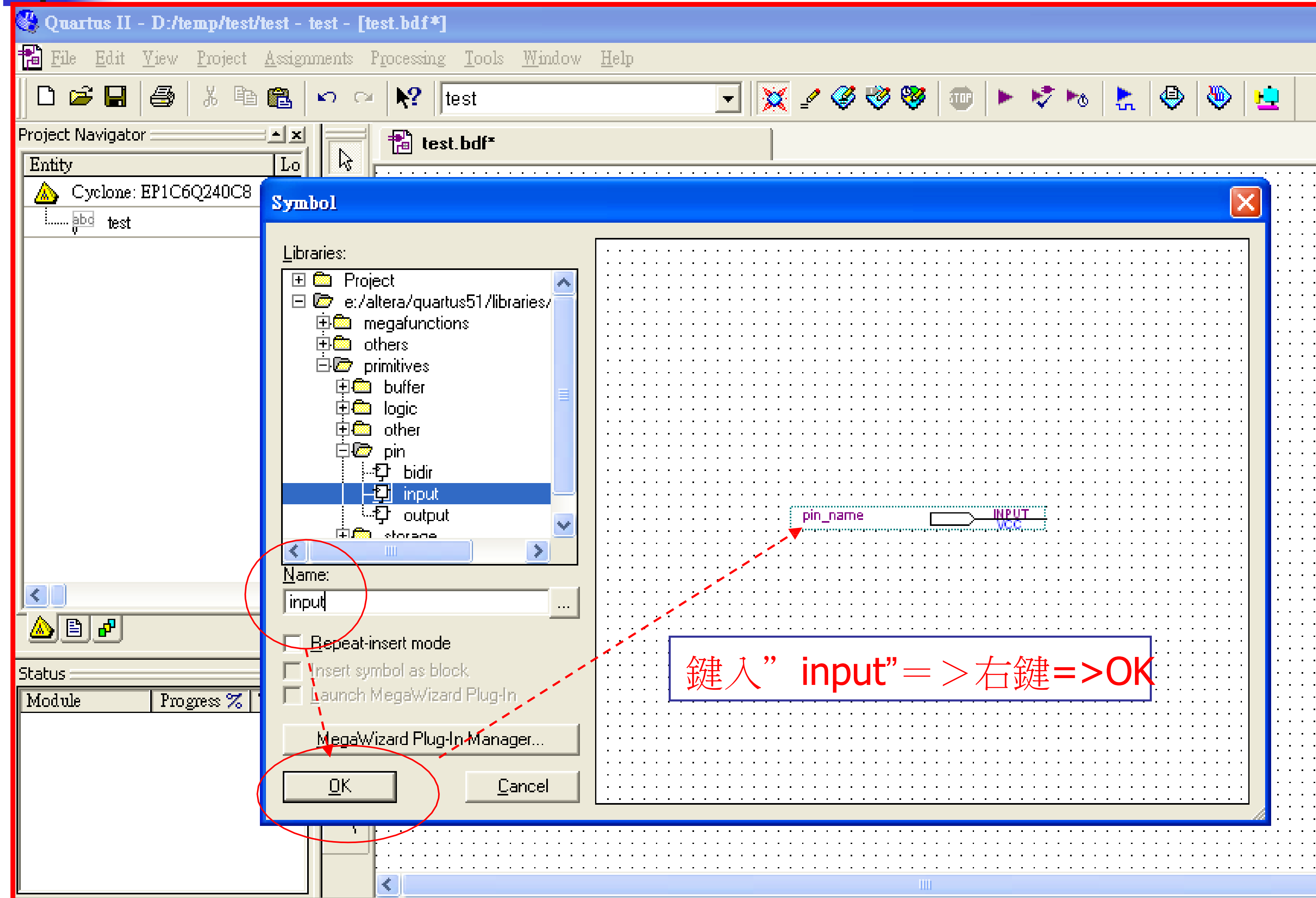
Bottom-Up Design (13)



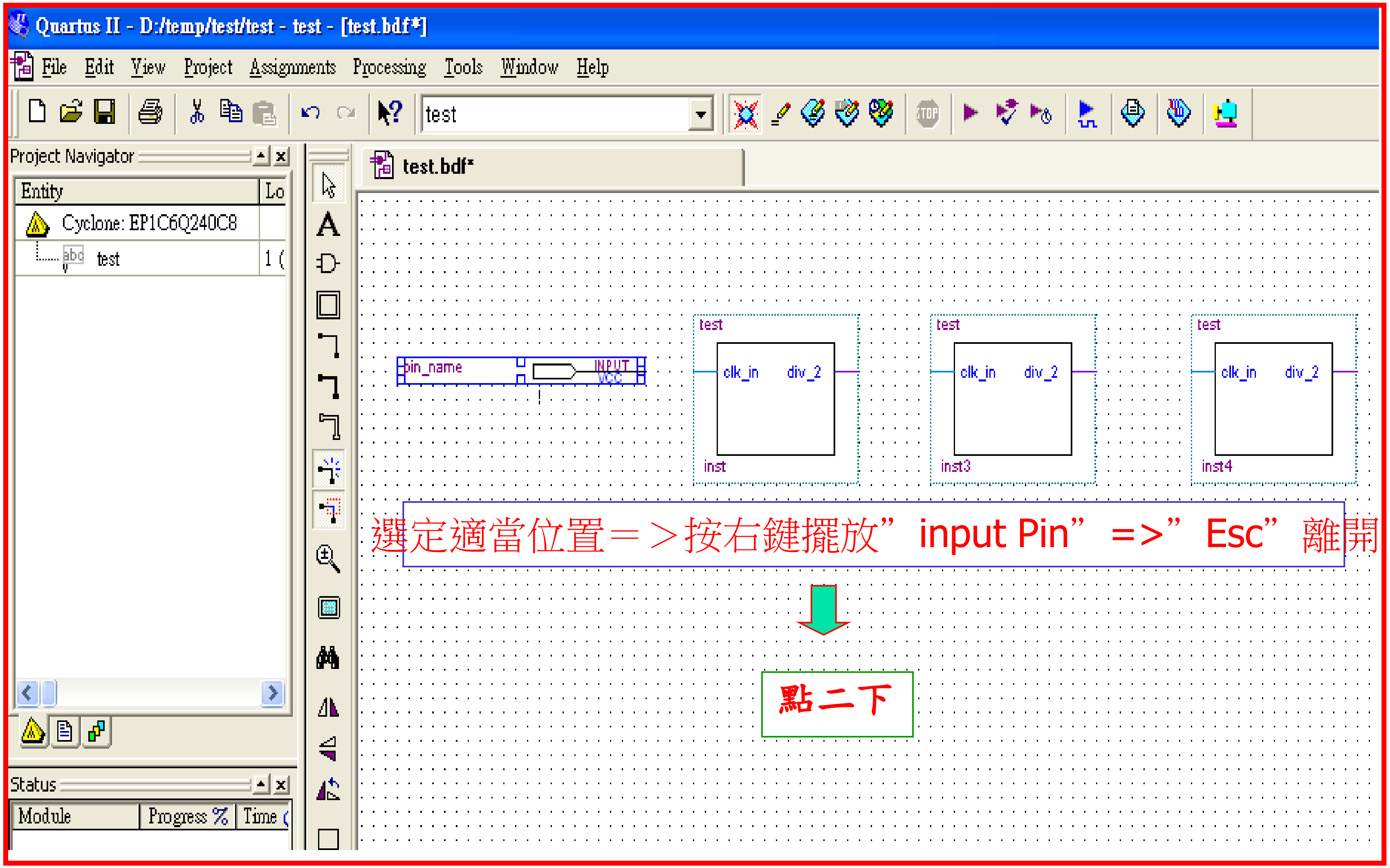
Bottom-Up Design (14)



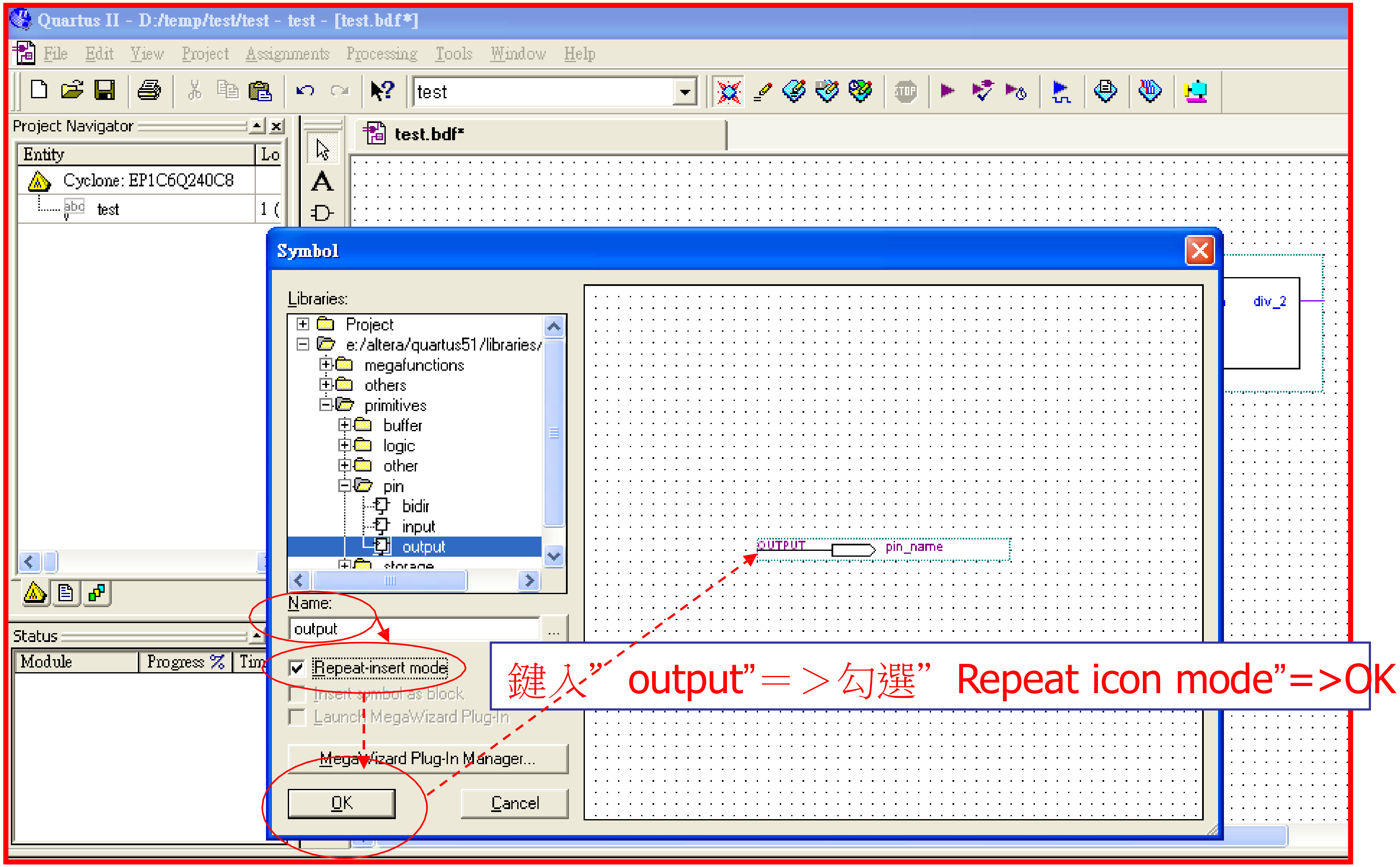
Bottom-Up Design (15)



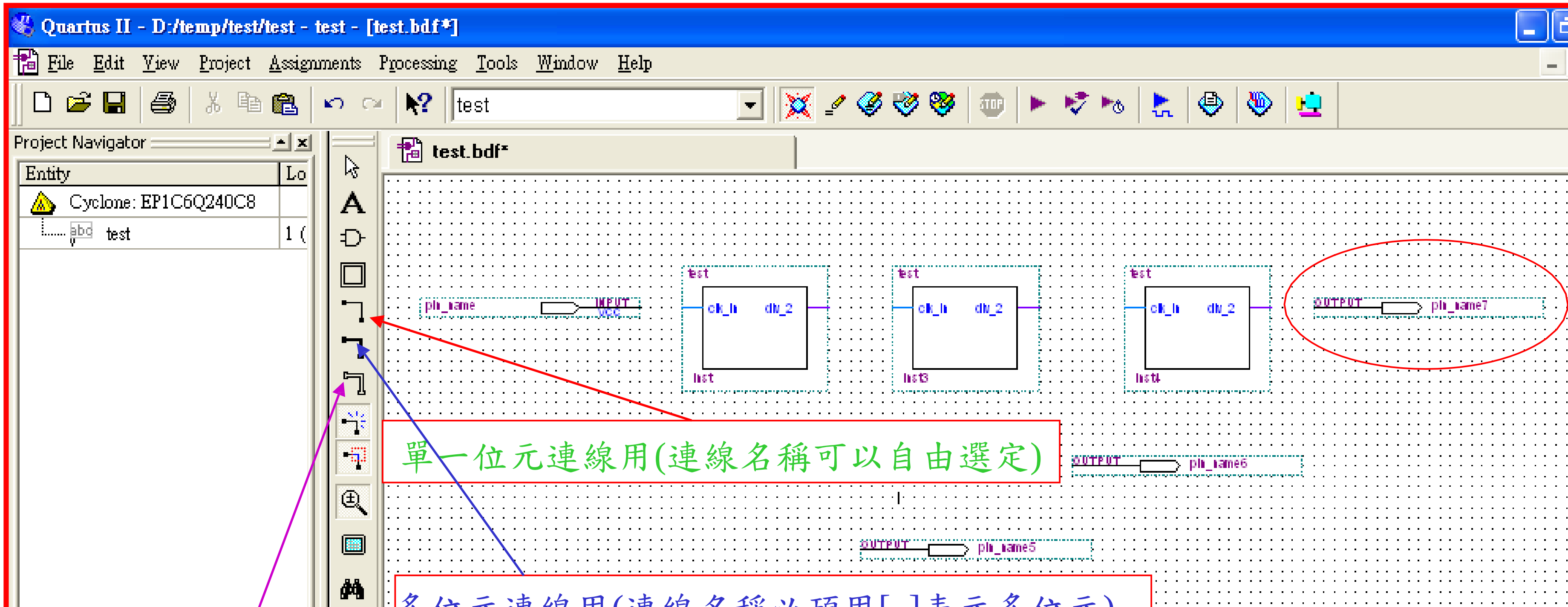
Bottom-Up Design (16)



Bottom-Up Design (17)



Bottom-Up Design (18)



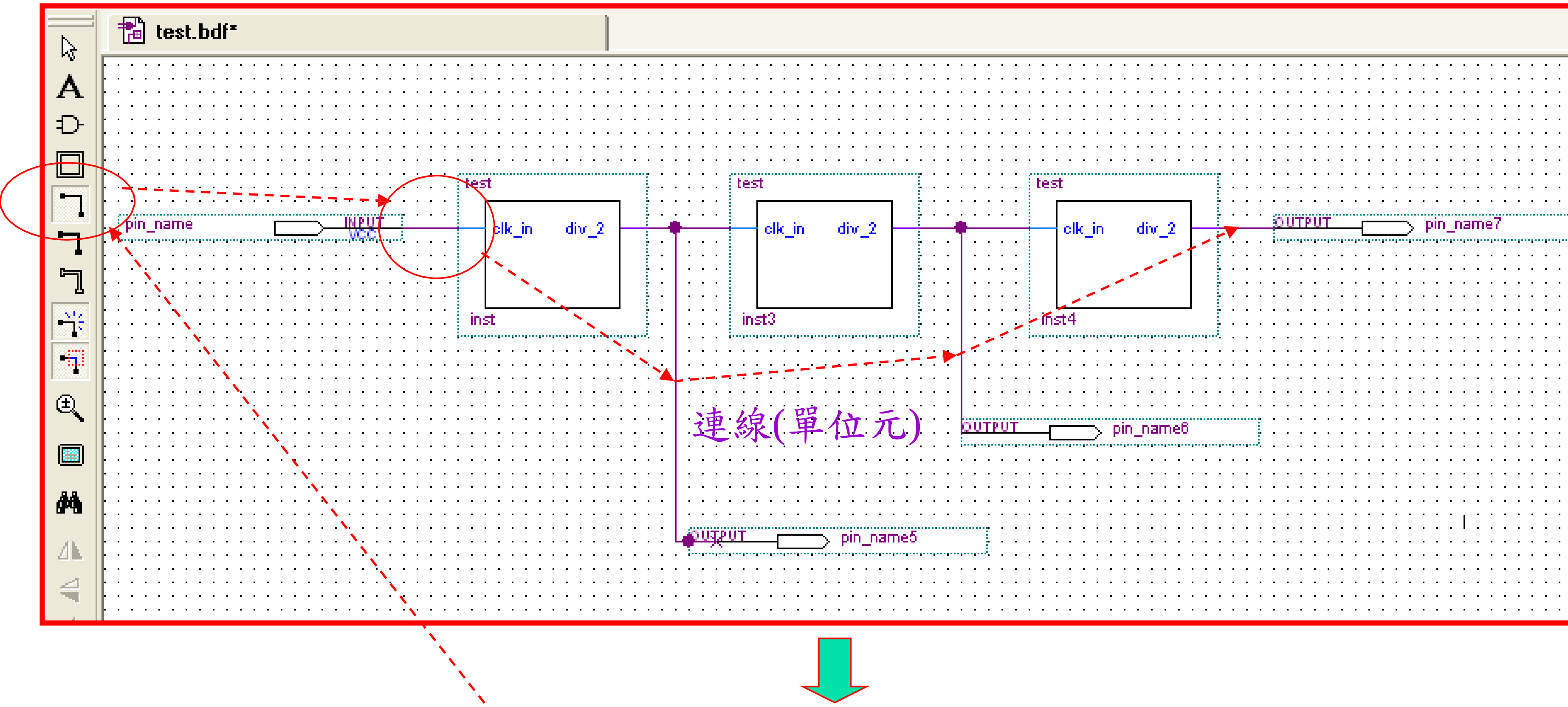
單一位元連線用(連線名稱可以自由選定)

多位元連線用(連線名稱必須用[..]表元多位元)

Top-down設計法則用(連線名稱必須指定)

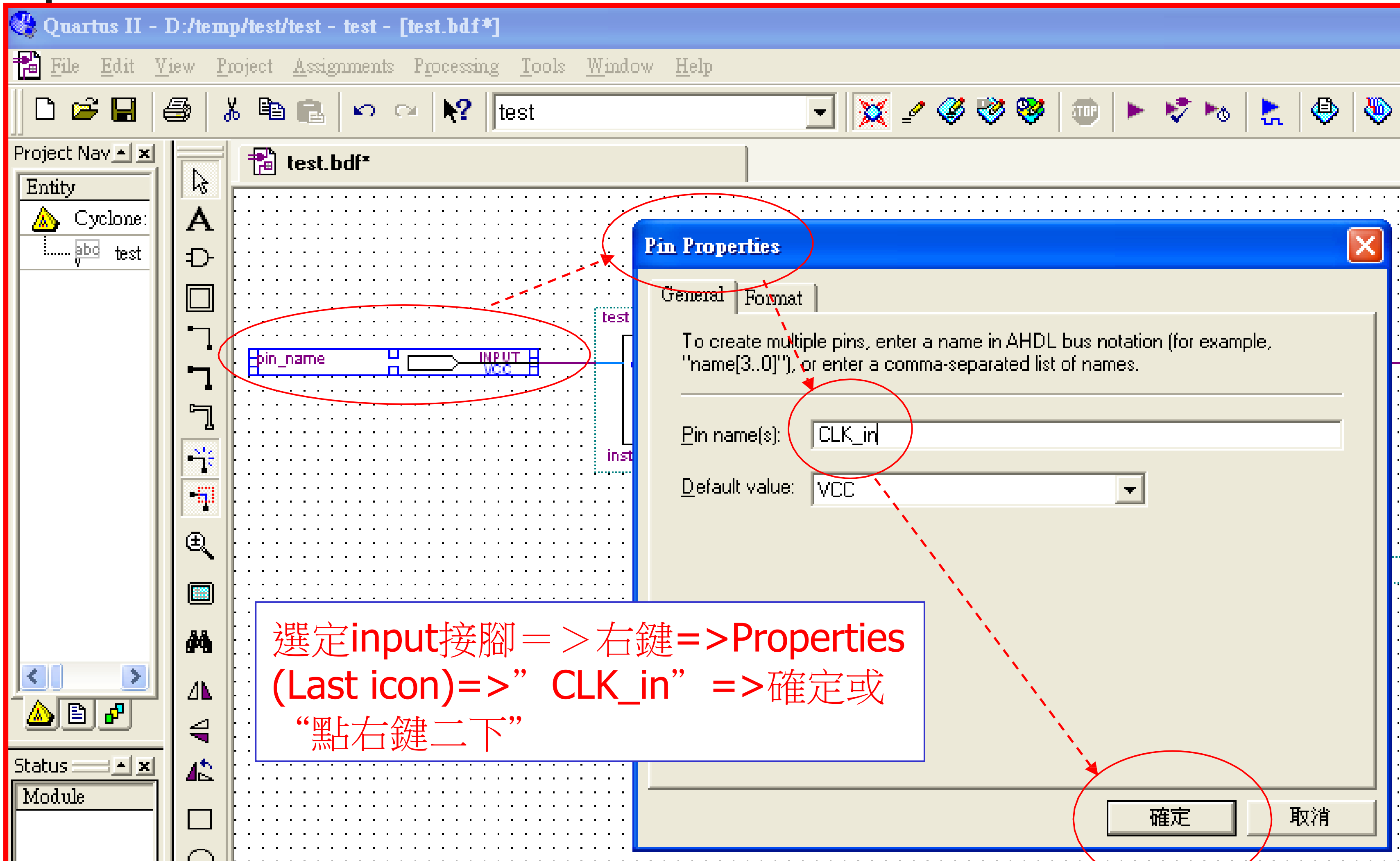
選定適當位置(三處)＝＞右鍵擺放=> ”Esc”離開. → 拉線

Bottom-Up Design (19)



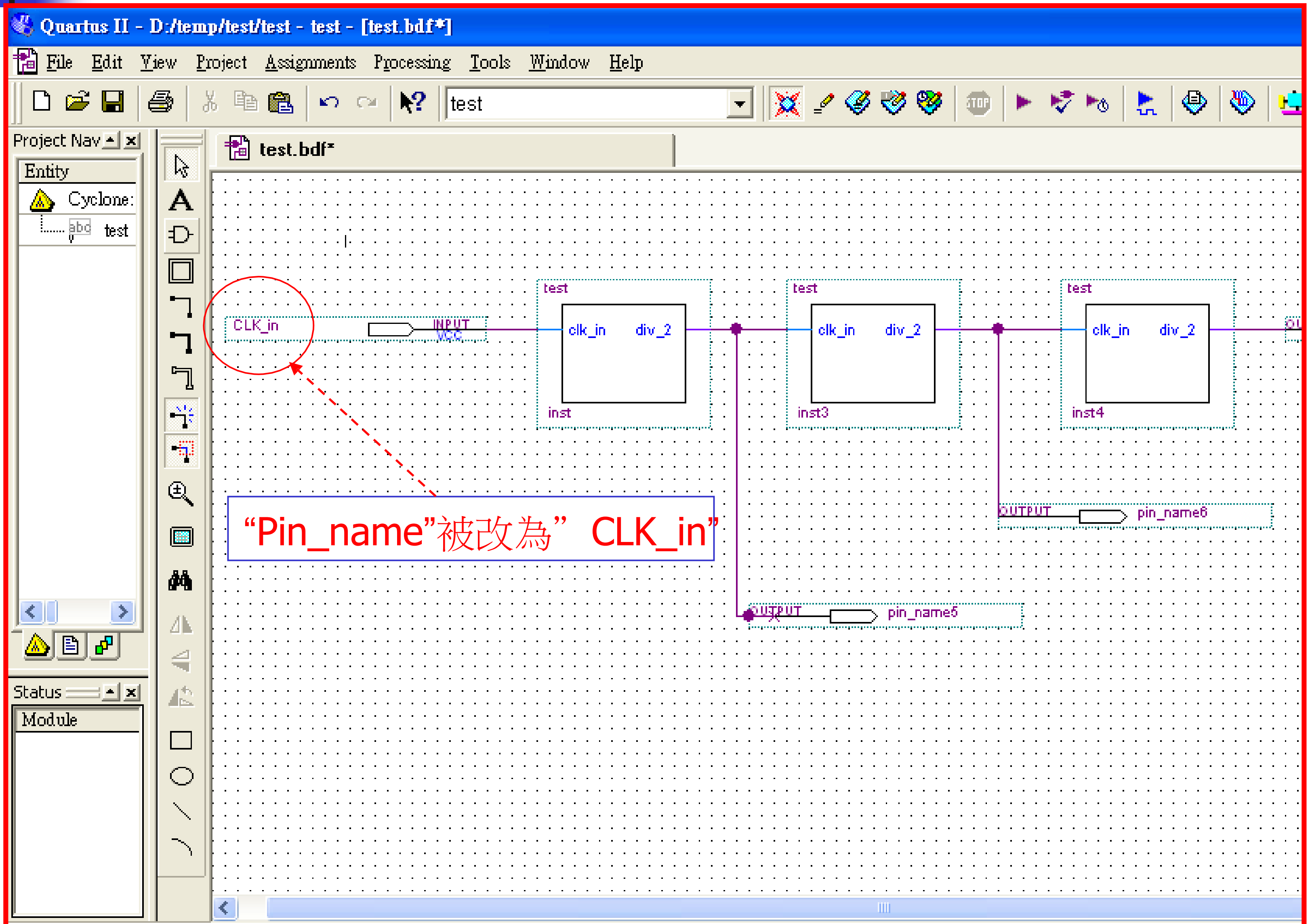
選定”連線icon”＝＞ 連線(單位元) =>更改輸出入接腳的名稱

Bottom-Up Design (20)

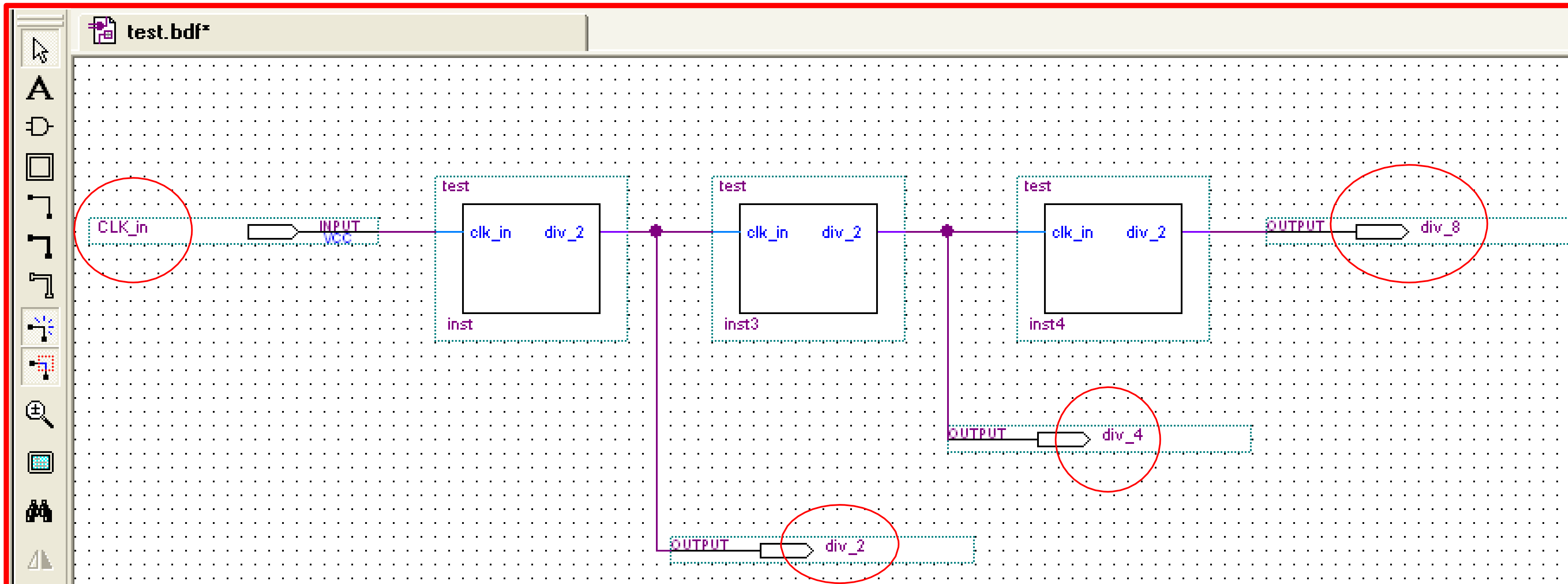


選定input接腳＝＞右鍵=>Properties (Last icon)=>” CLK_in” =>確定或 “點右鍵二下”

Bottom-Up Design (21)



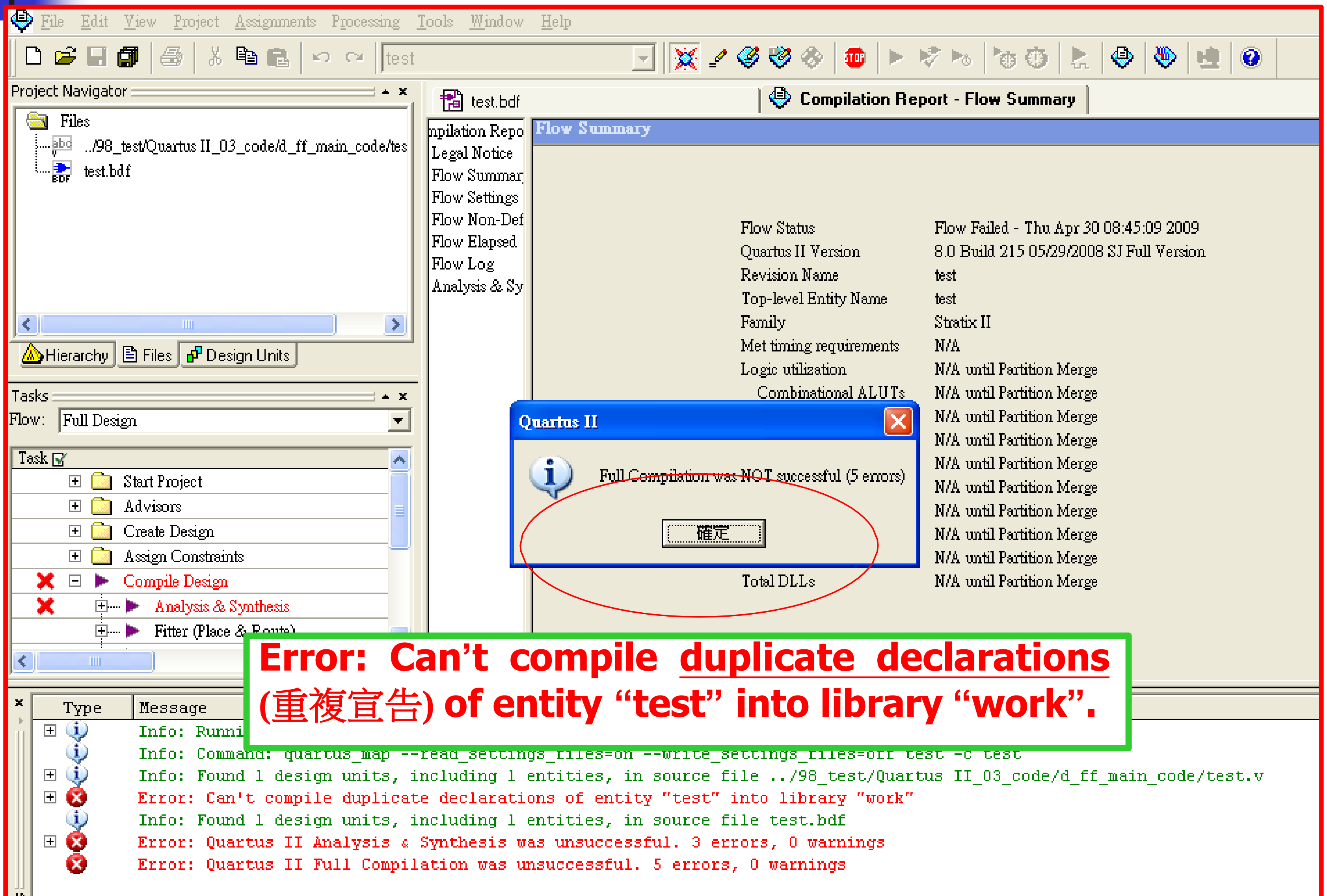
Bottom-Up Design (22)



"Pin_name"全部更改為:
(1). Input pin: 被改為" clk_in"
(2). Output pin: 被改為" div_2"
(3). Output pin: 被改為" div_4"
(4). Output pin: 被改為" div_8"

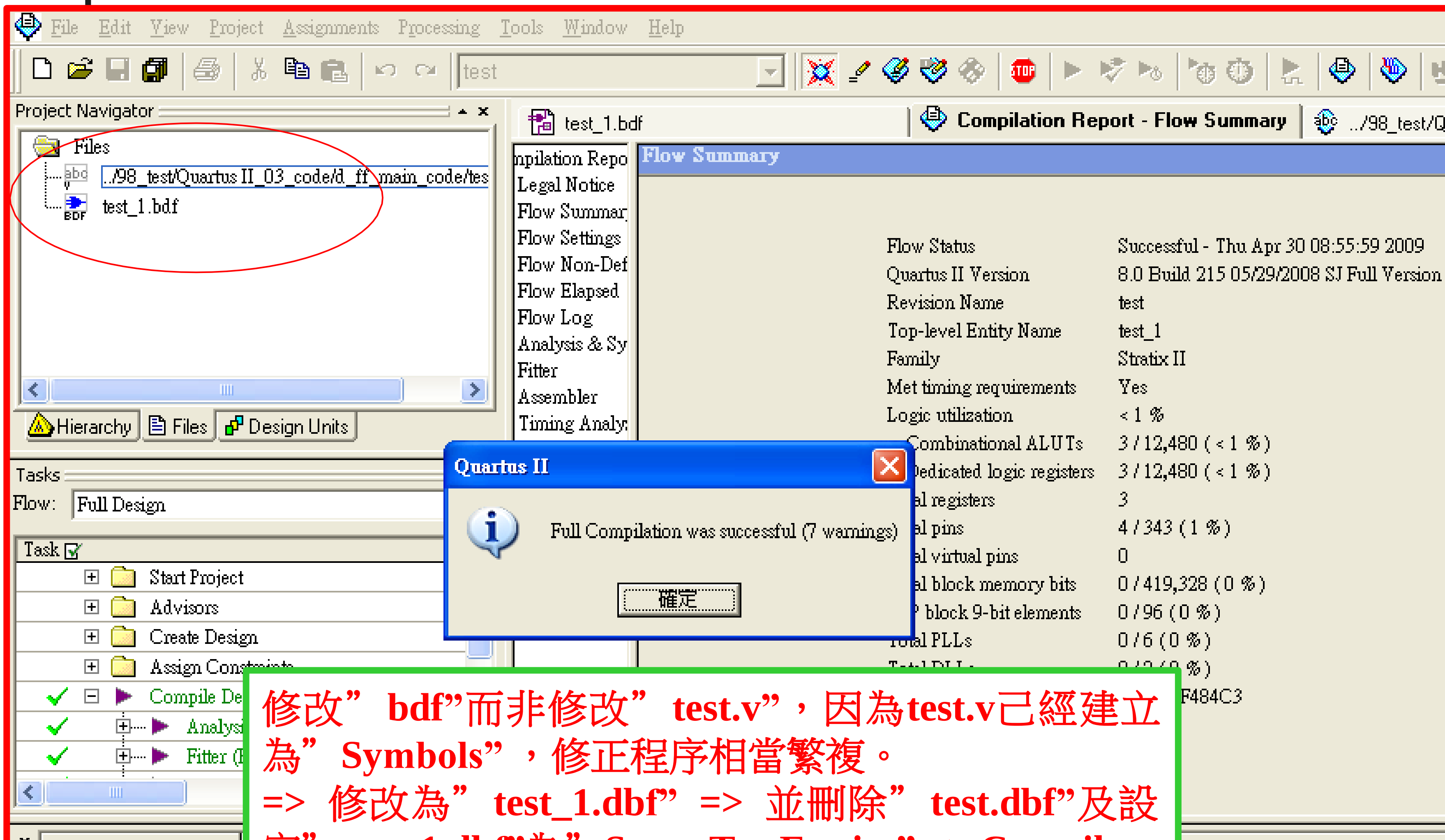
"Save" => "Compiler icon"(" ►").

Bottom-Up Design (23)



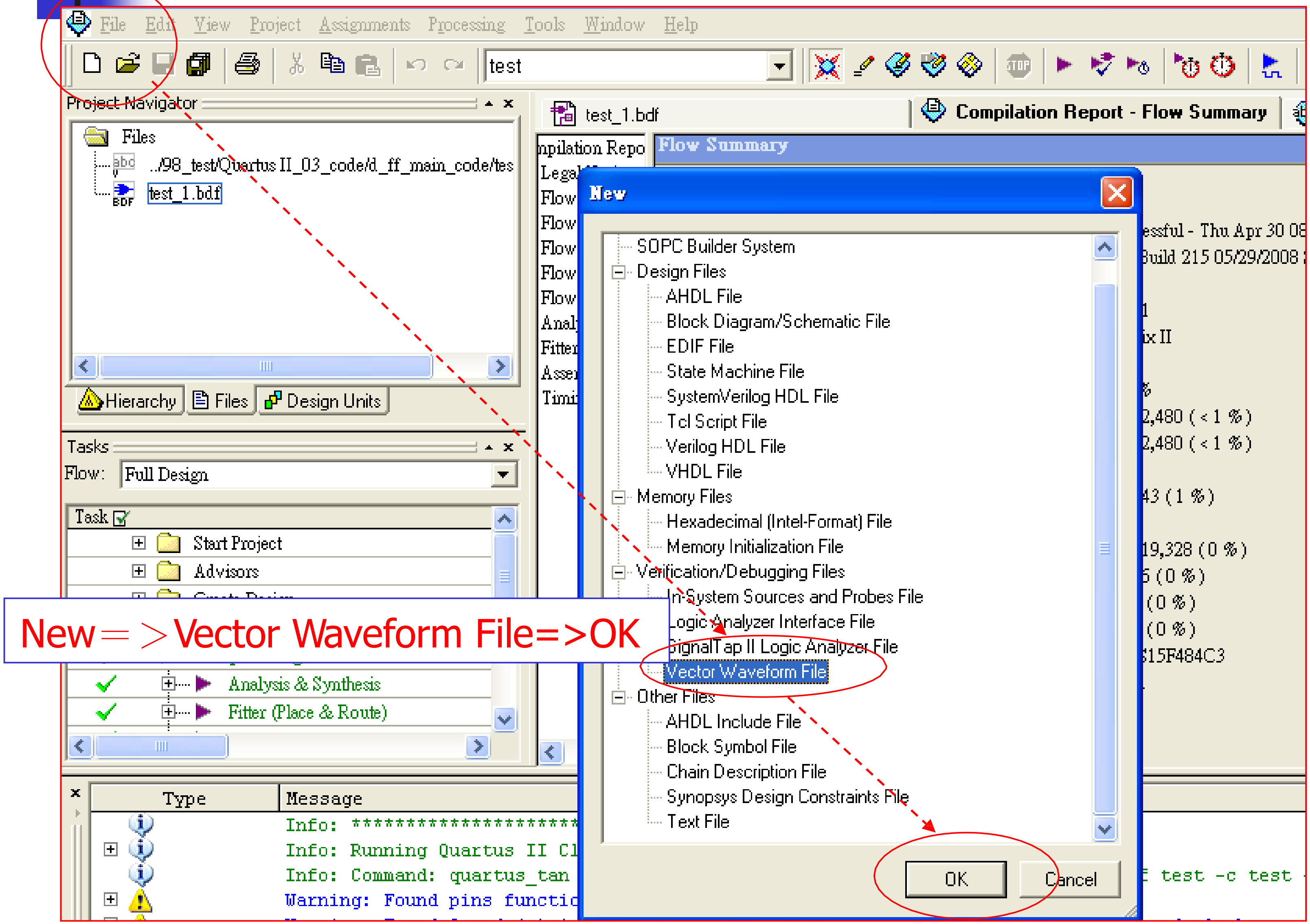
Error: Can't compile duplicate declarations (重複宣告) of entity "test" into library "work".

Bottom-Up Design (24)

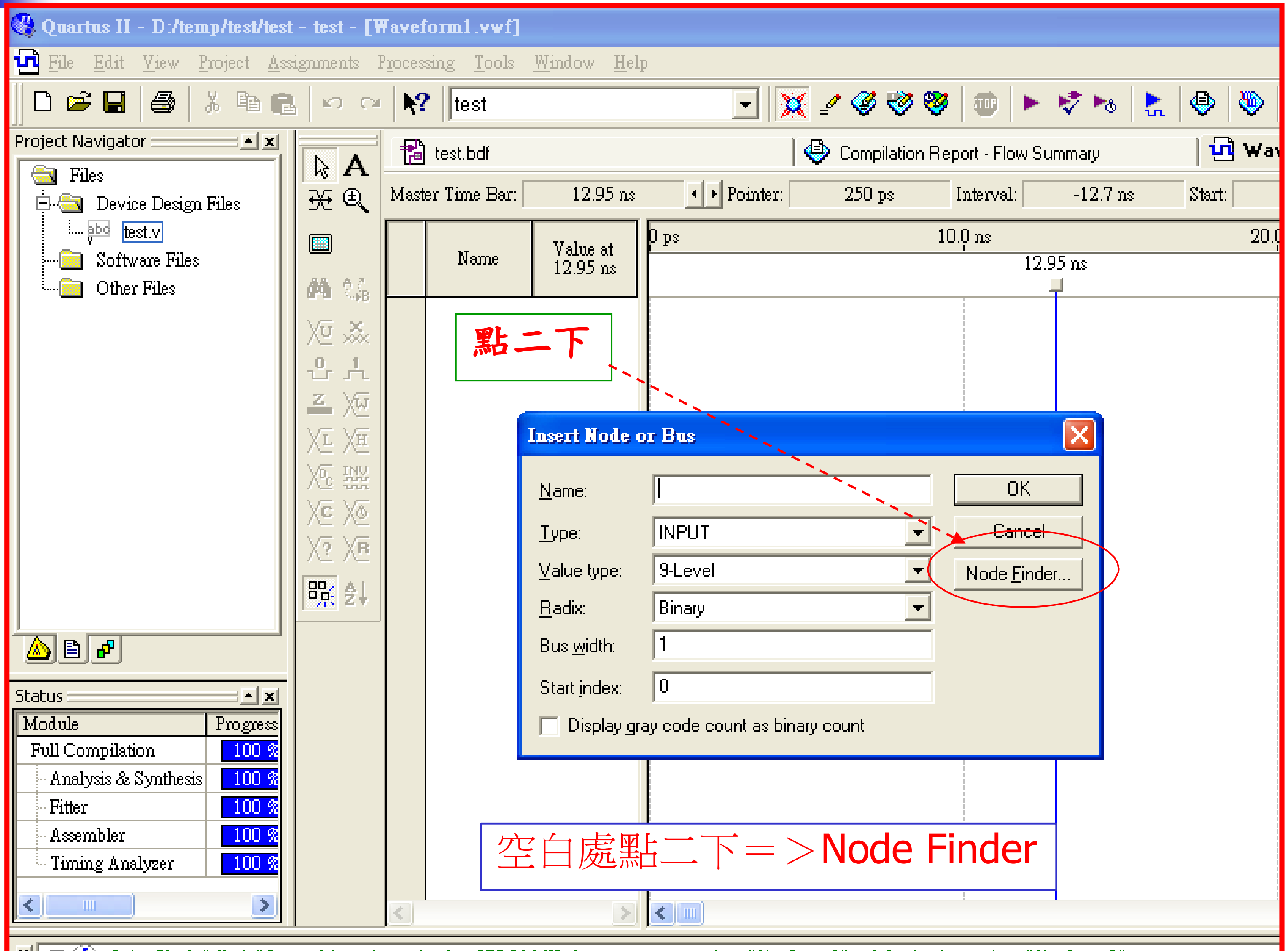


修改”bdf”而非修改”test.v”，因為test.v已經建立為”Symbols”，修正程序相當繁複。
=> 修改為”test_1.dbf” => 並刪除”test.dbf”及設定”test_1.dbf”為”Set as Top Emtity” => Compiler.

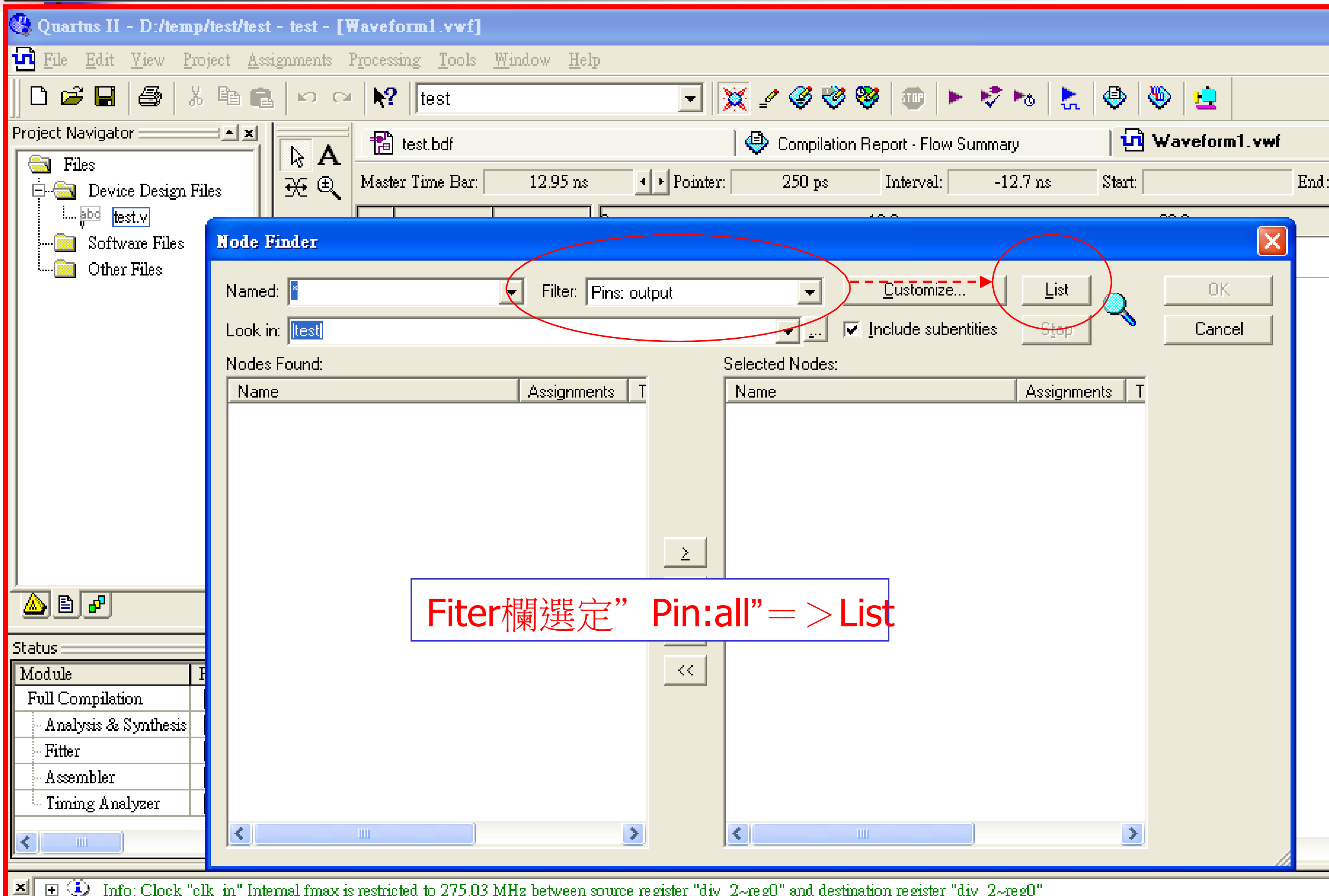
Bottom-Up Design (25)



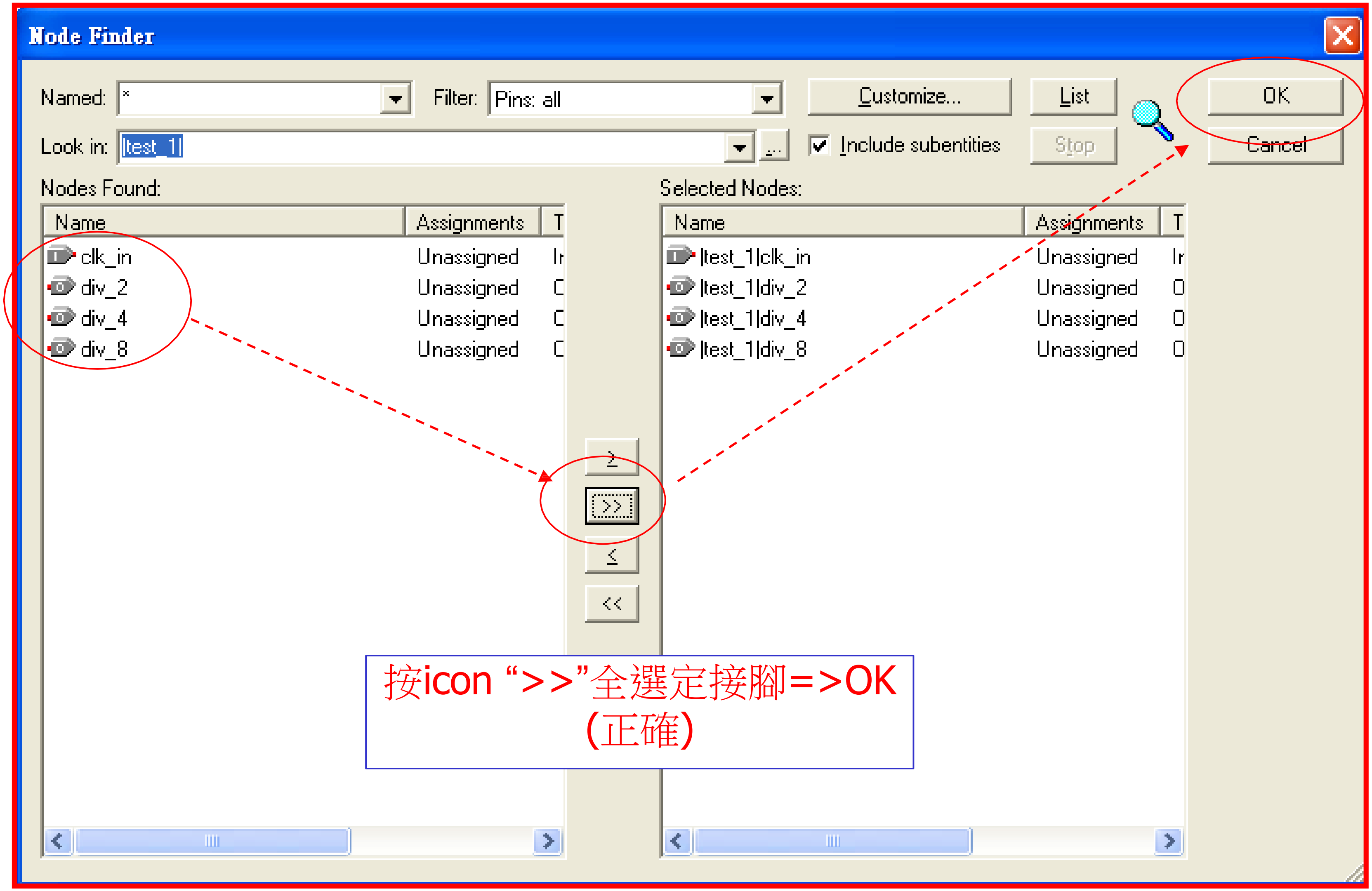
Bottom-Up Design (26)



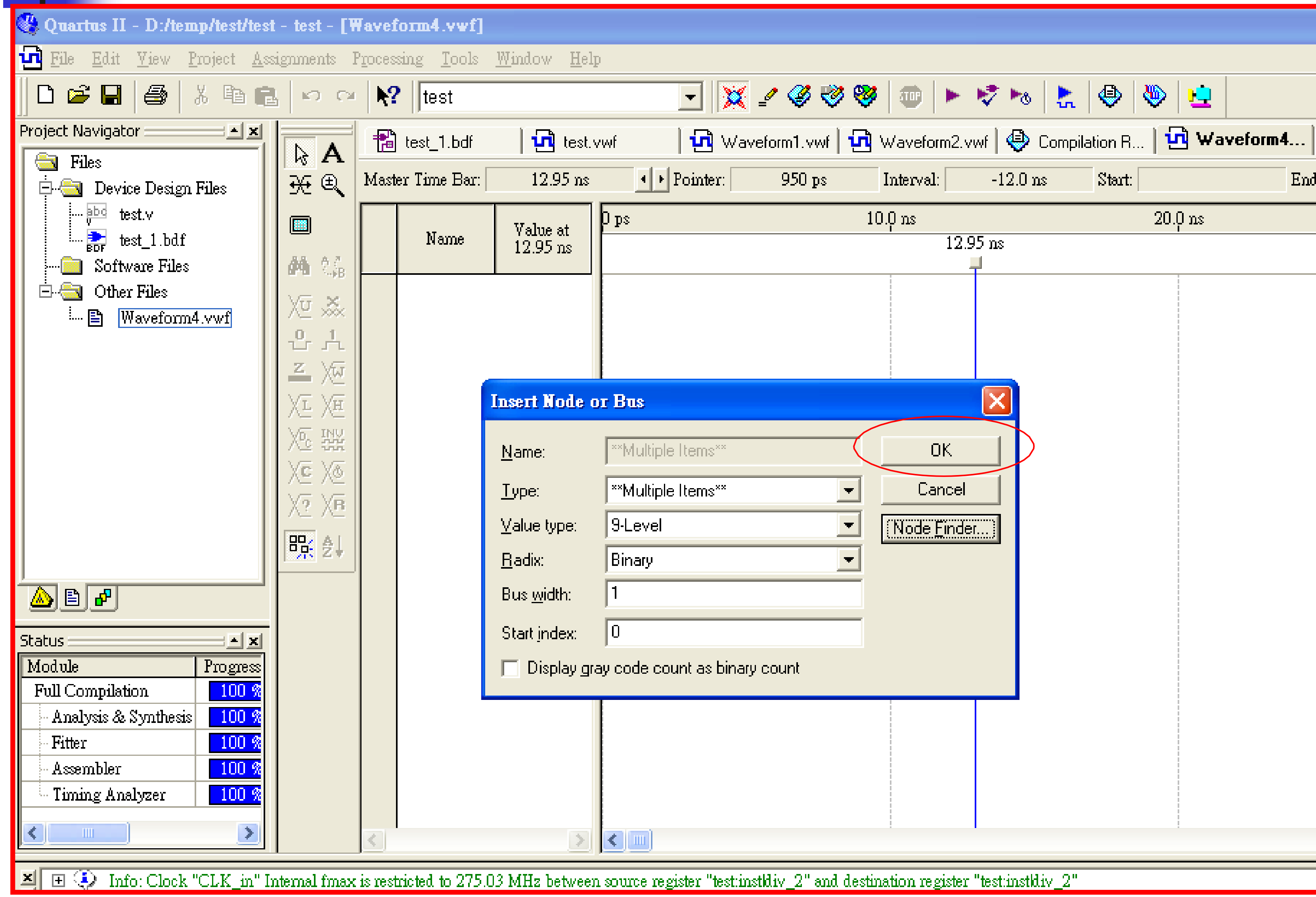
Bottom-Up Design (27)



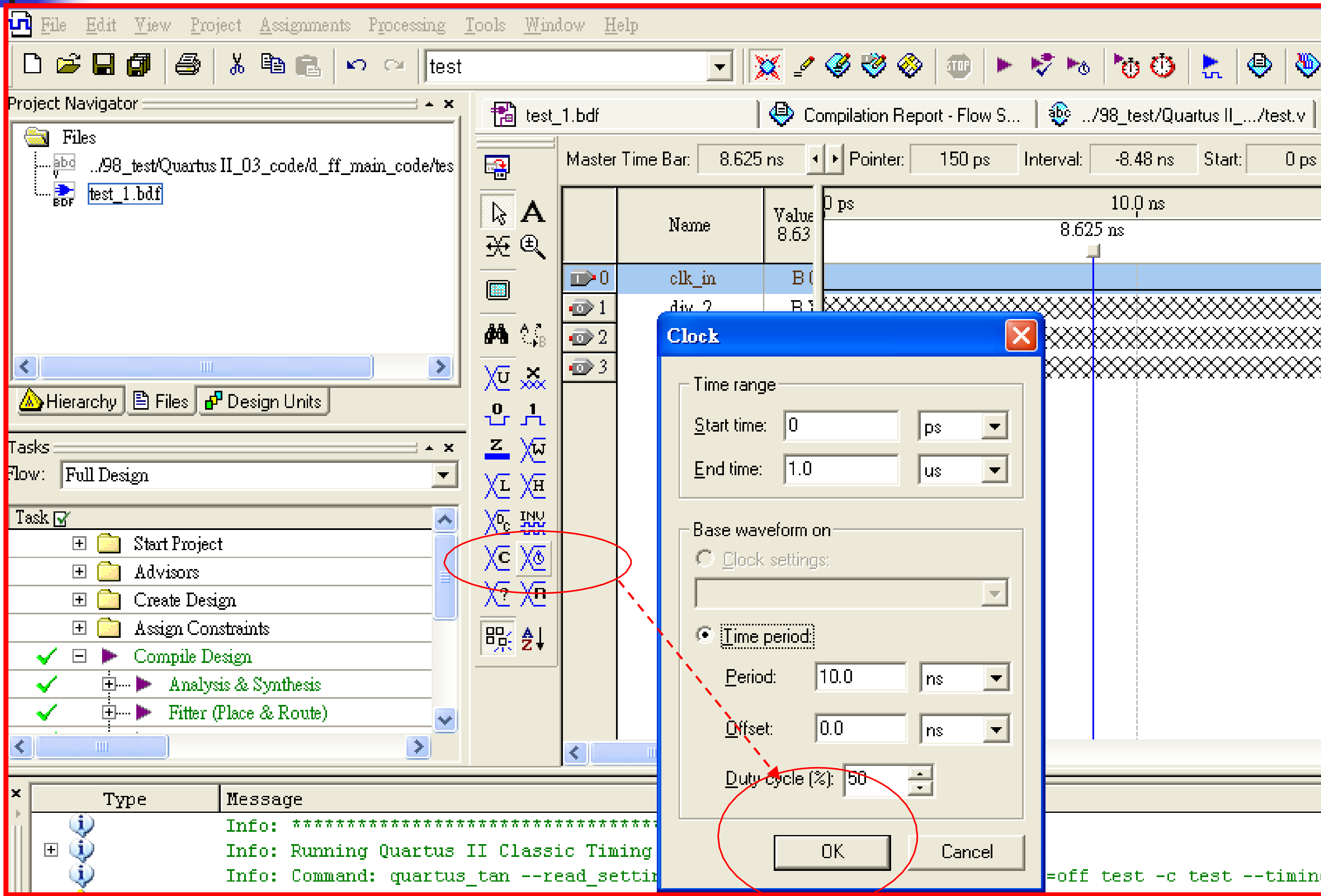
Bottom-Up Design (28)



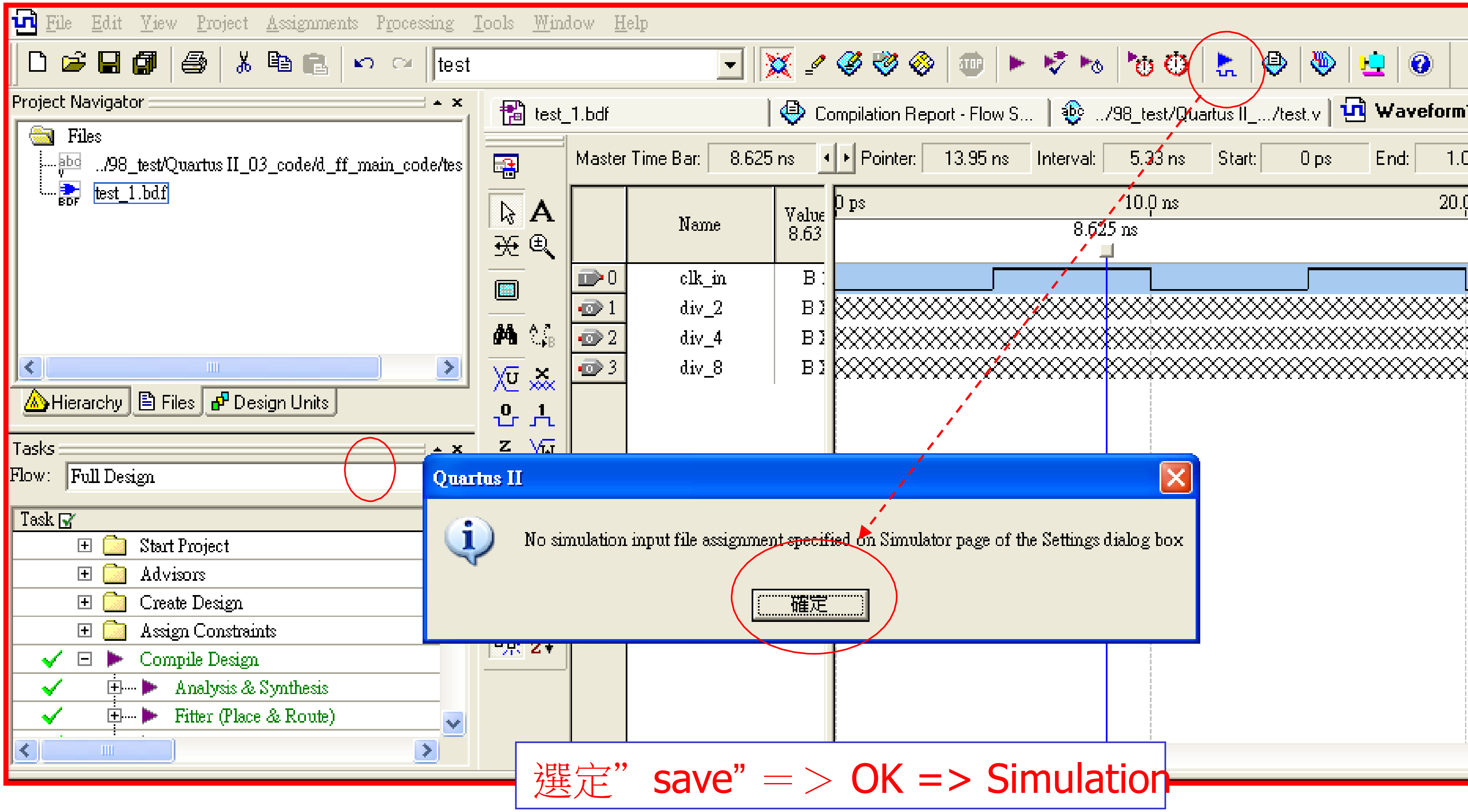
Bottom-Up Design (29)



Bottom-Up Design (30)

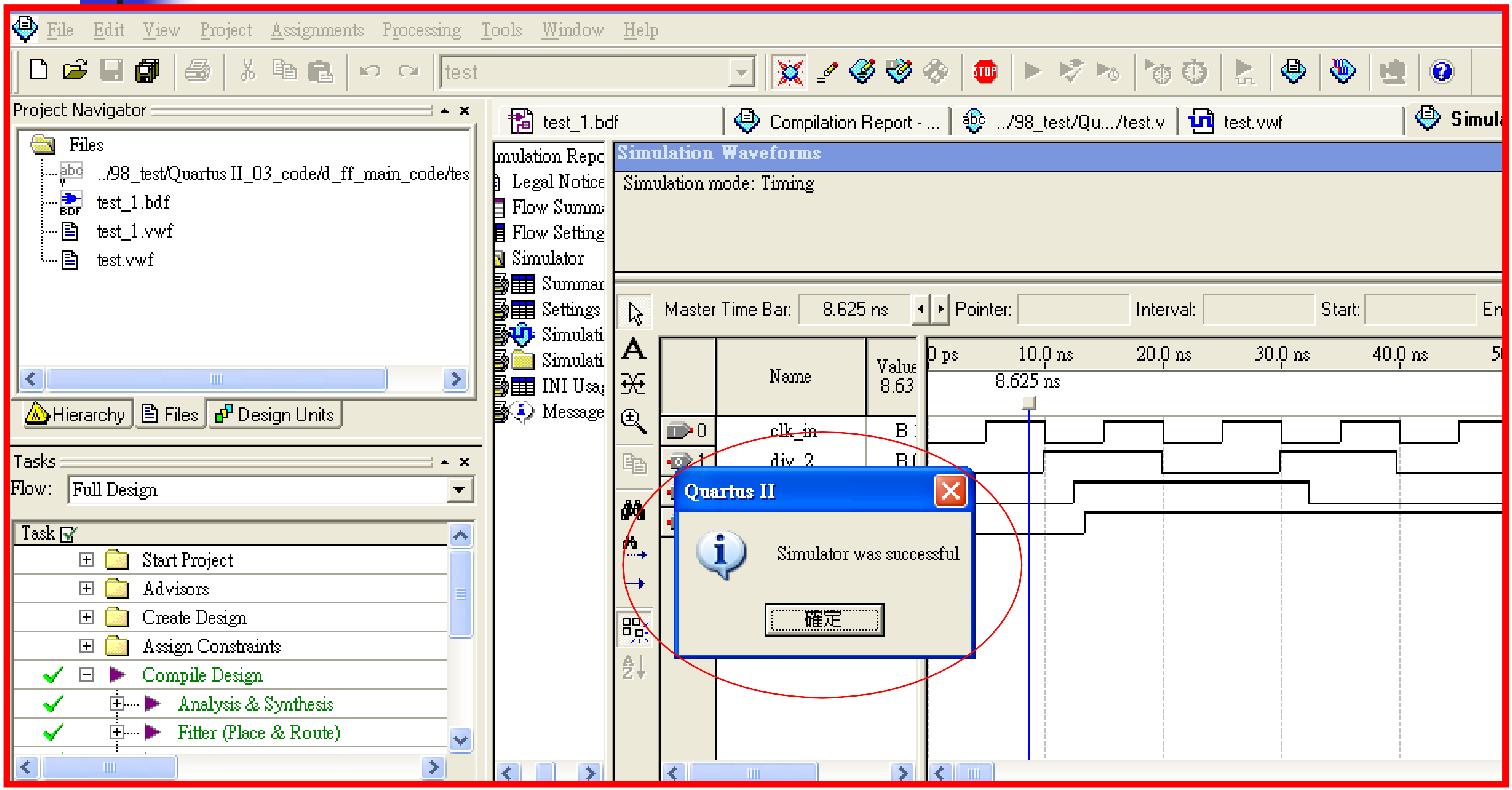


Bottom-Up Design (31)

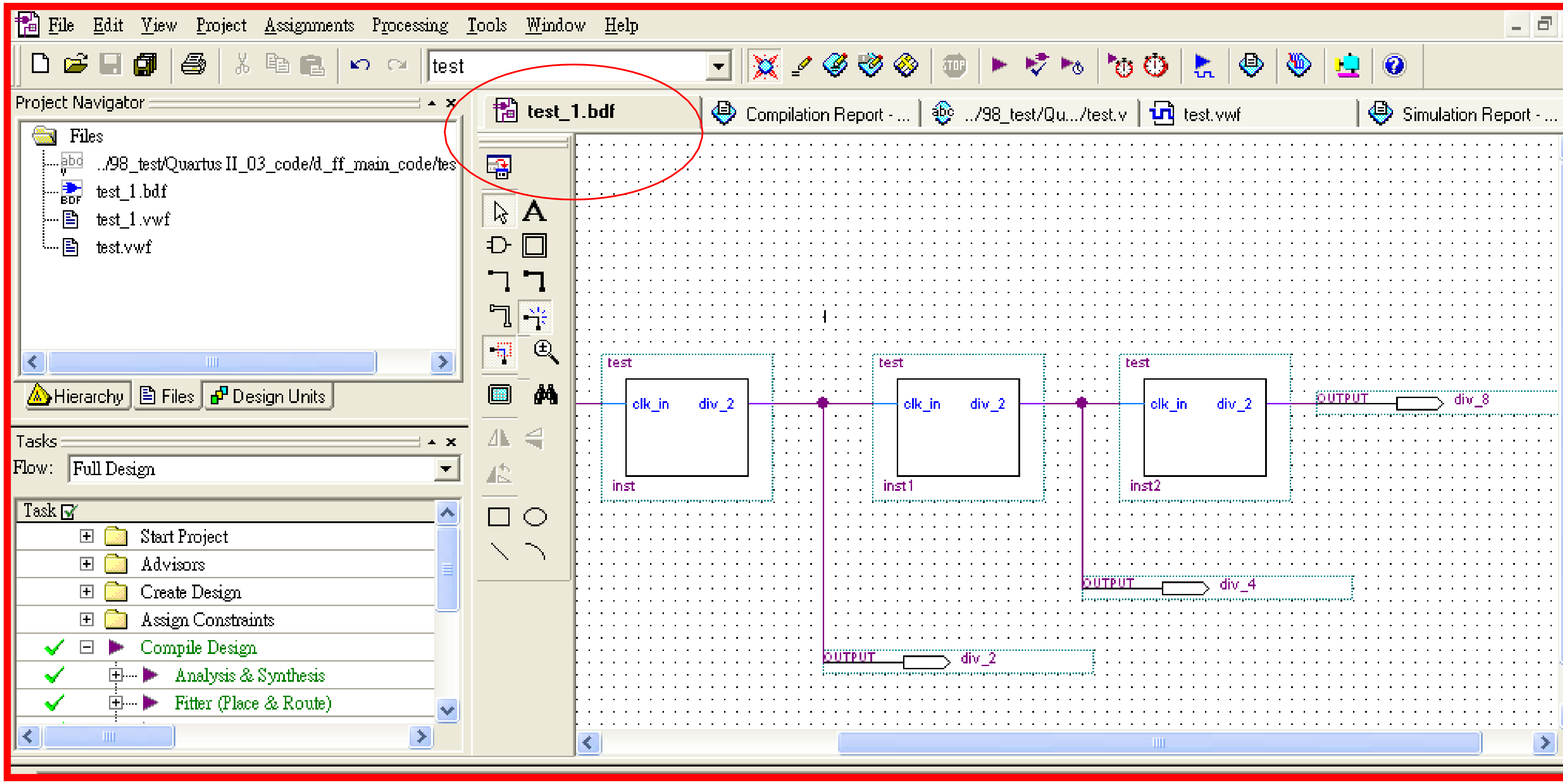


若出現”讀不到檔案,表示其**VWF(test_1.vwf)**檔名與**test_1.bdf**相同,改為不同即可。

Bottom-Up Design (32)

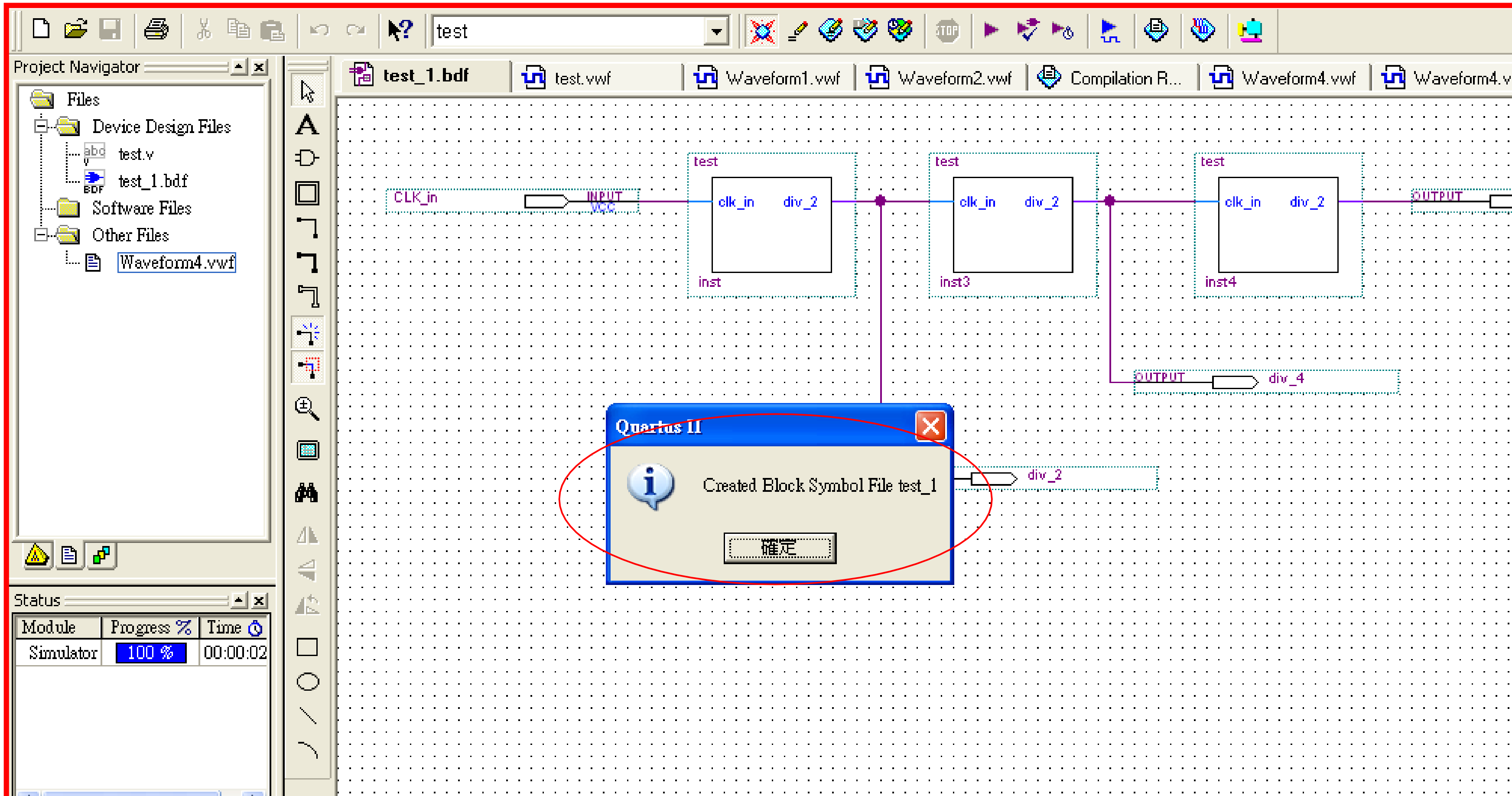


Bottom-Up Design (33)

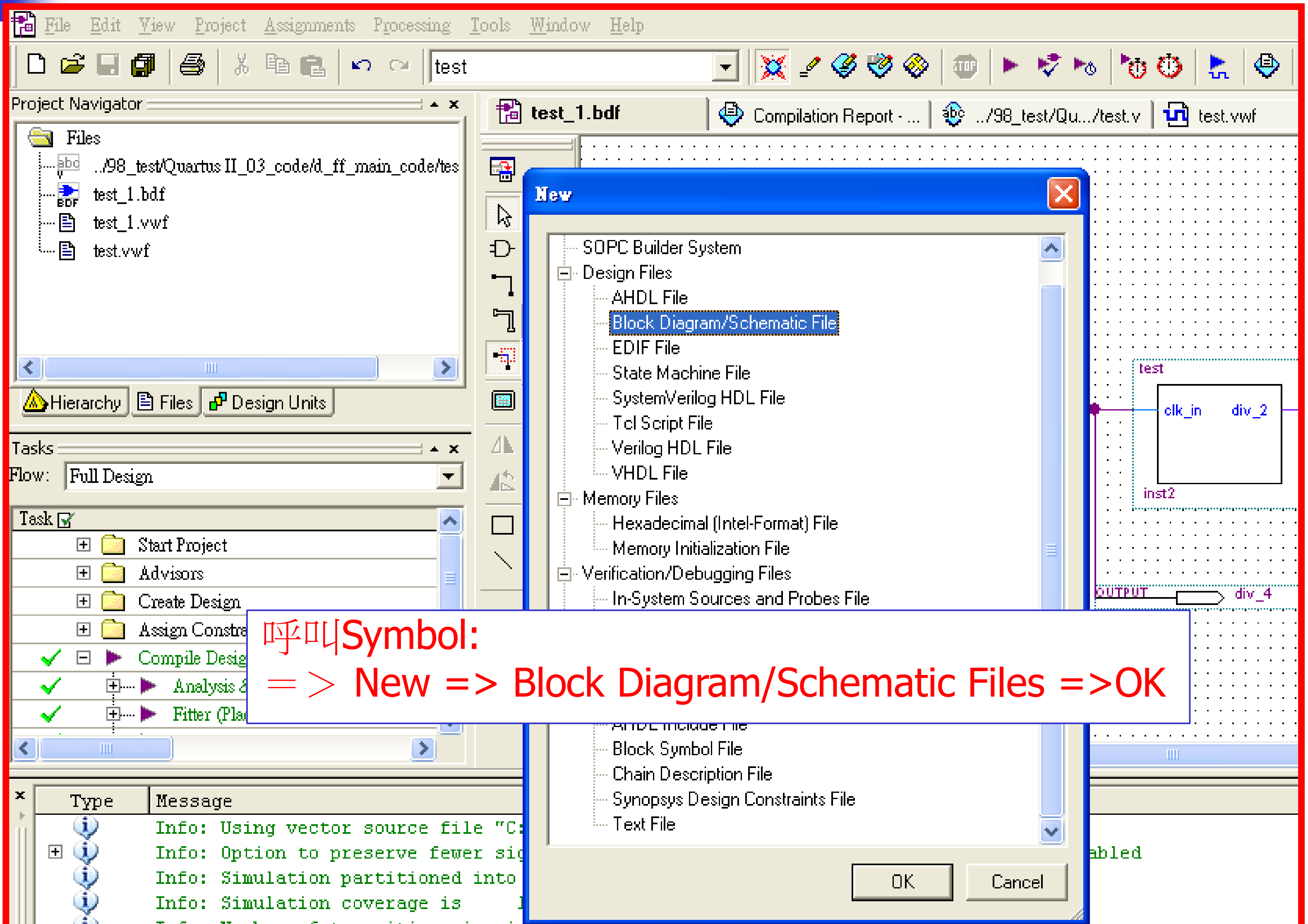


選定” test_1.bdf” =>New＝＞Create/Update=>**Create “Symbol” Files** for current file.

Bottom-Up Design (34)

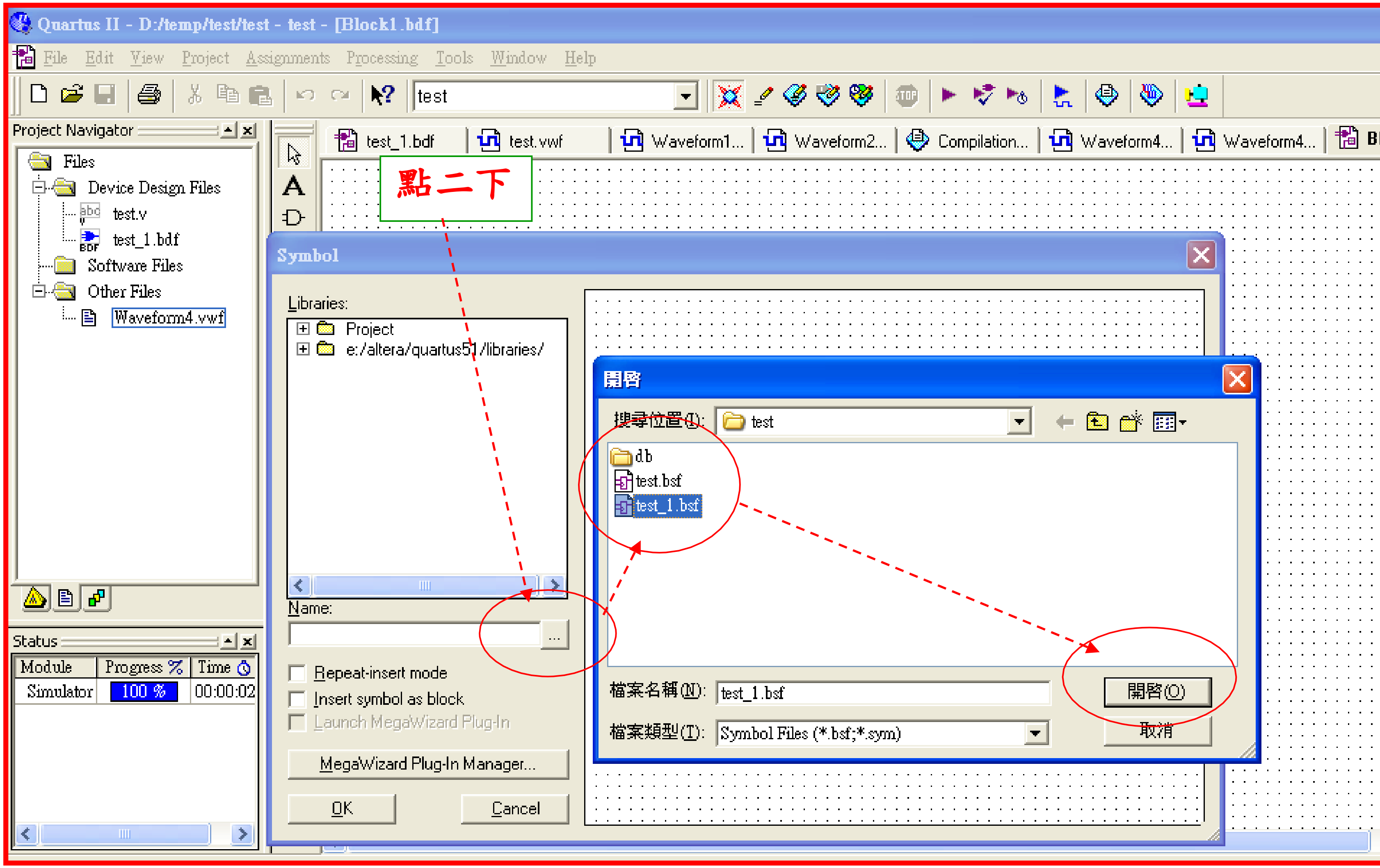


Bottom-Up Design (35)

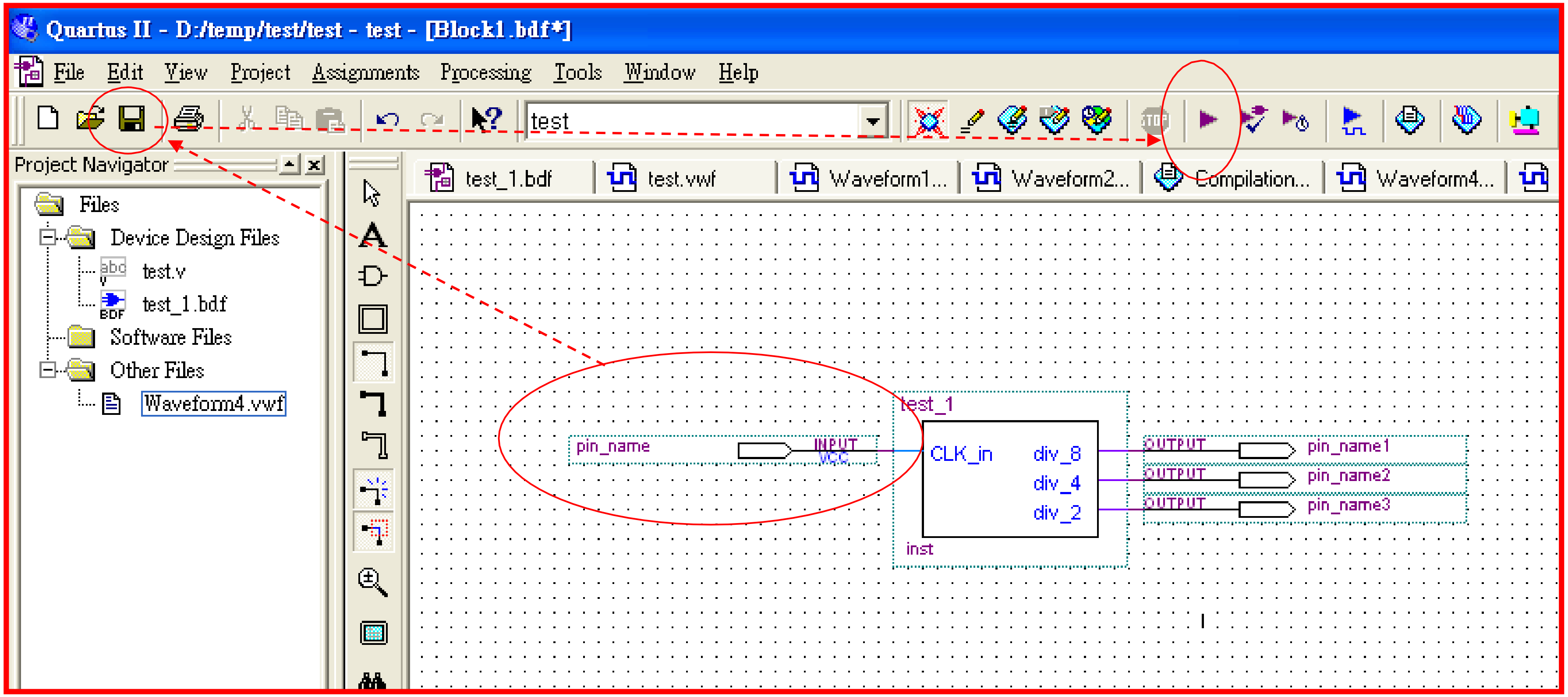


呼叫Symbol:
＝＞ New => Block Diagram/Schematic Files =>OK

Bottom-Up Design (36)

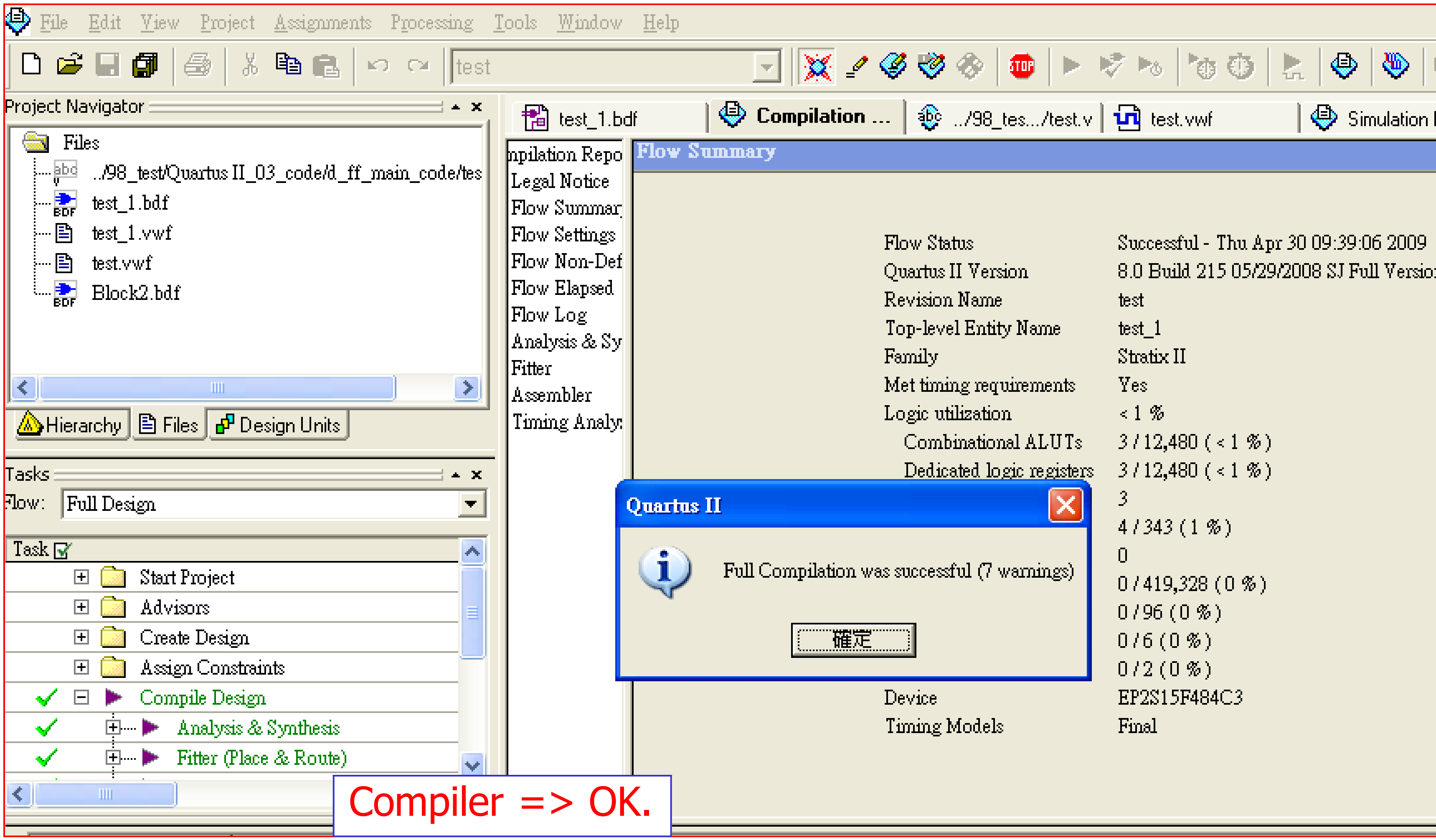


Bottom-Up Design (37)



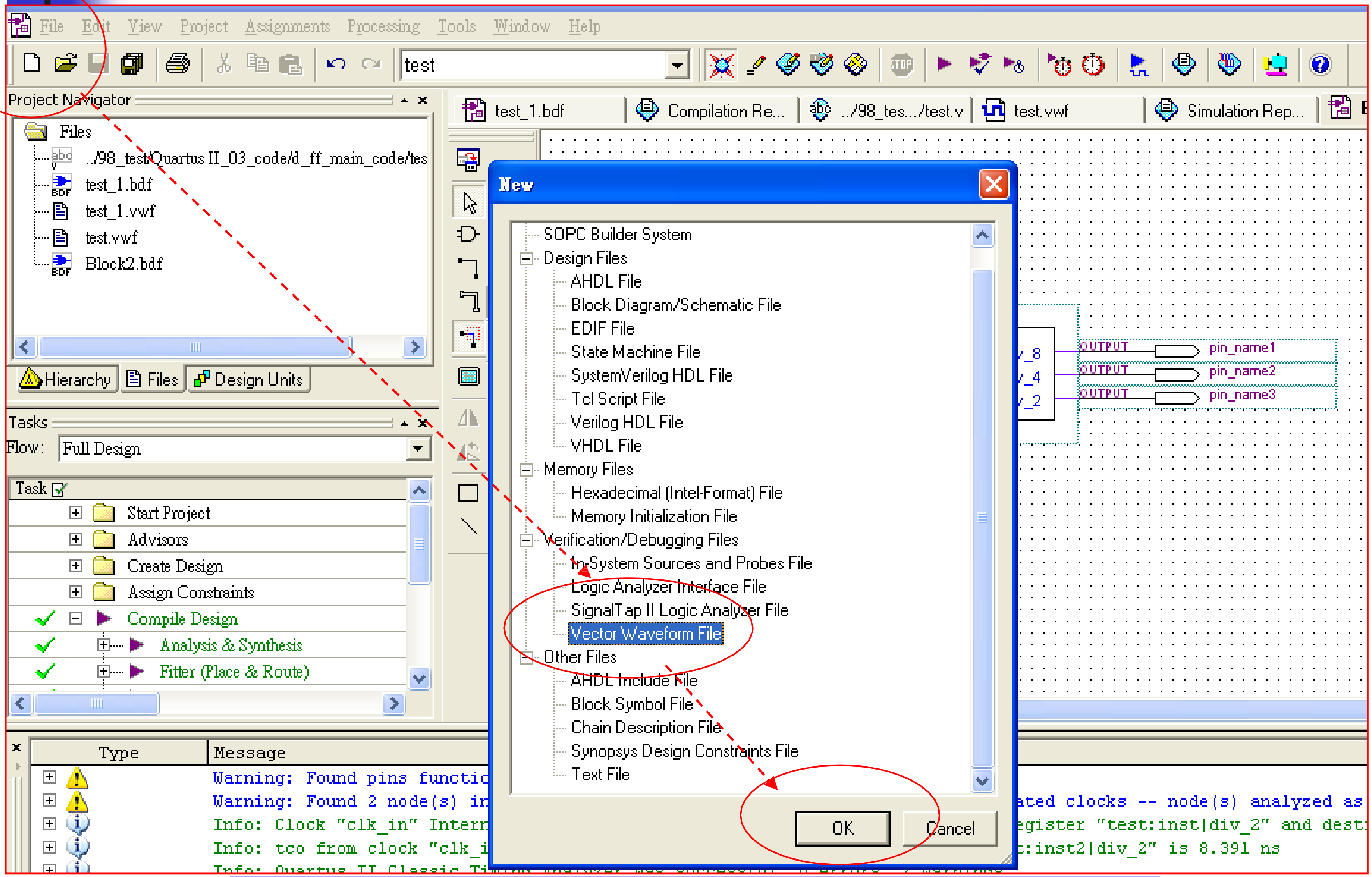
載入輸出入接腳＝＞連線=>Save=>Compiler

Bottom-Up Design (38)



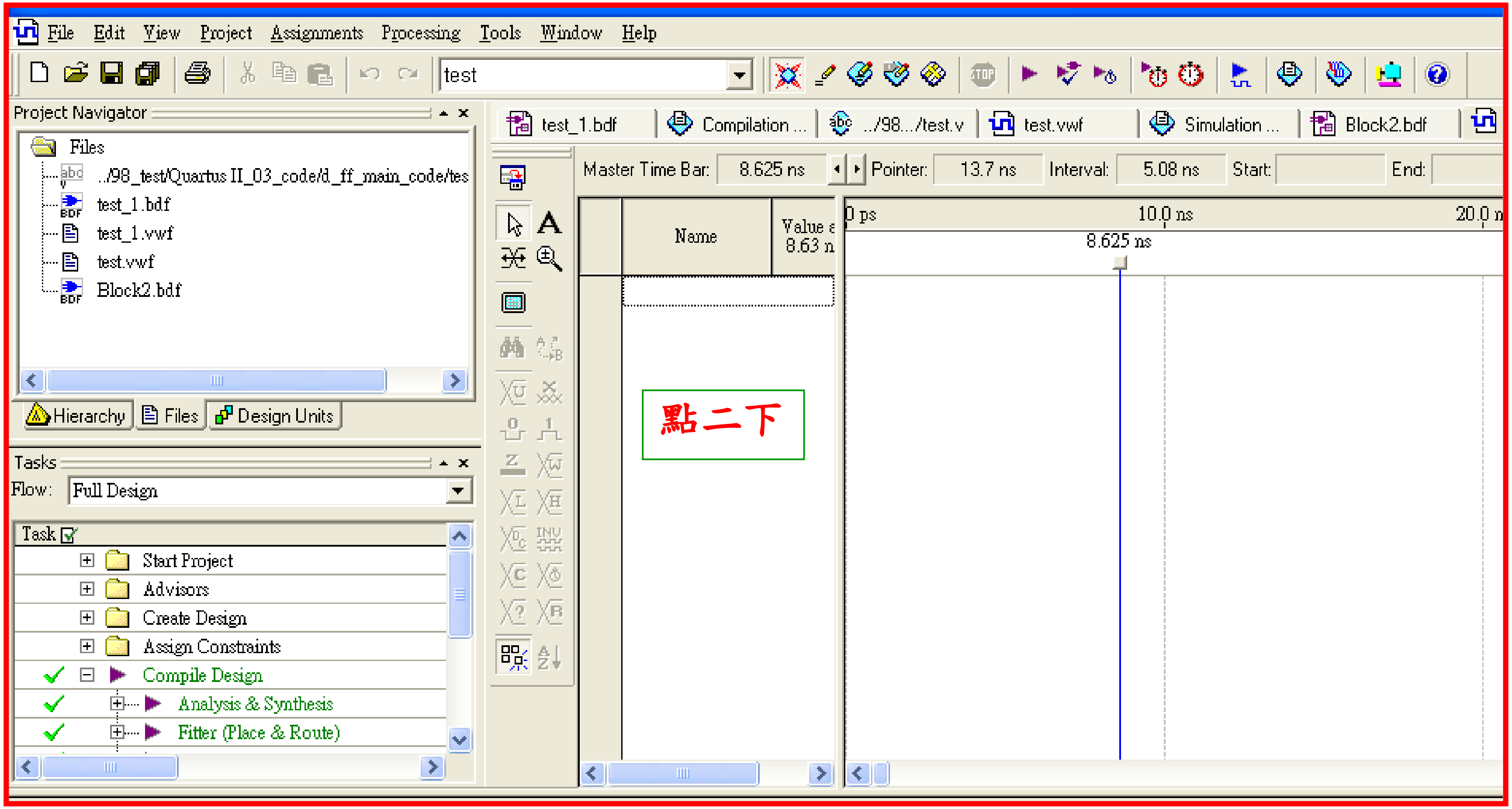
Compiler => OK.

Bottom-Up Design (39)

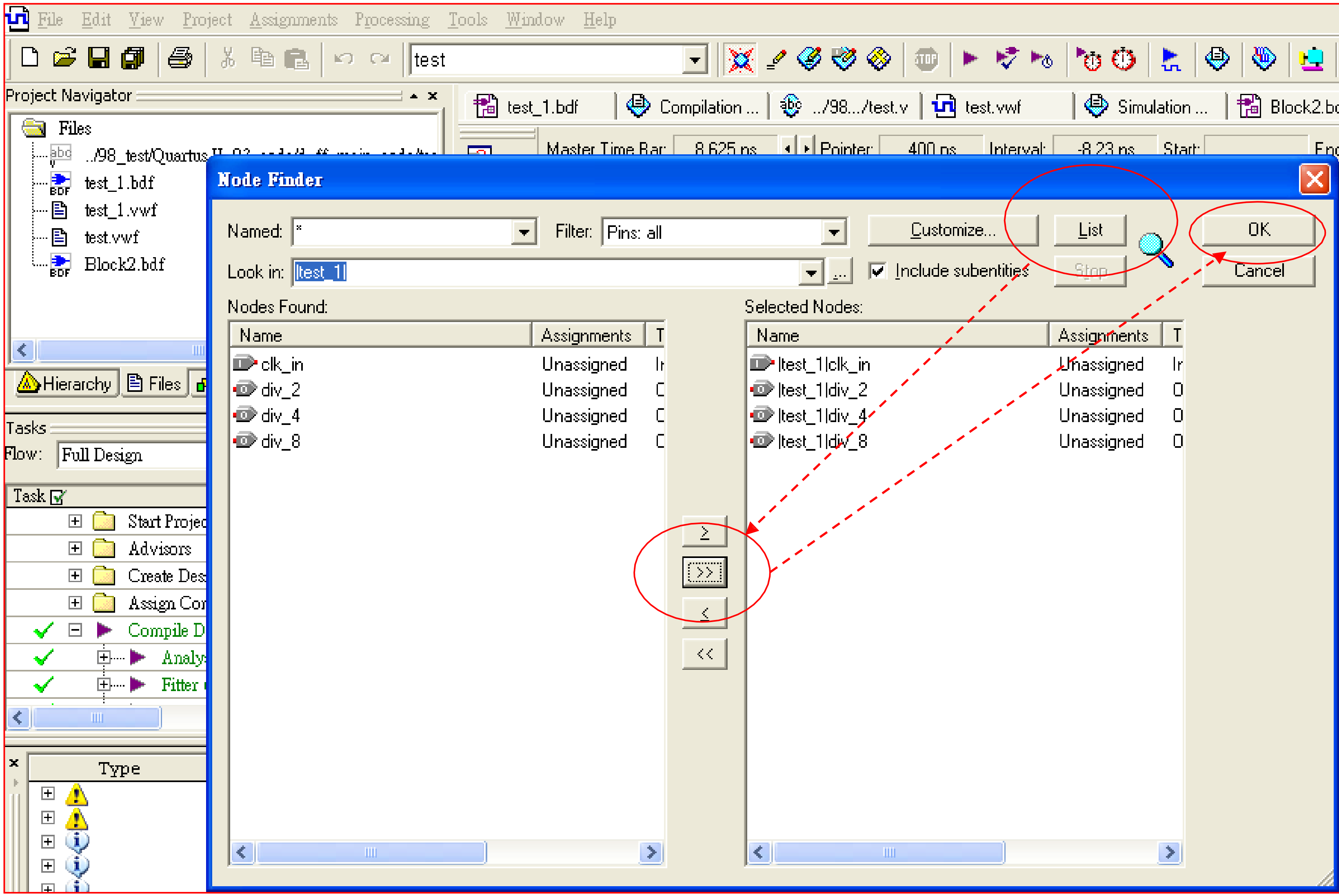


New＝＞Other Files=>Vector Waveform File=>OK.

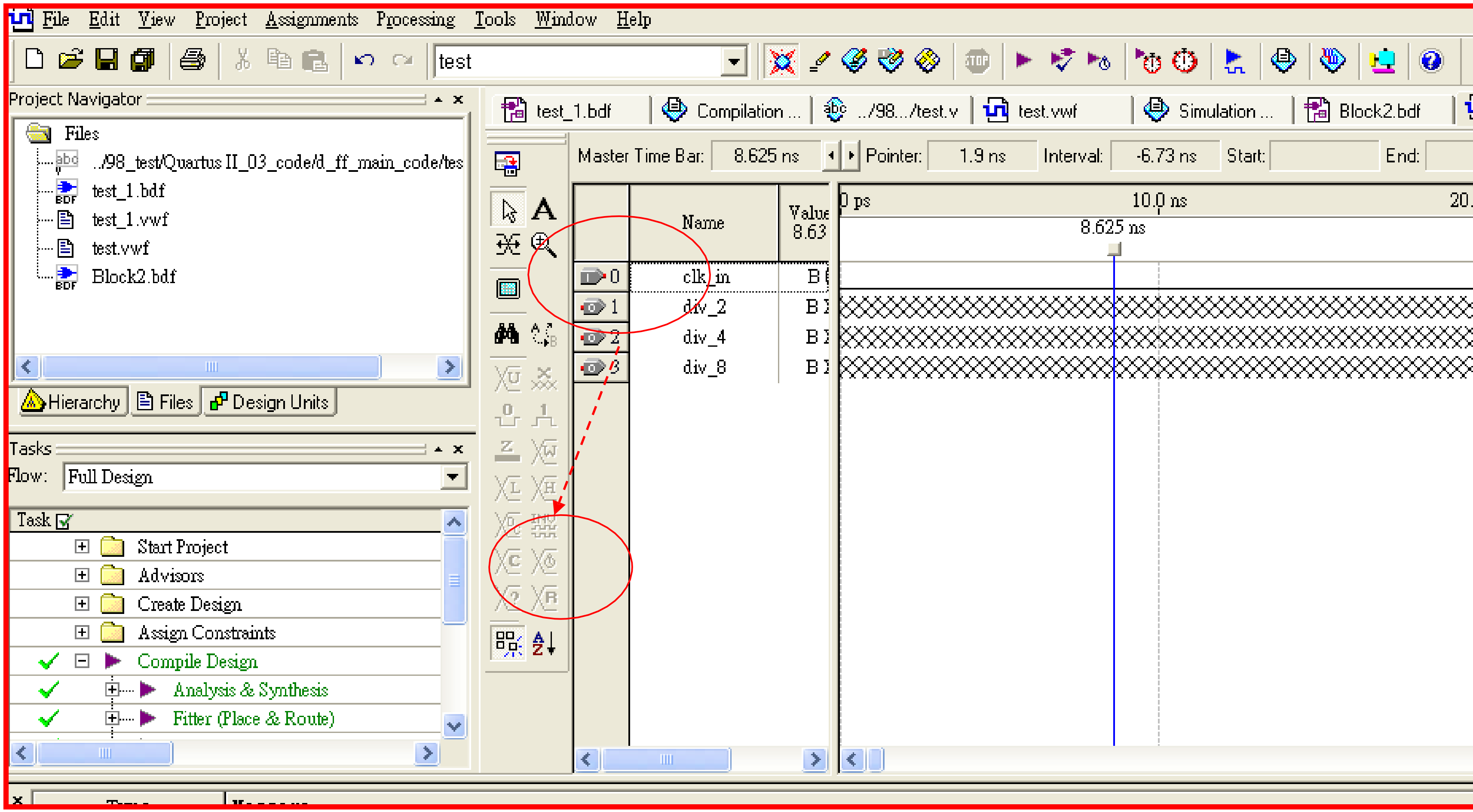
Bottom-Up Design (40)



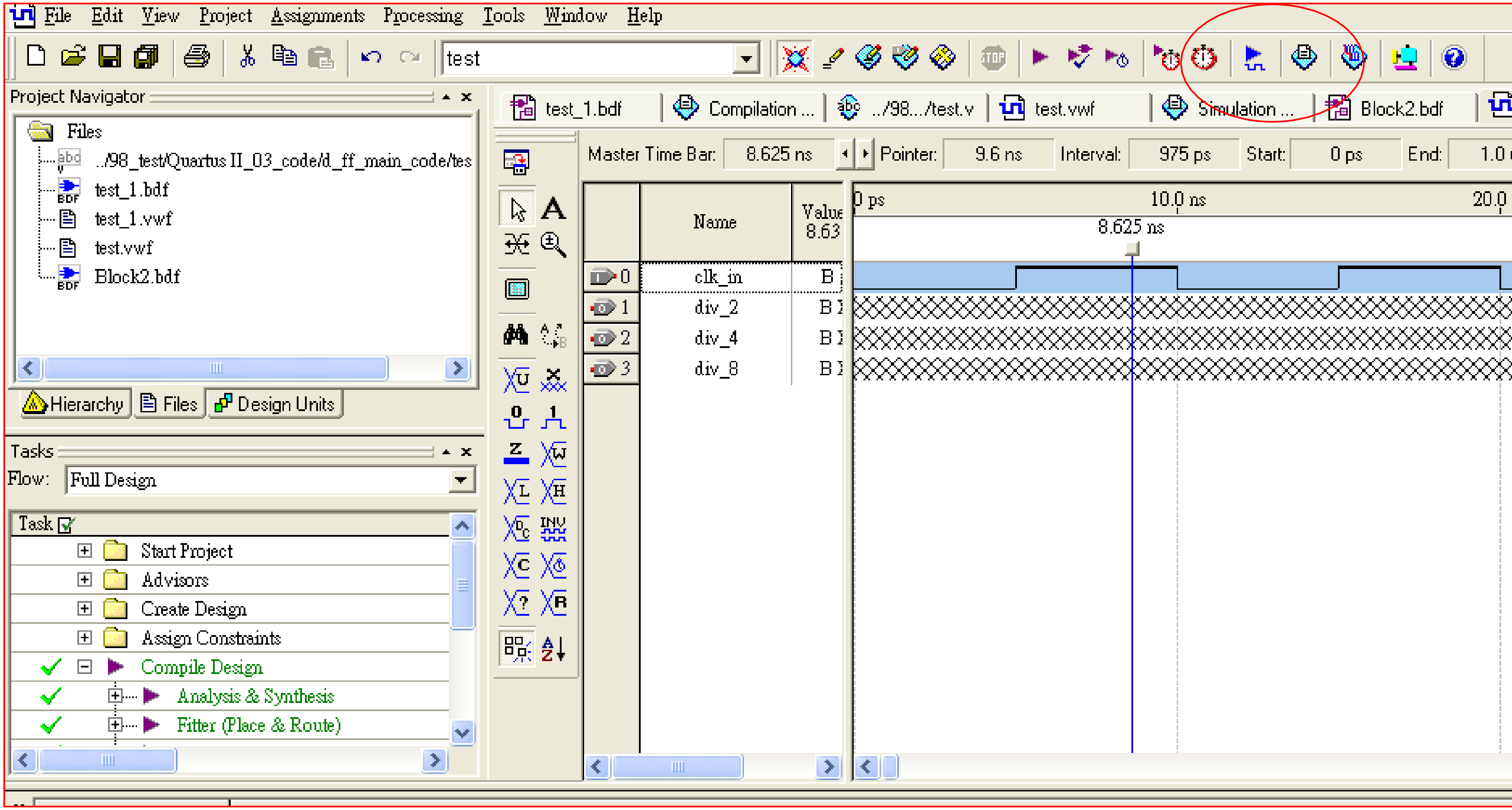
Bottom-Up Design (41)



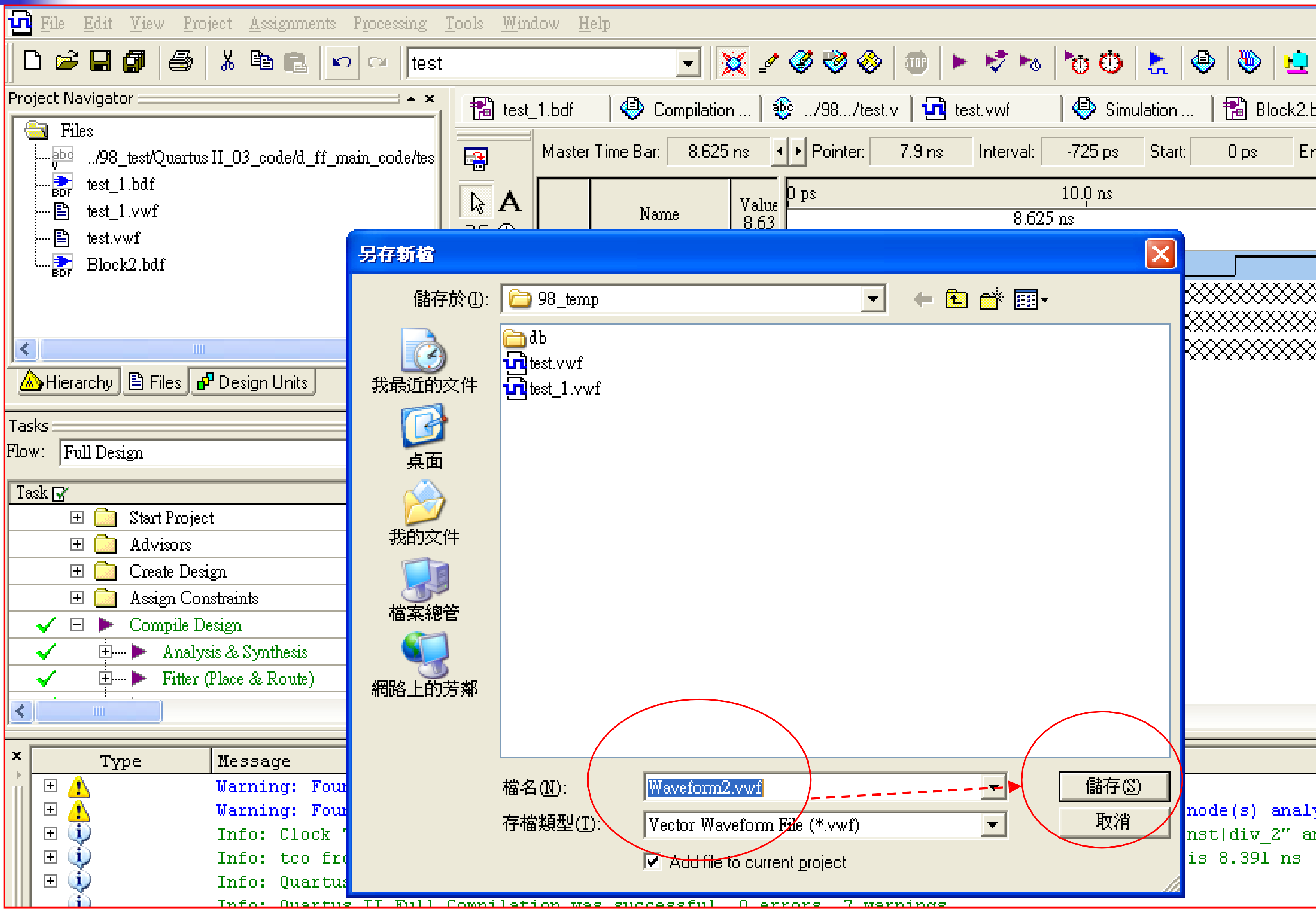
Bottom-Up Design (42)



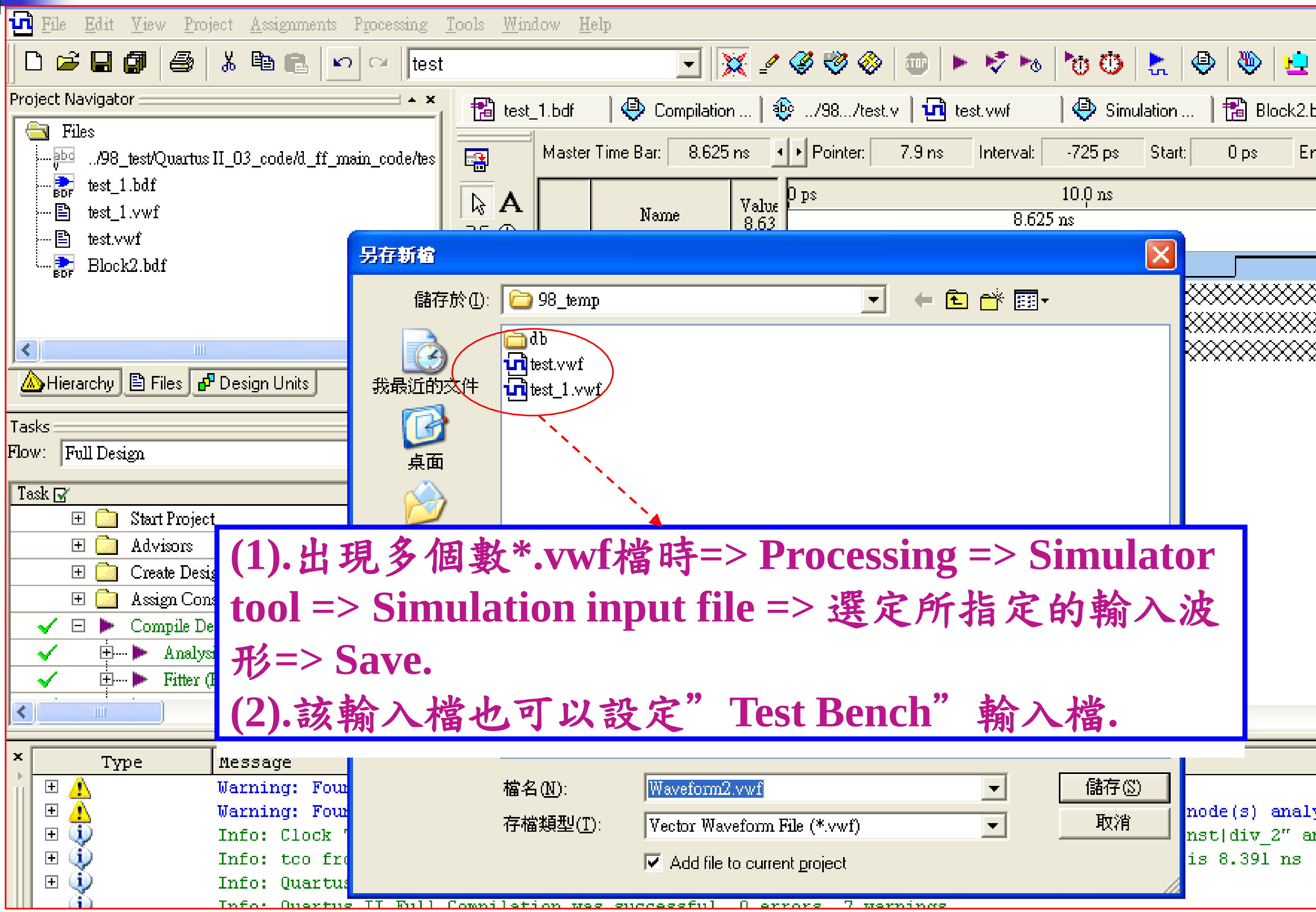
Bottom-Up Design (43)



Bottom-Up Design (44)



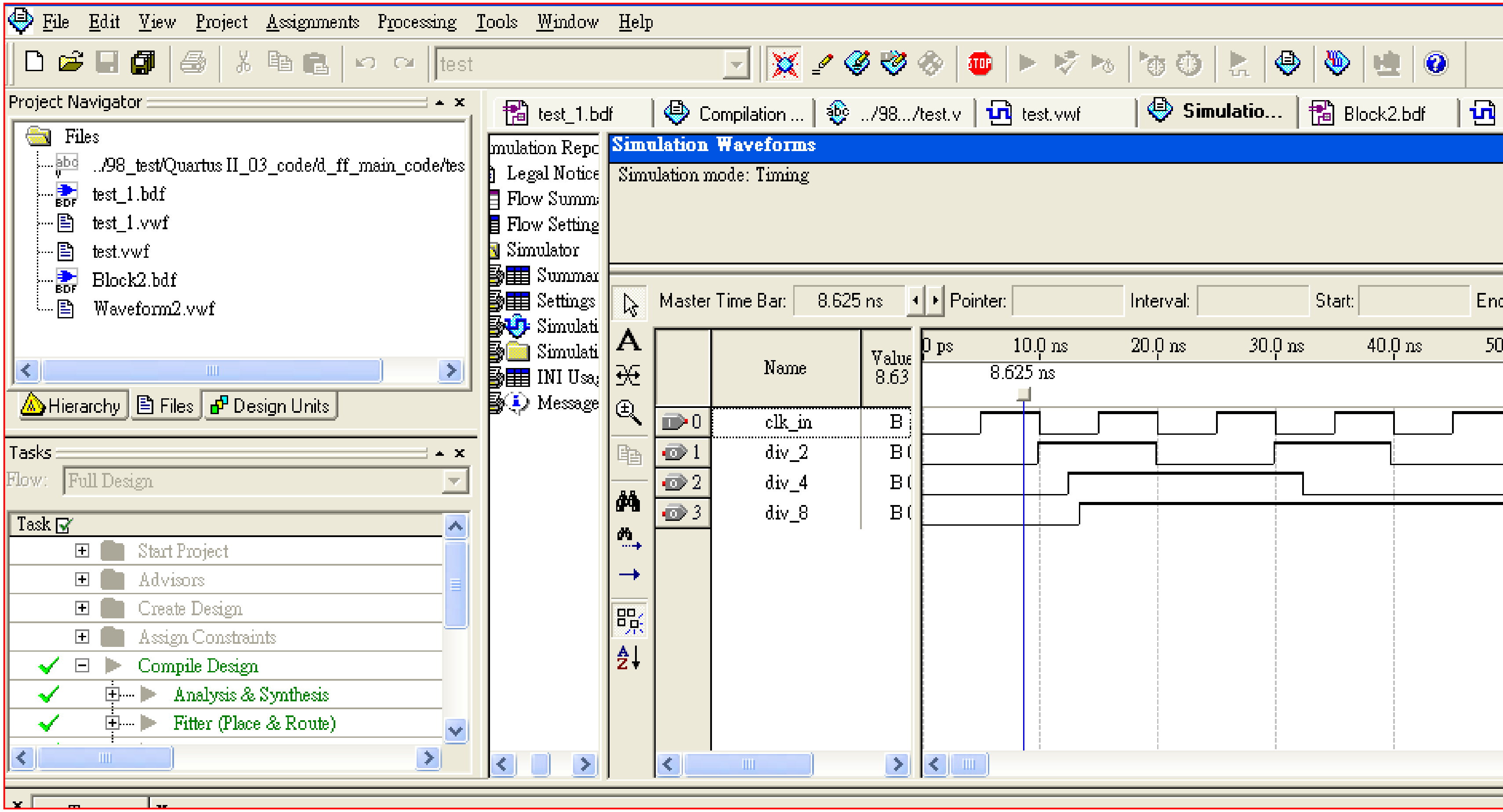
Appendix: Waveform Selection



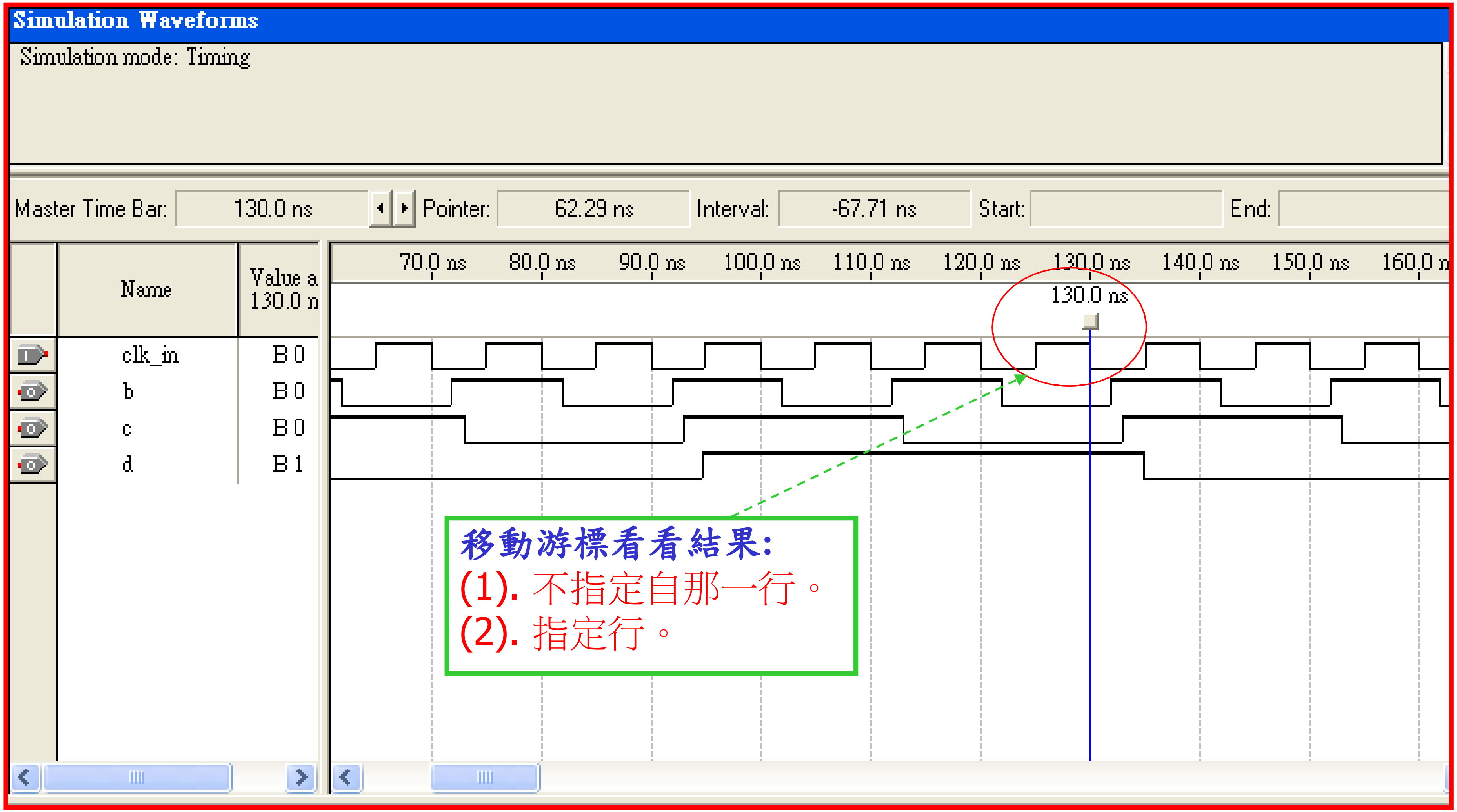
(1).出現多個數*.vwf檔時=> Processing => Simulator tool => Simulation input file => 選定所指定的輸入波形=> Save.

(2).該輸入檔也可以設定”Test Bench”輸入檔.

Bottom-Up Design (45)



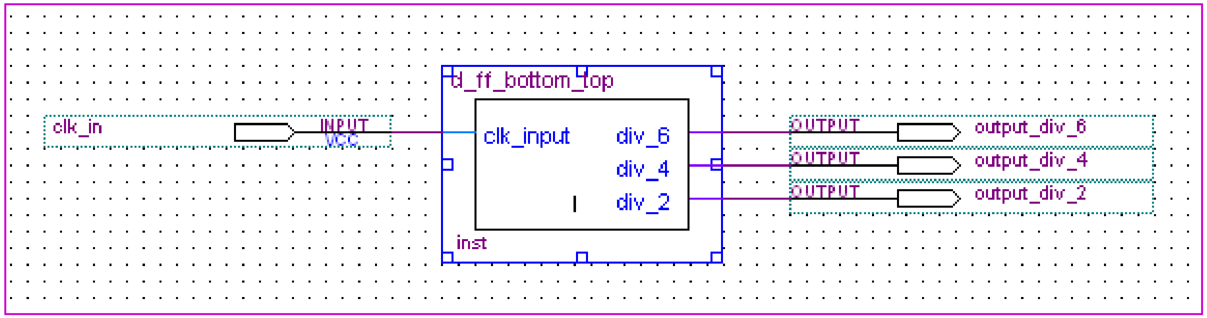
Bottom-Up Design (46)

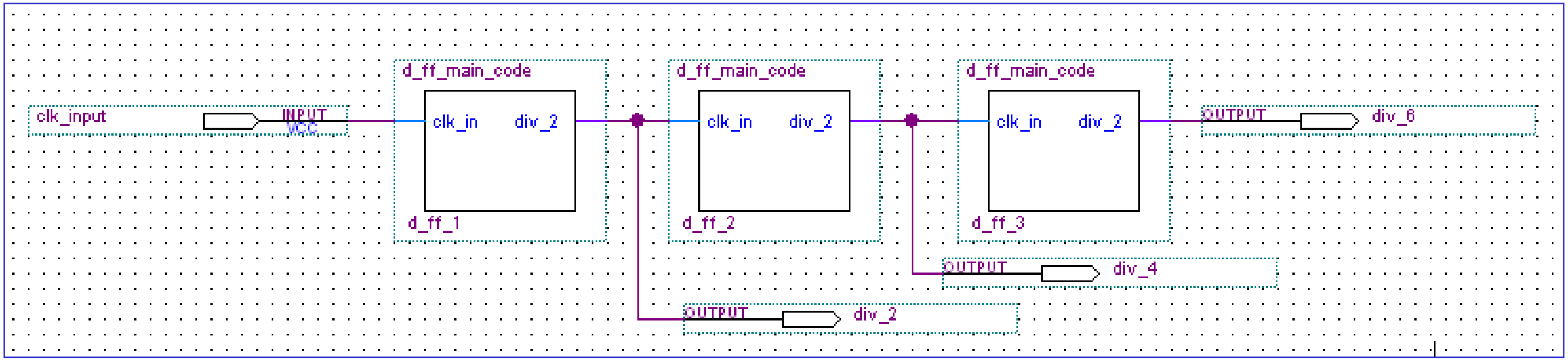


Bottom-Up Design (47)

Output

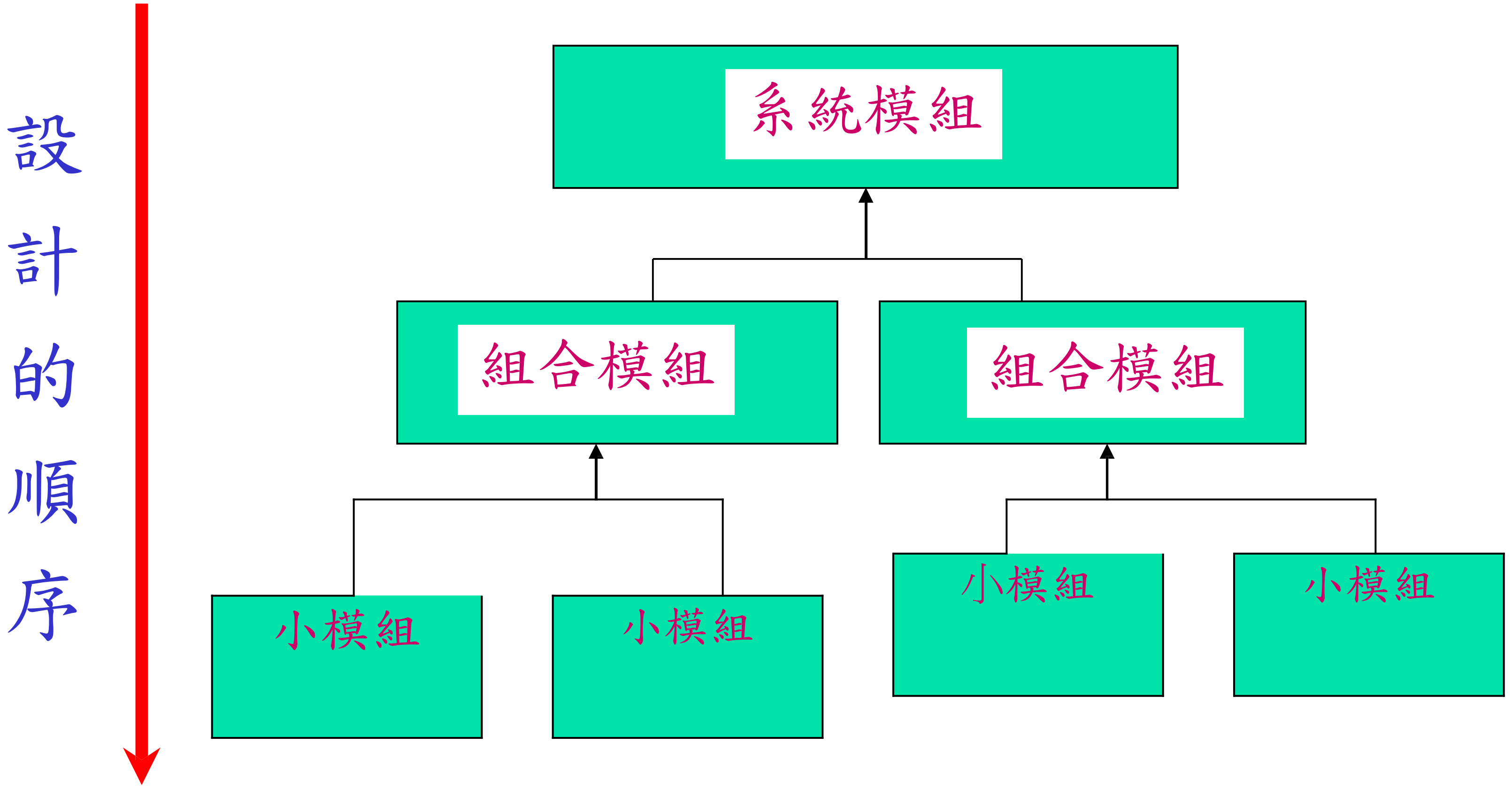
```
module d_ff_main_code(clk_in,div_2);

input clk_in;
output div_2;

reg div_2;

always@(posedge clk_in)
begin
   div_2 = div_2 ^ 1;
end

endmodule
```





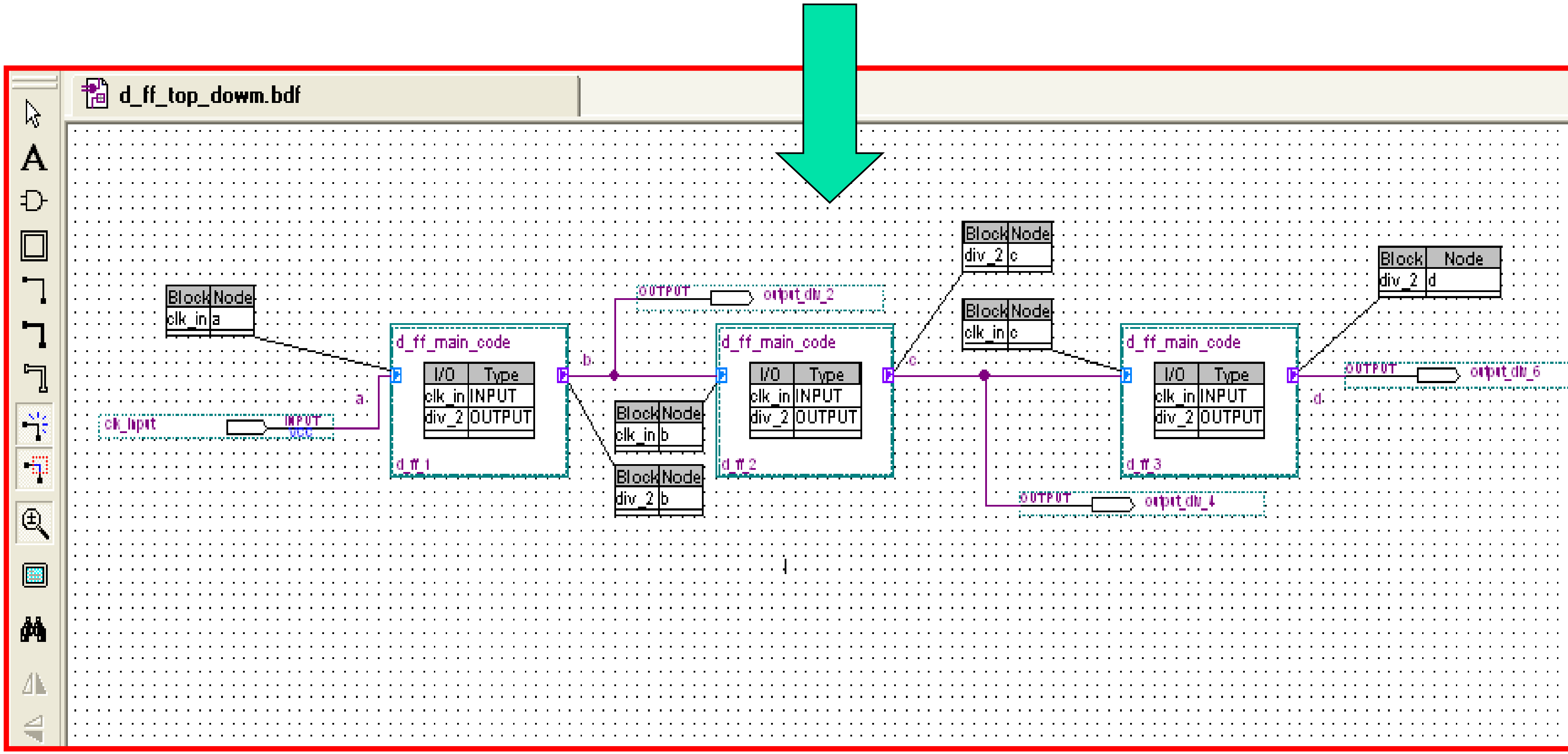
Top-Down Design Flow

Top-Down Design

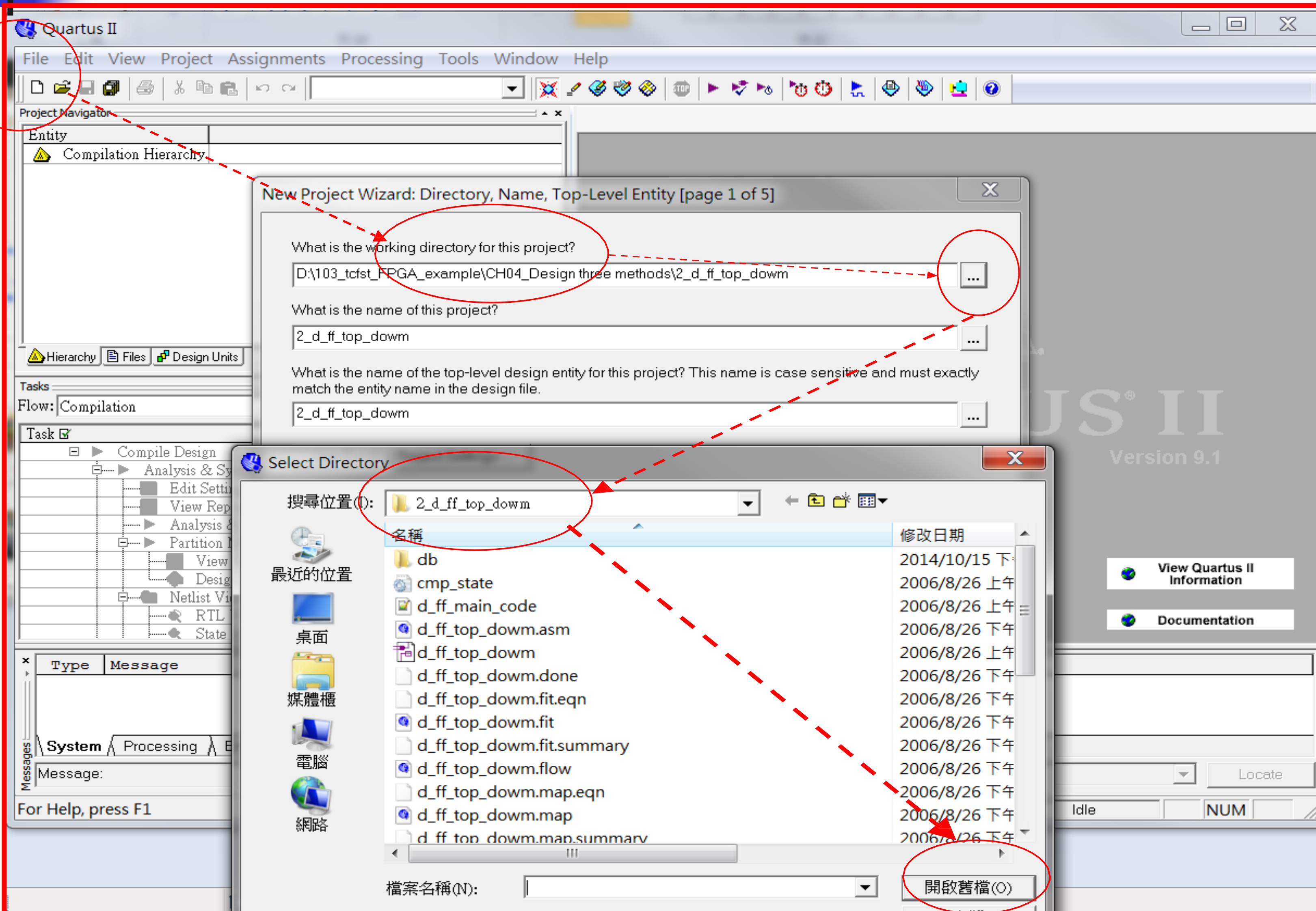


Top-Down Design (1)

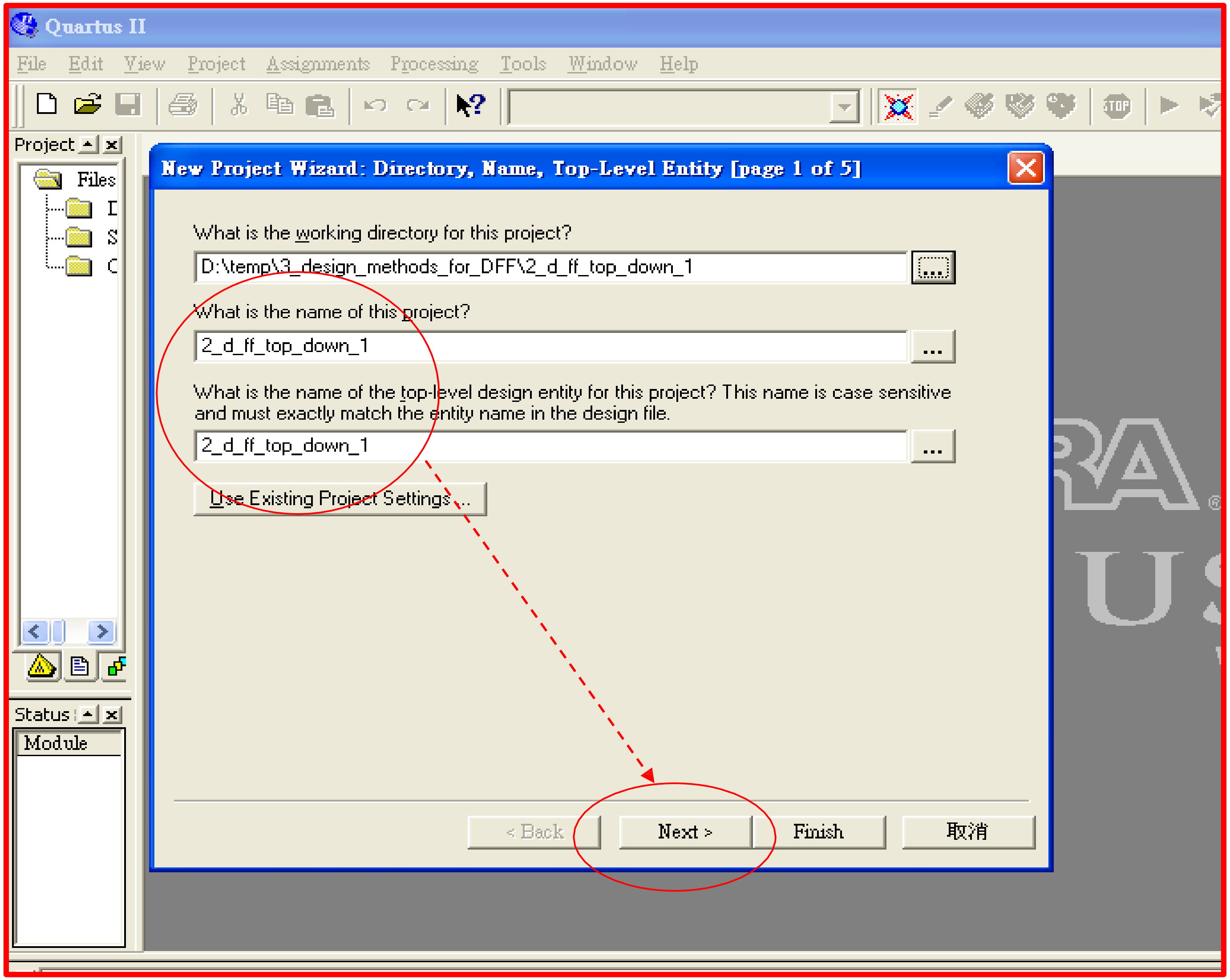
目標



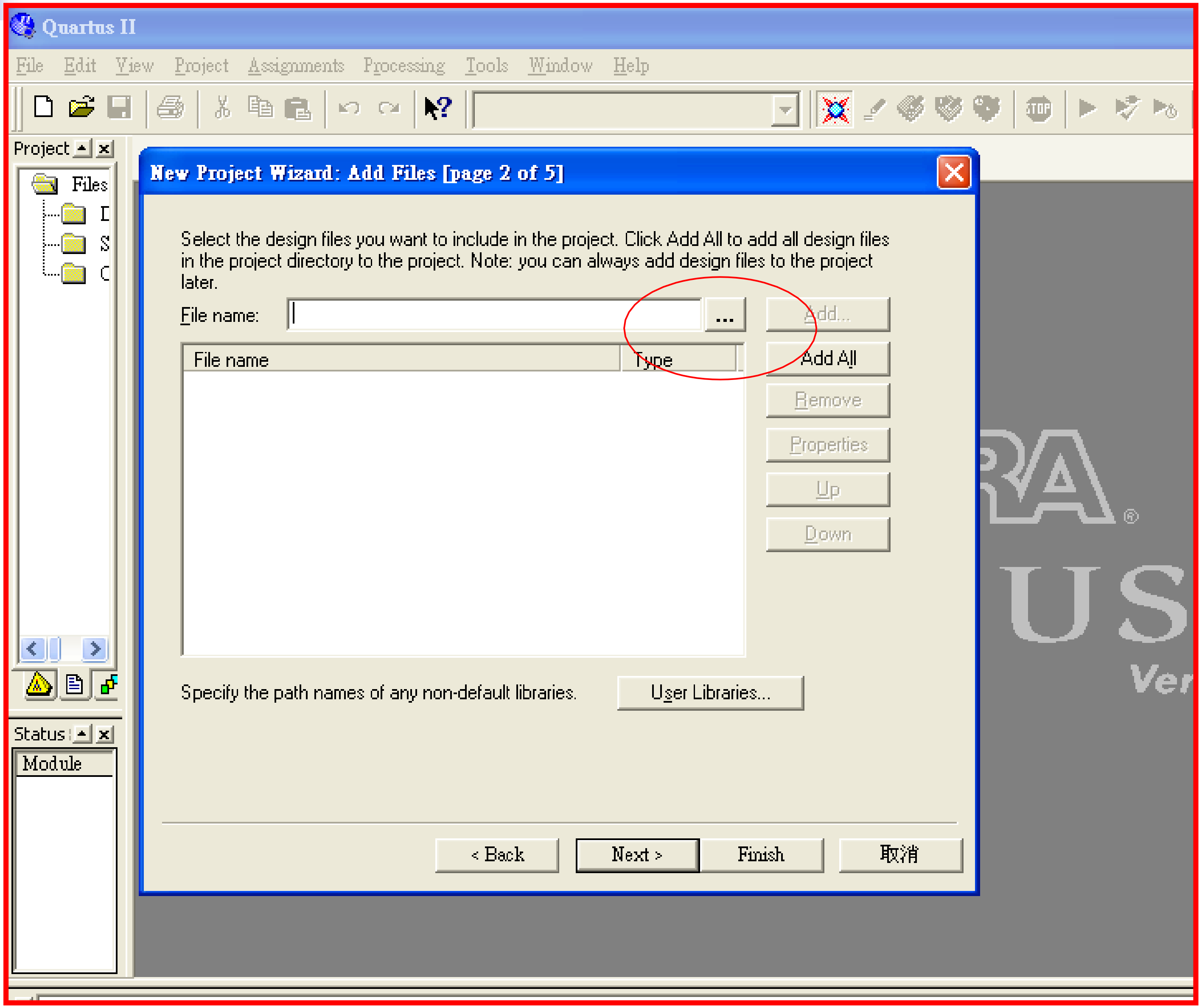
Top-Down Design (2)



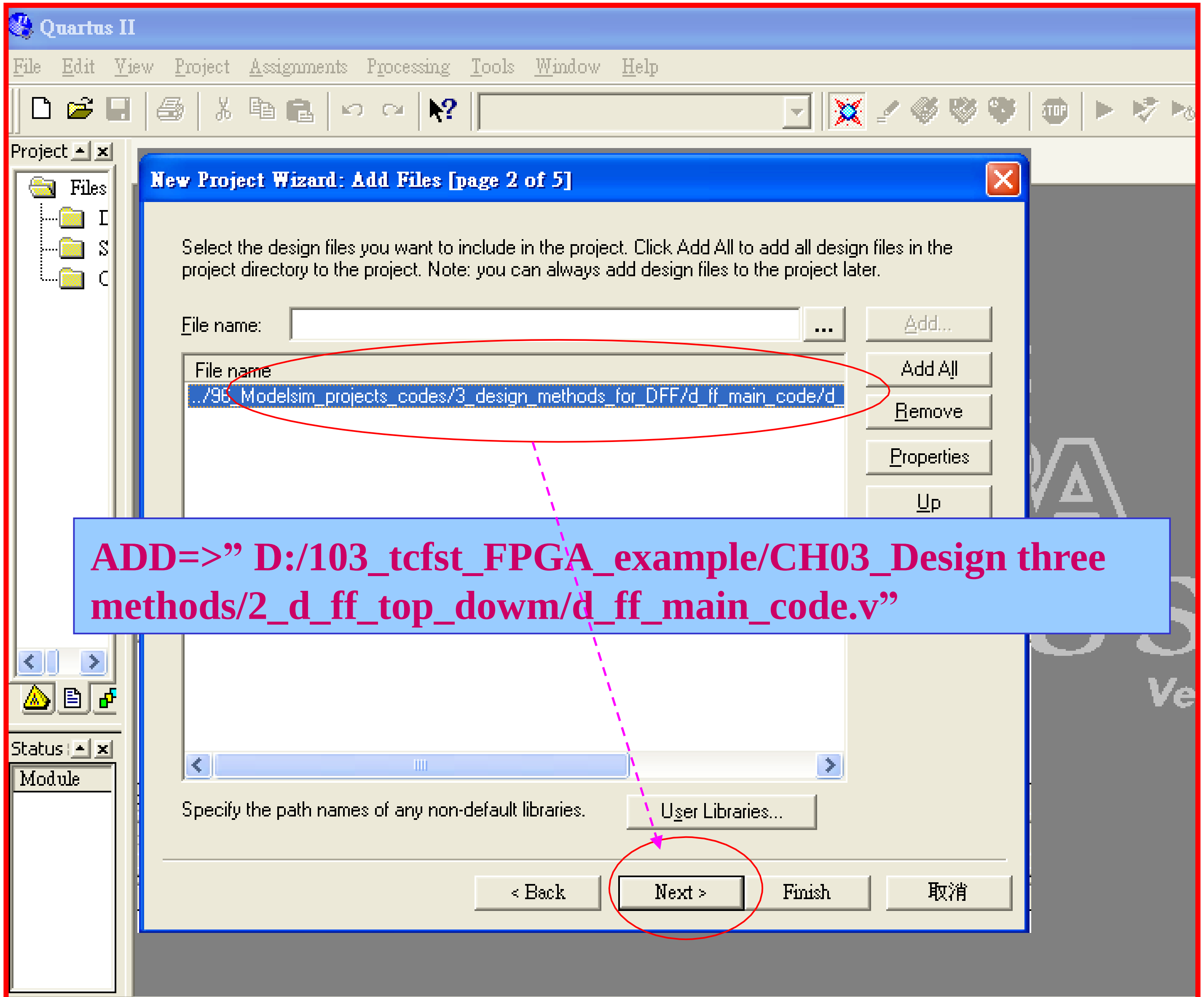
Top-Down Design (3)



Top-Down Design (4)

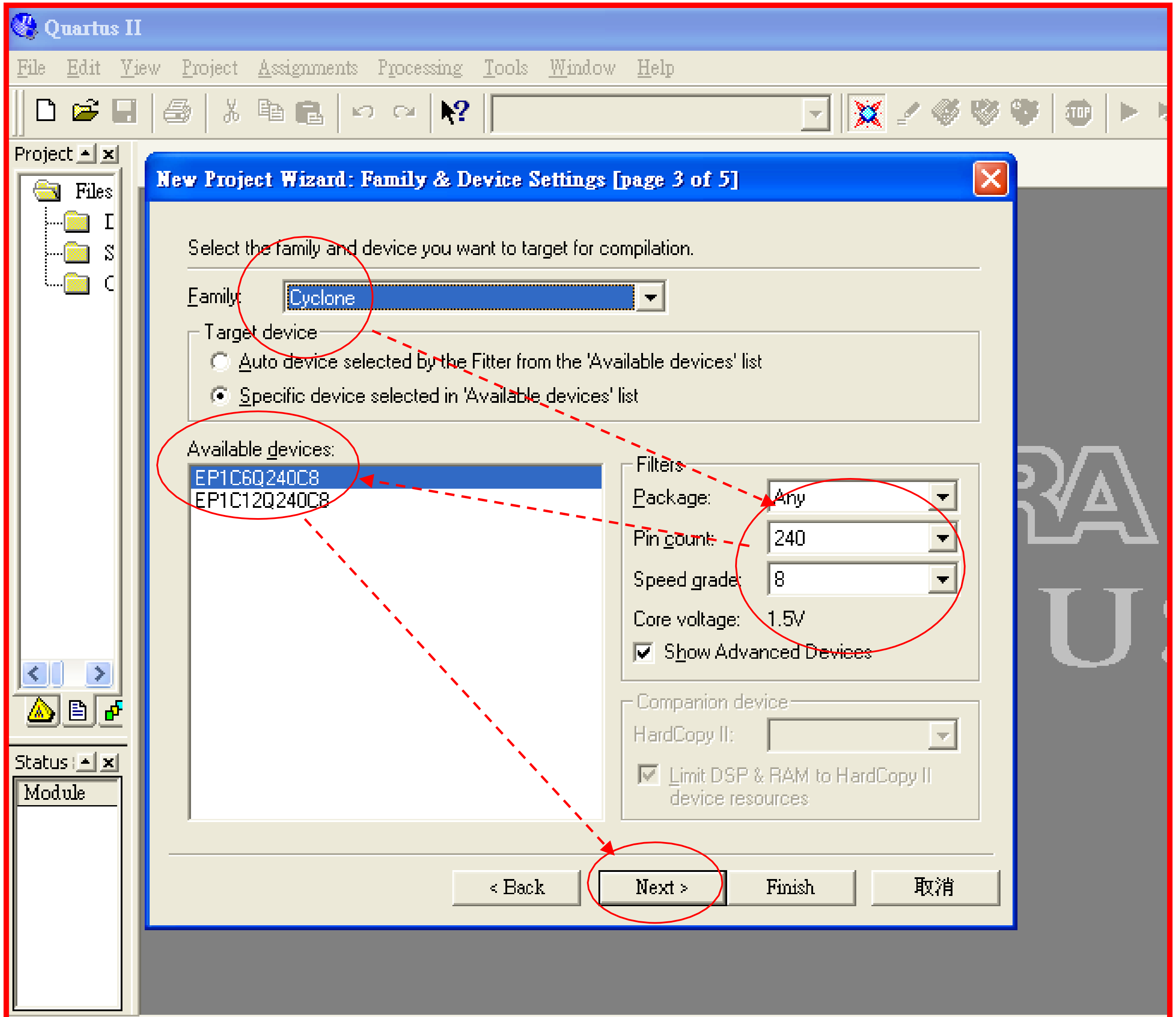


Top-Down Design (5)

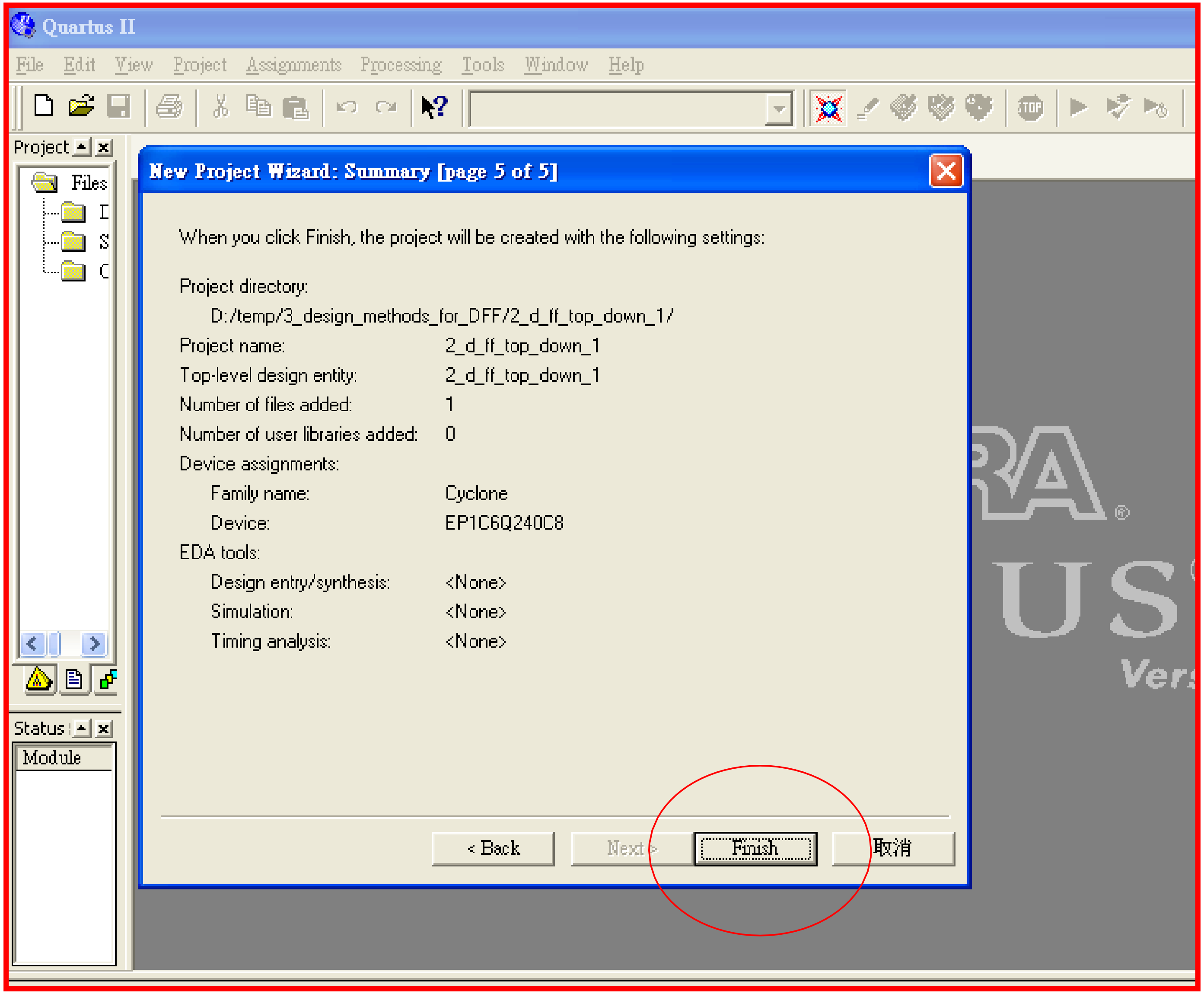


ADD=>” D:/103_tcfst_FPGA_example/CH03_Design three methods/2_d_ff_top_dowm/d_ff_main_code.v”

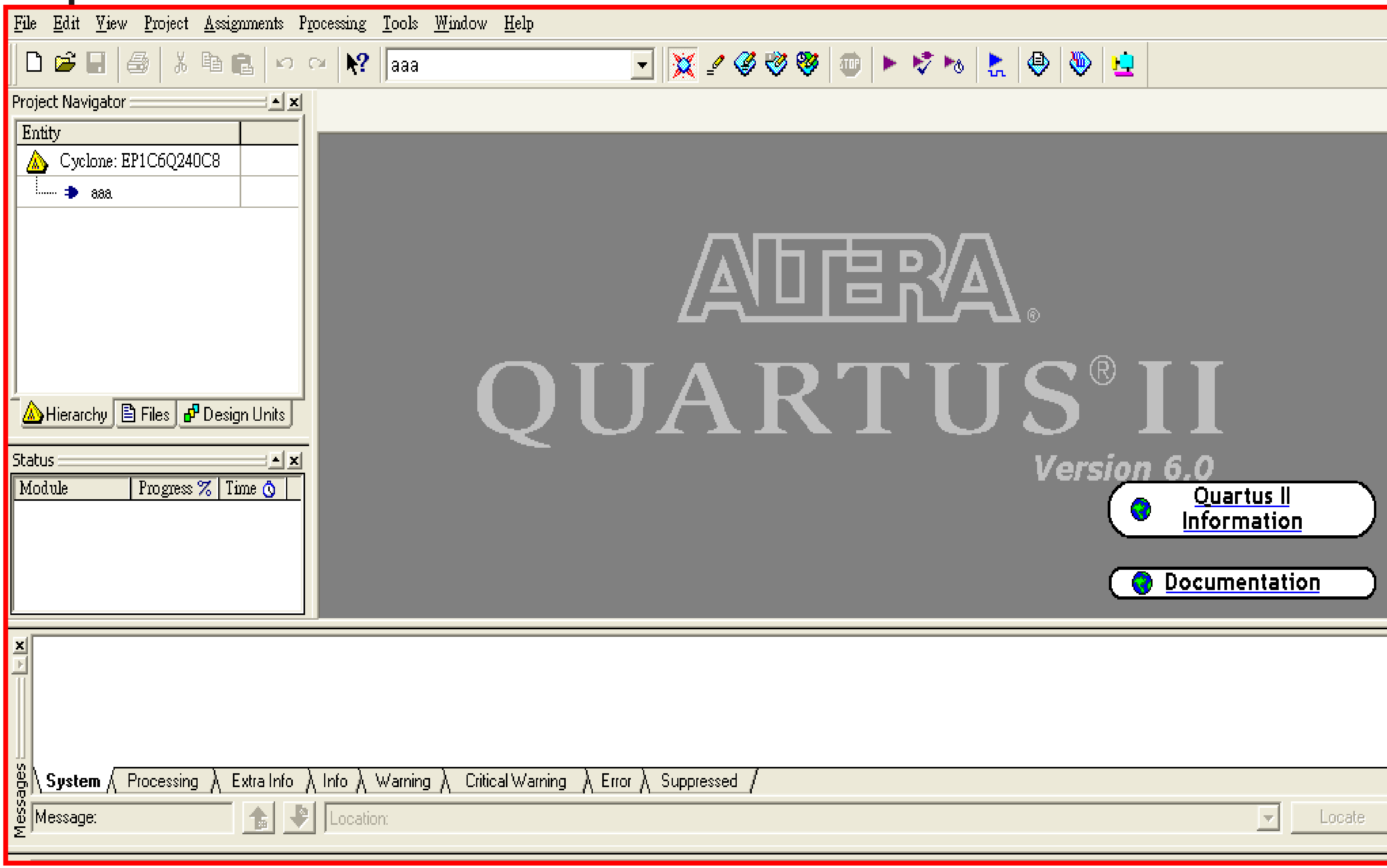
Top-Down Design (6)



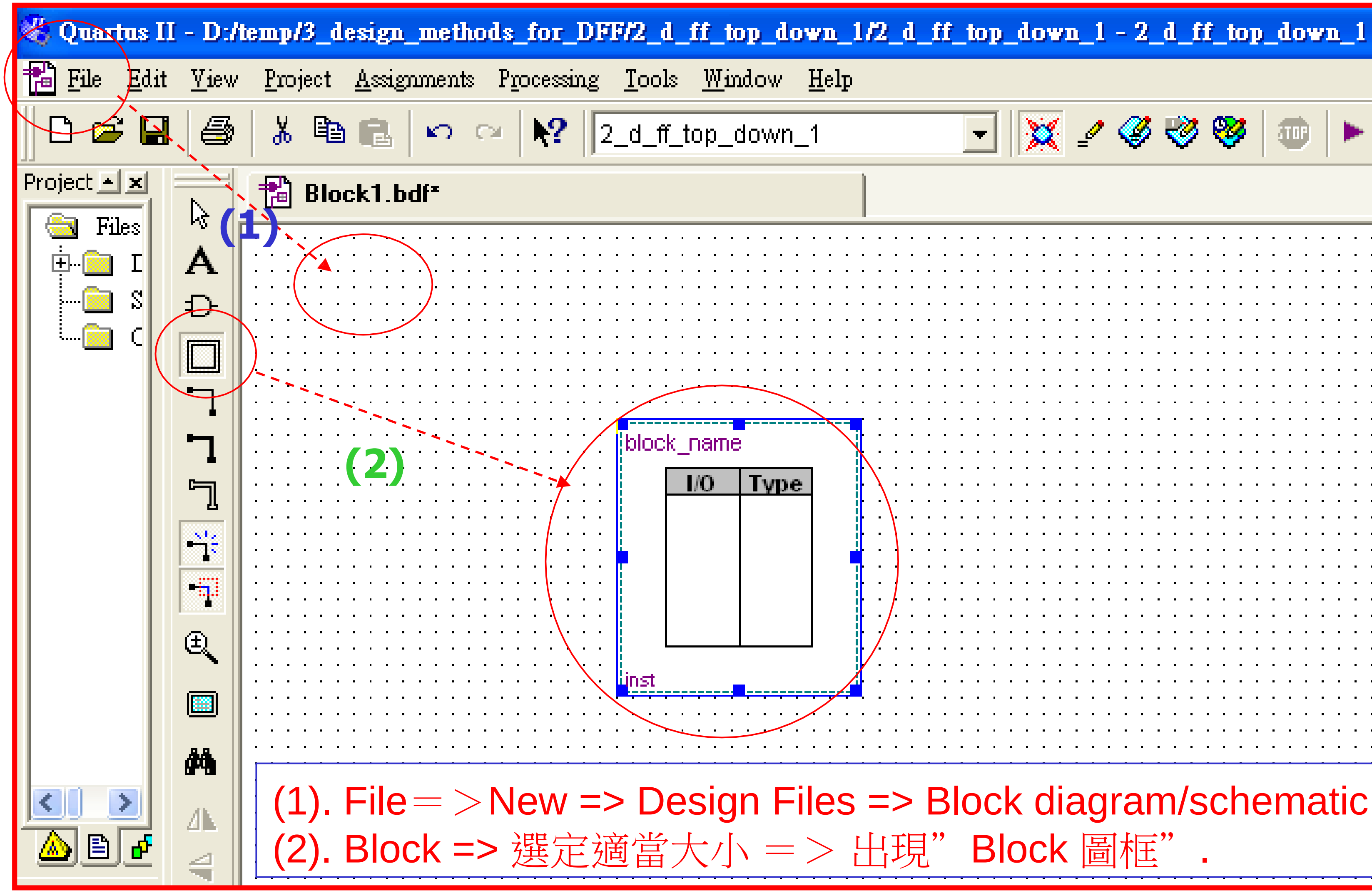
Top-Down Design (7)



Top-Down Design (8)

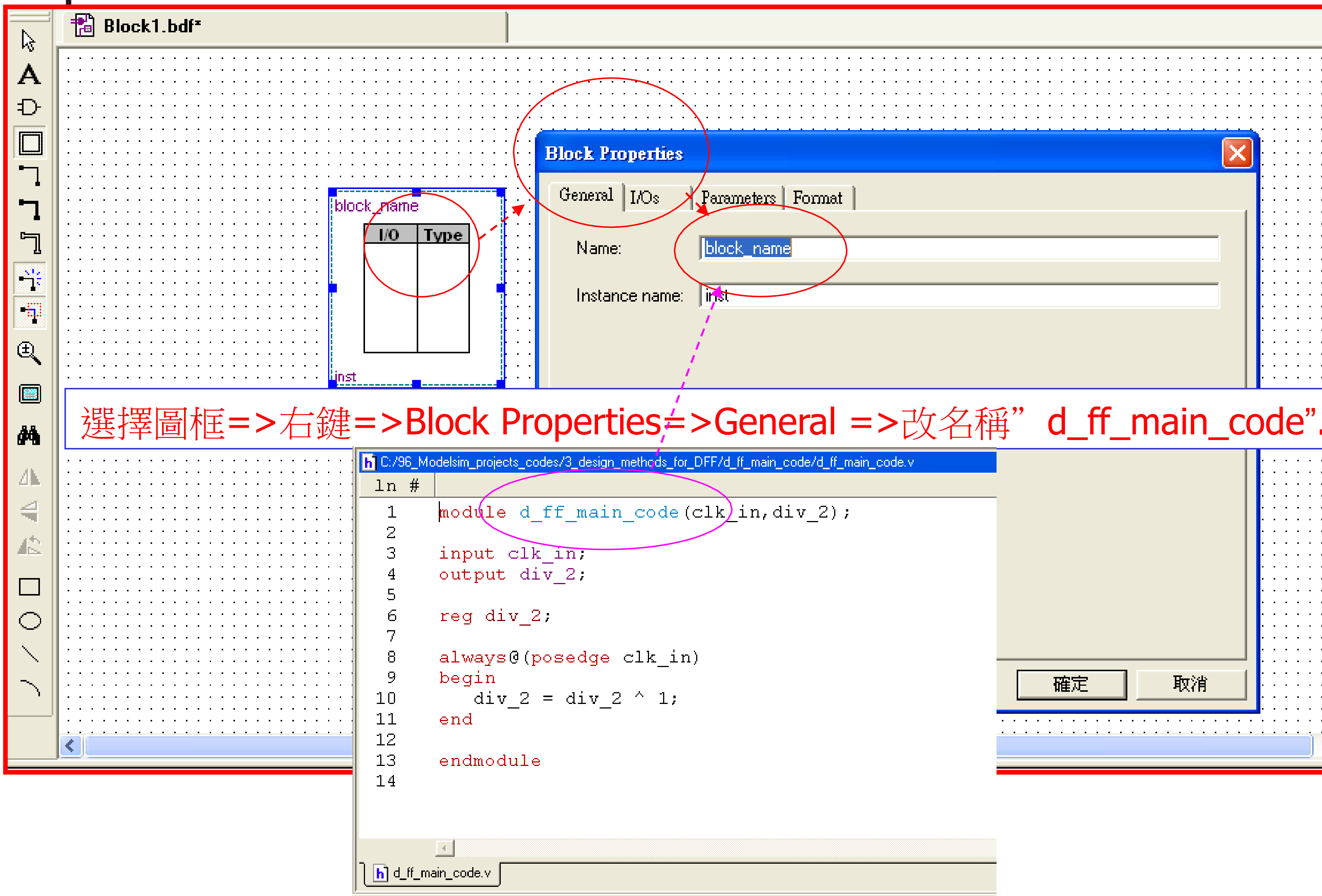


Top-Down Design (9)



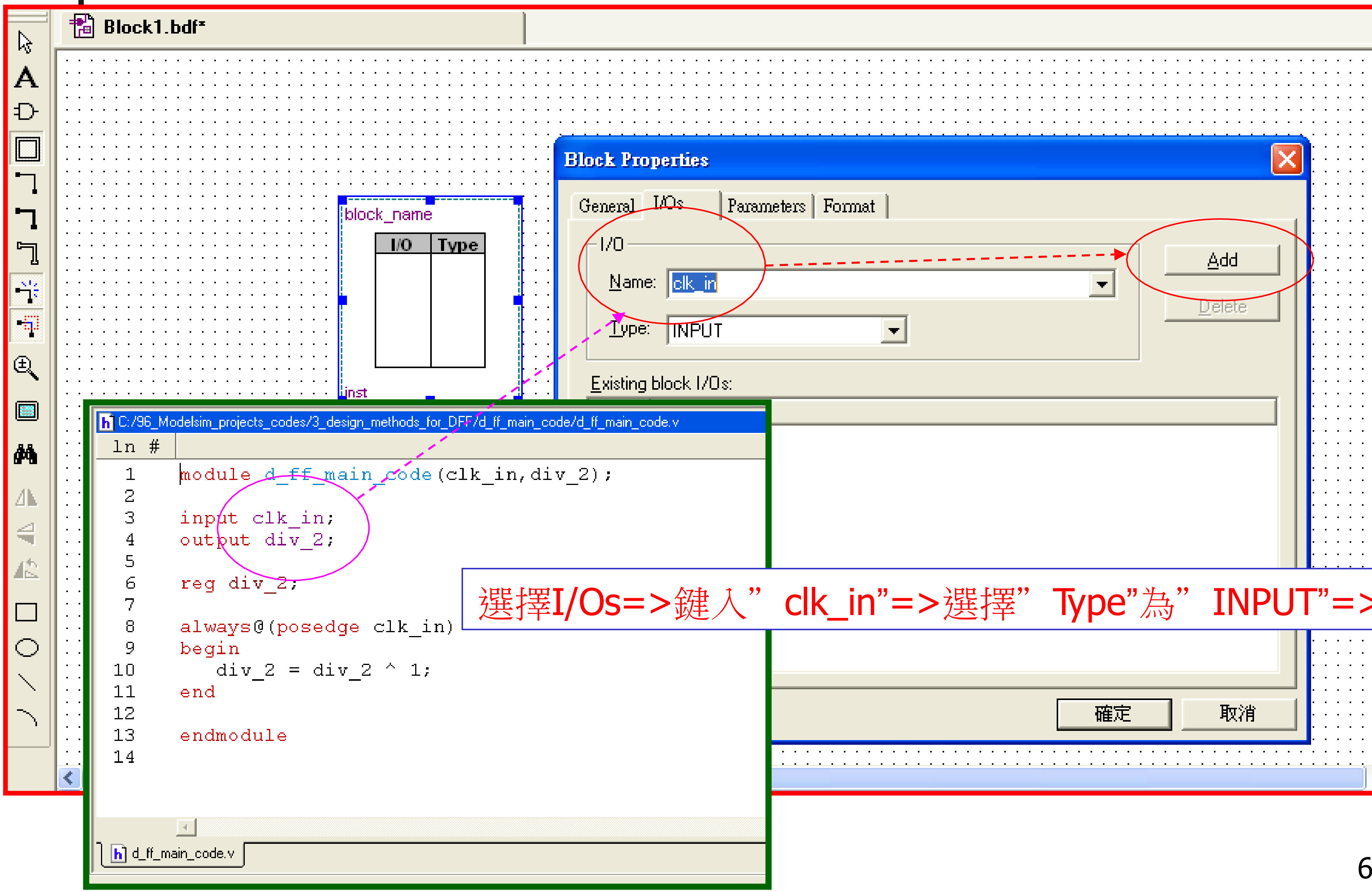
(1). File＝＞New => Design Files => Block diagram/schematic file.
(2). Block => 選定適當大小 ＝＞ 出現”Block 圖框” .

Top-Down Design (10)



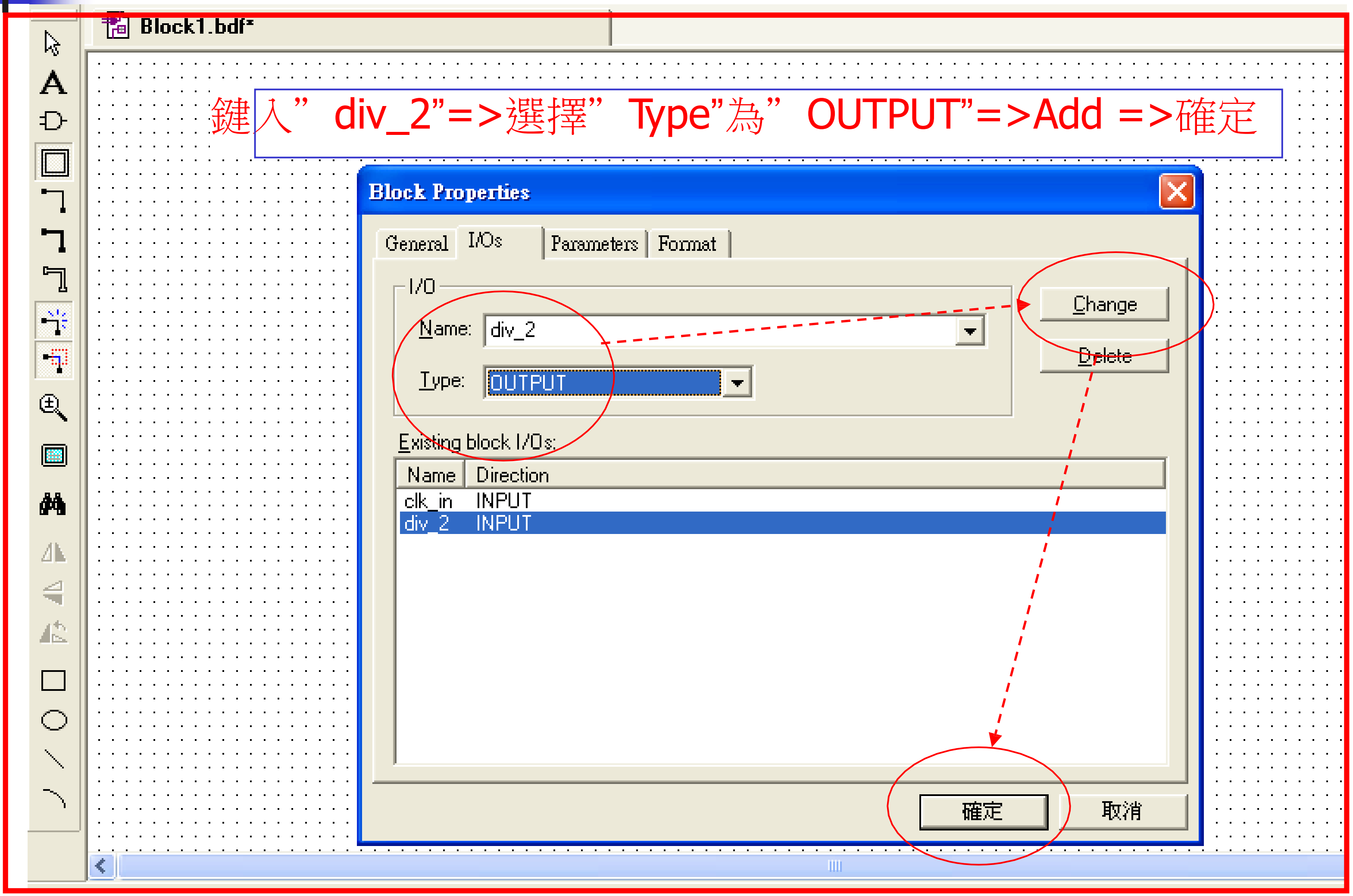
選擇圖框=>右鍵=>Block Properties=>General =>改名稱”d_ff_main_code”.

Top-Down Design (11)

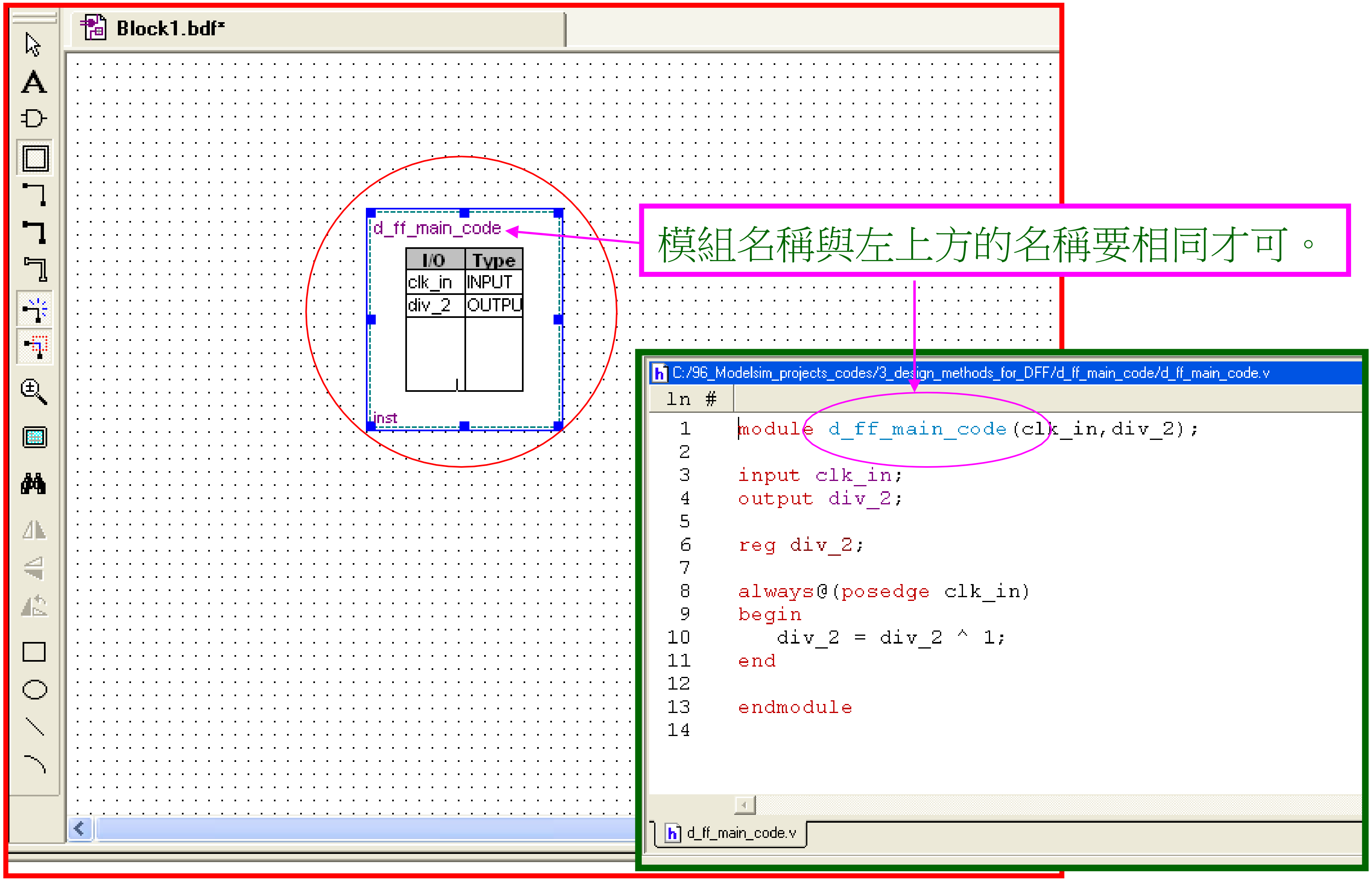


選擇I/Os=>鍵入” clk_in”=>選擇” Type”為” INPUT”=>Ad

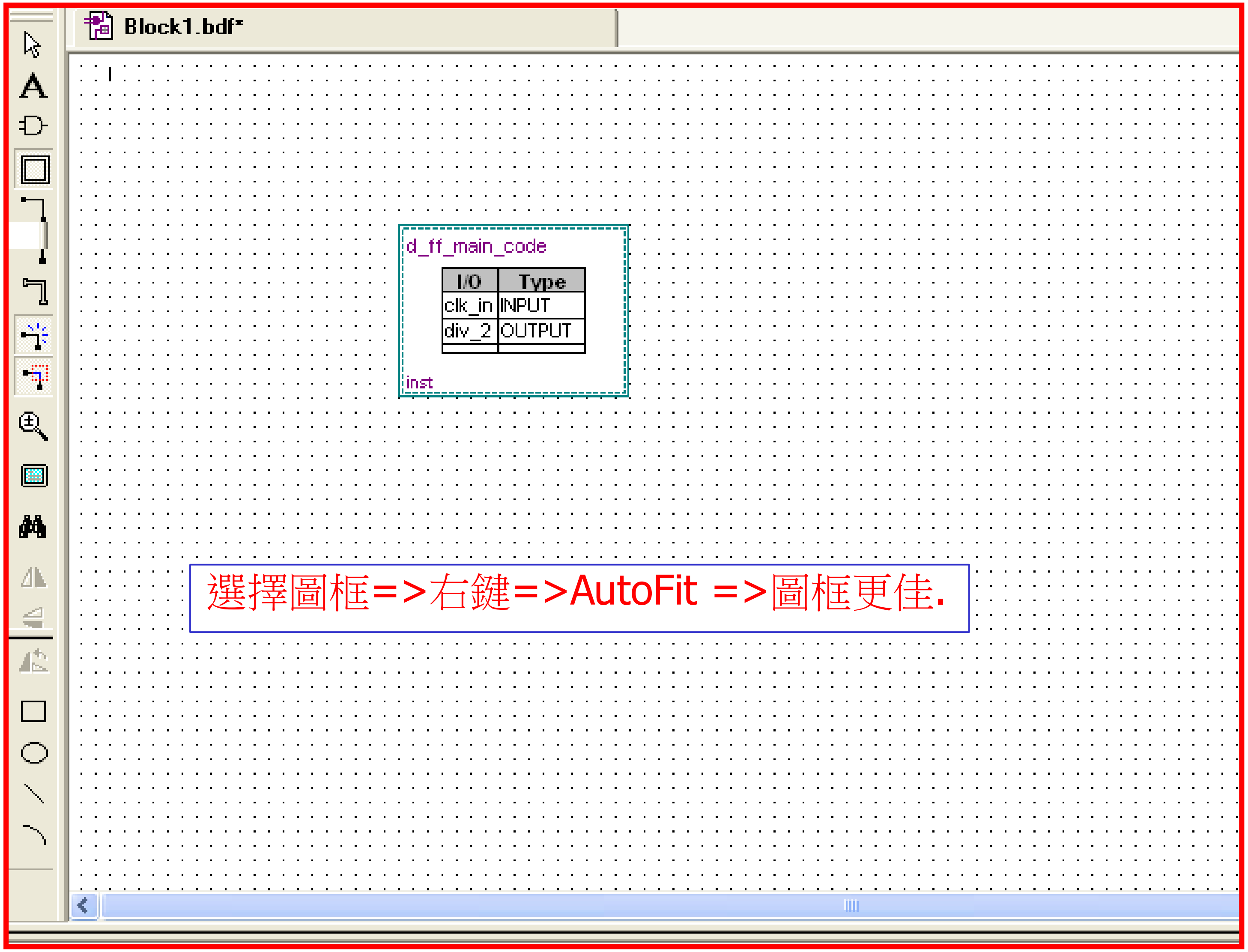
Top-Down Design (12)



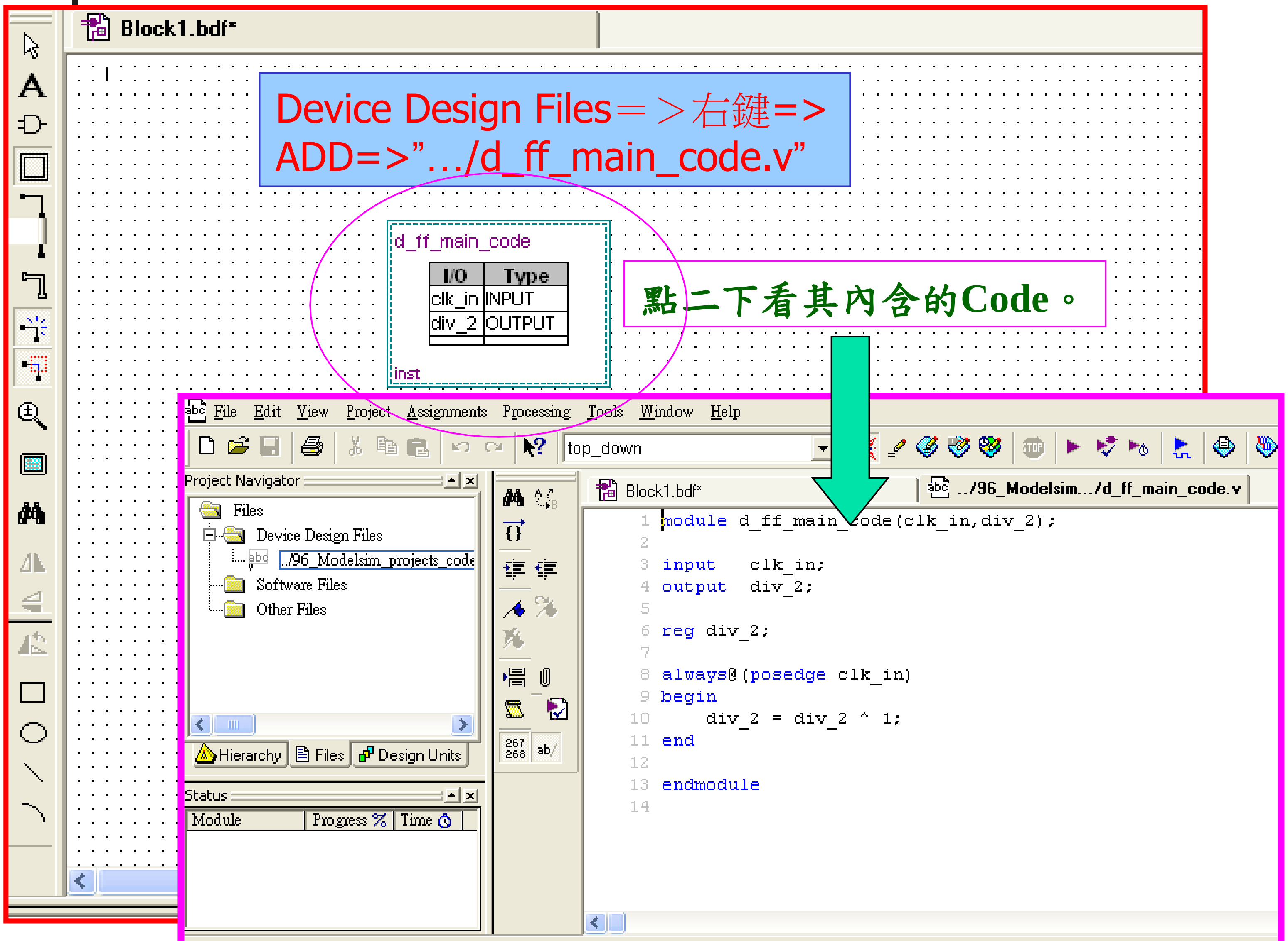
Top-Down Design (13)



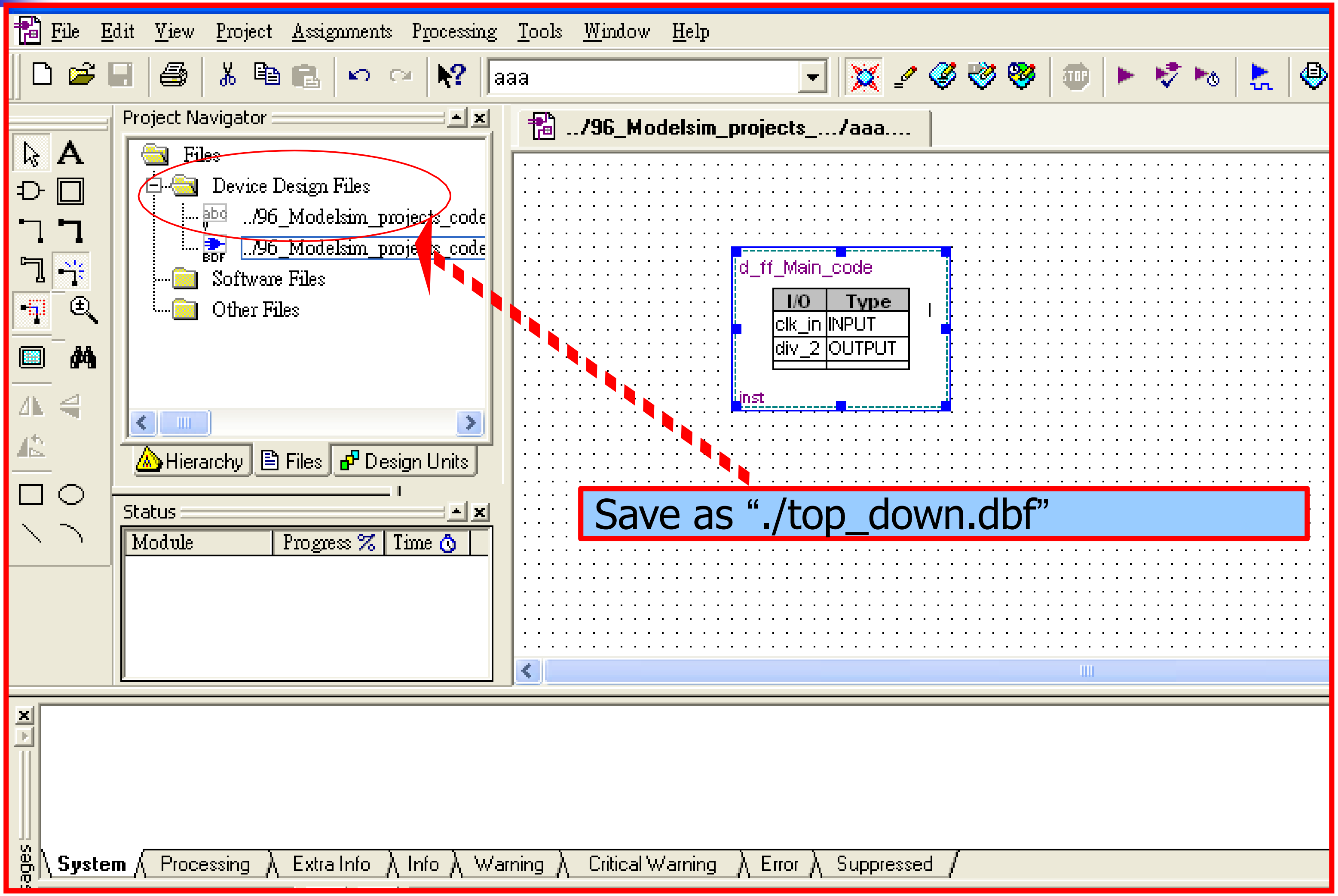
Top-Down Design (14)



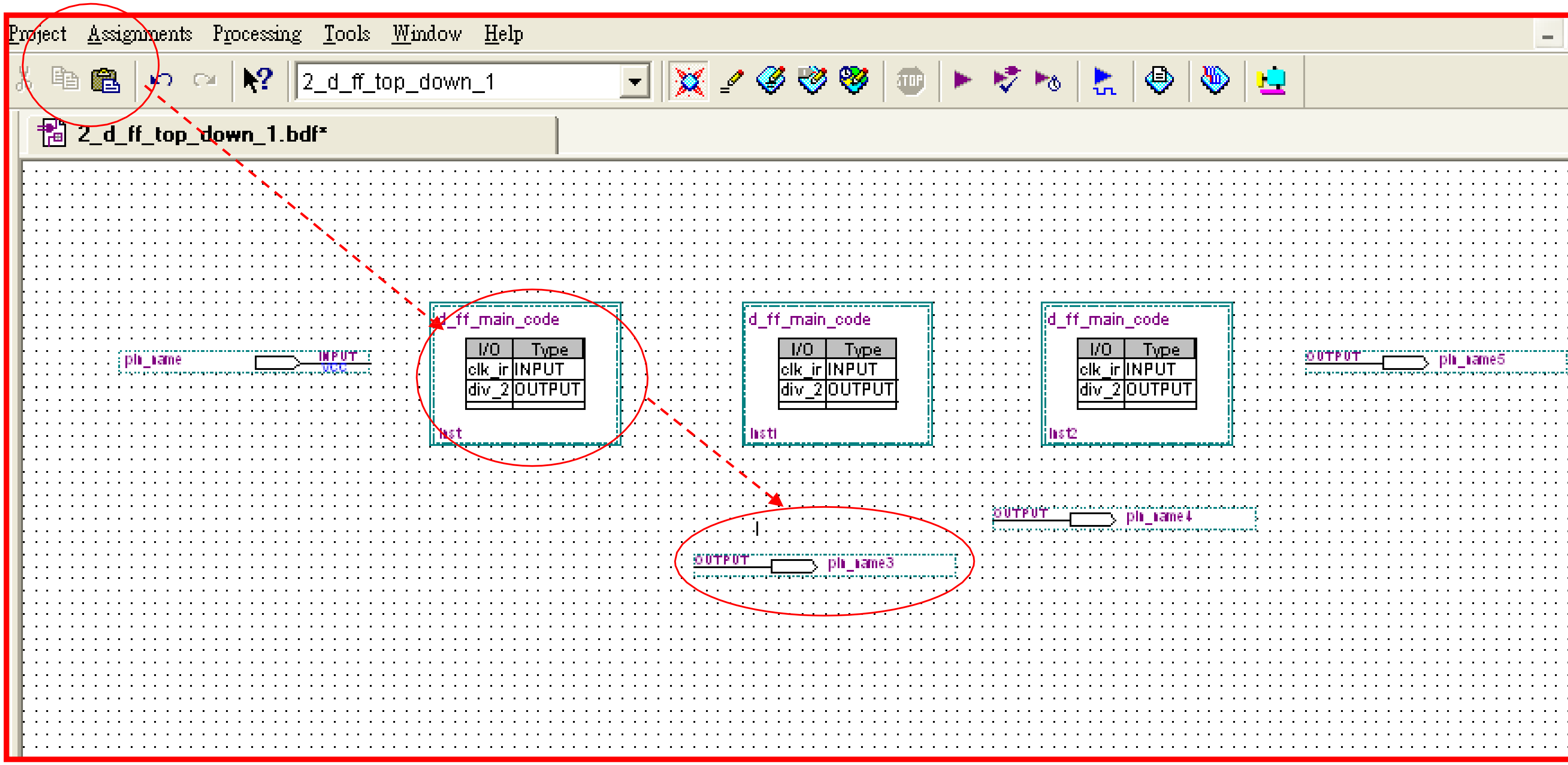
Top-Down Design (15)



Top-Down Design (16)

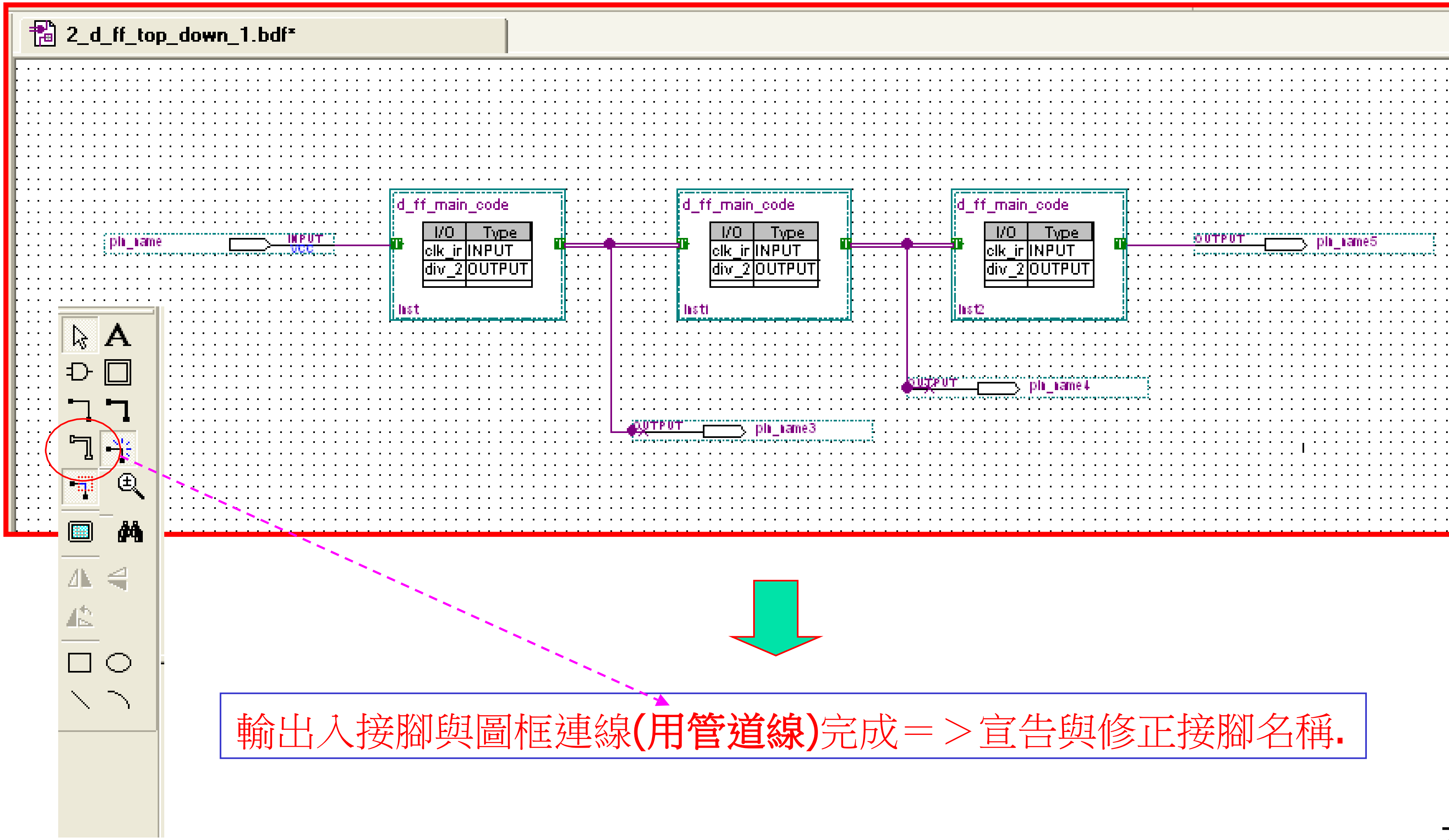


Top-Down Design (17)



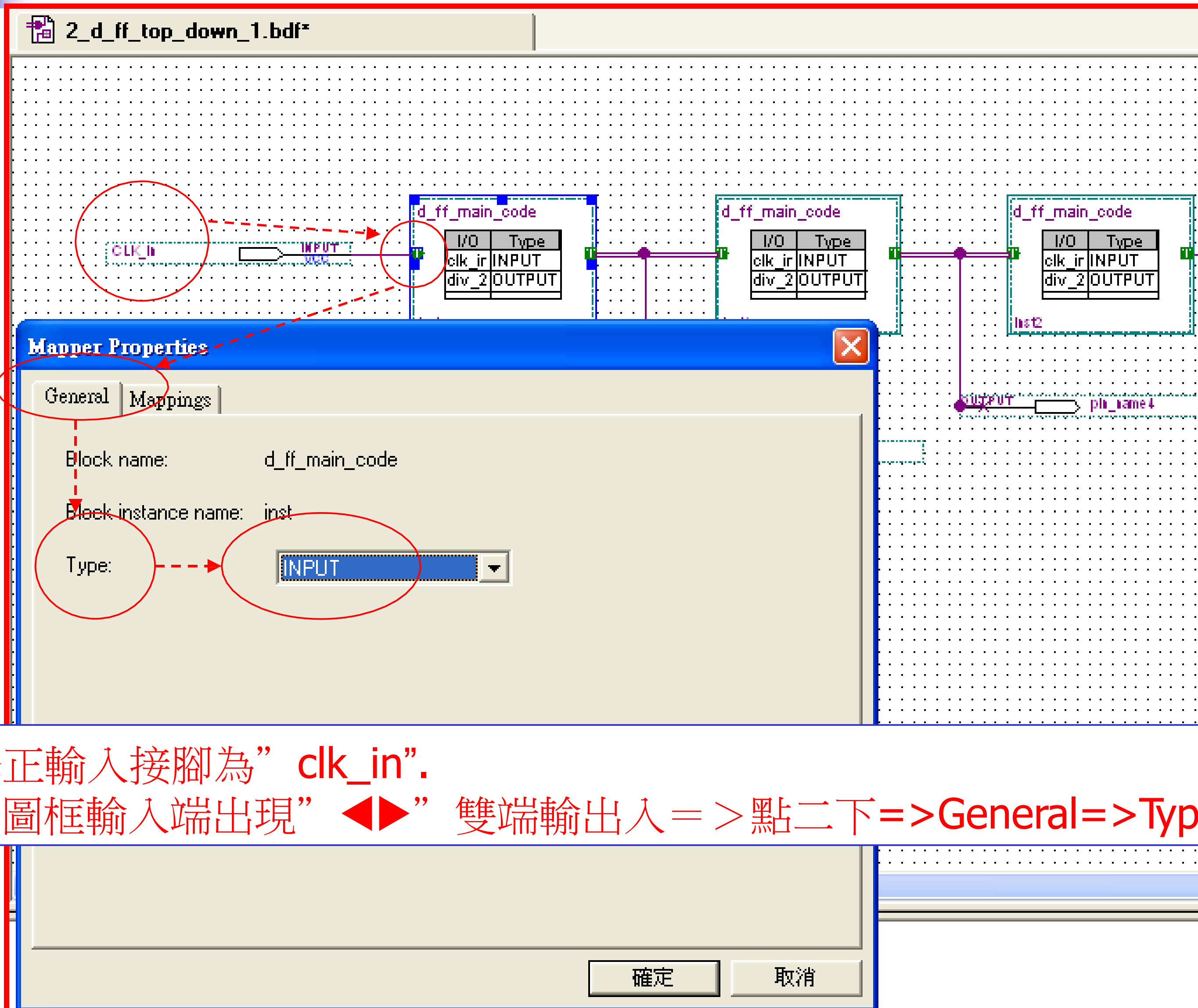
Copy 2 Figures＝＞Insert “Input and Output Pins”=>連線

Top-Down Design (18)



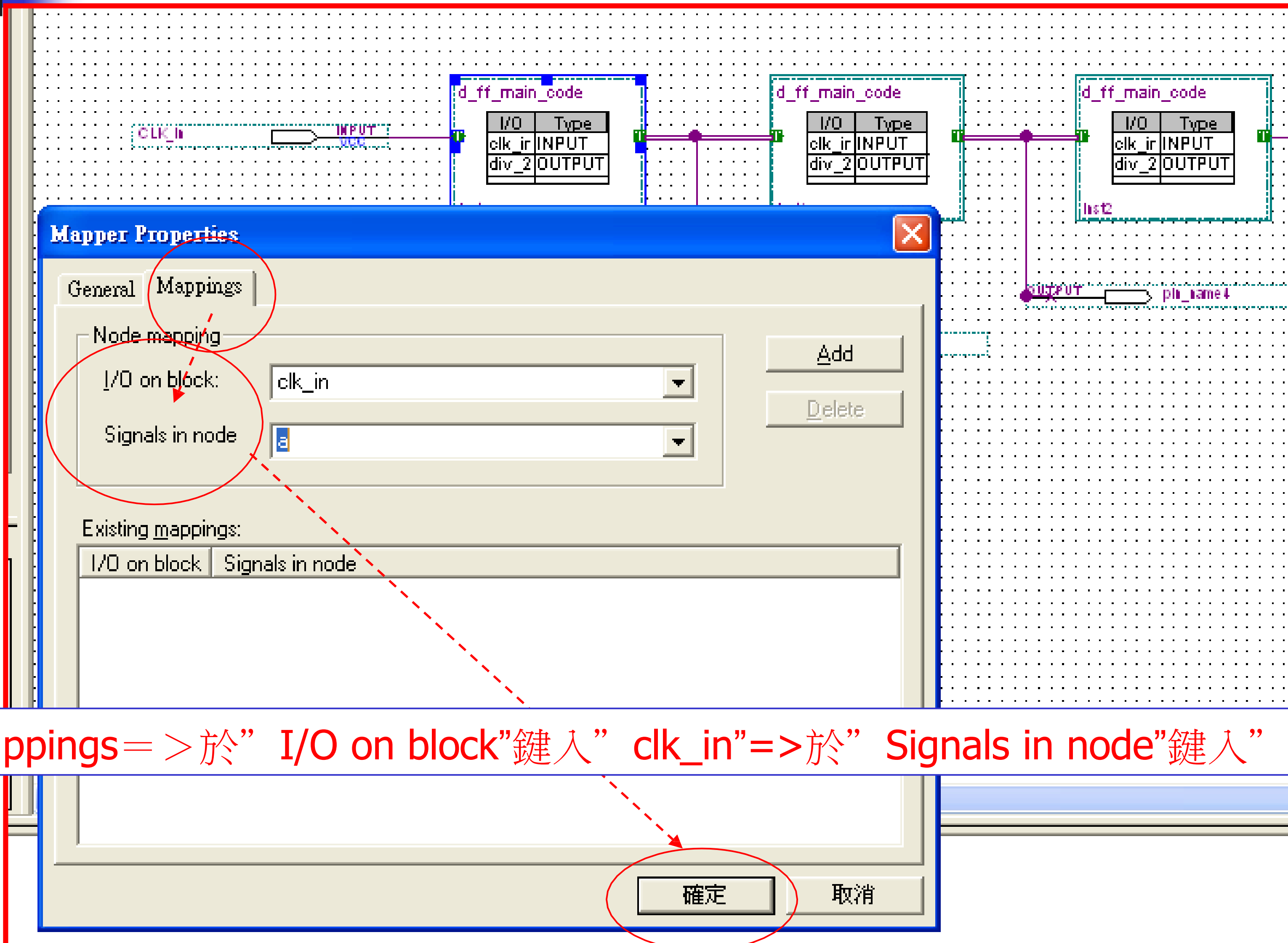
輸出入接腳與圖框連線(用管道線)完成＝＞宣告與修正接腳名稱.

Top-Down Design (19)



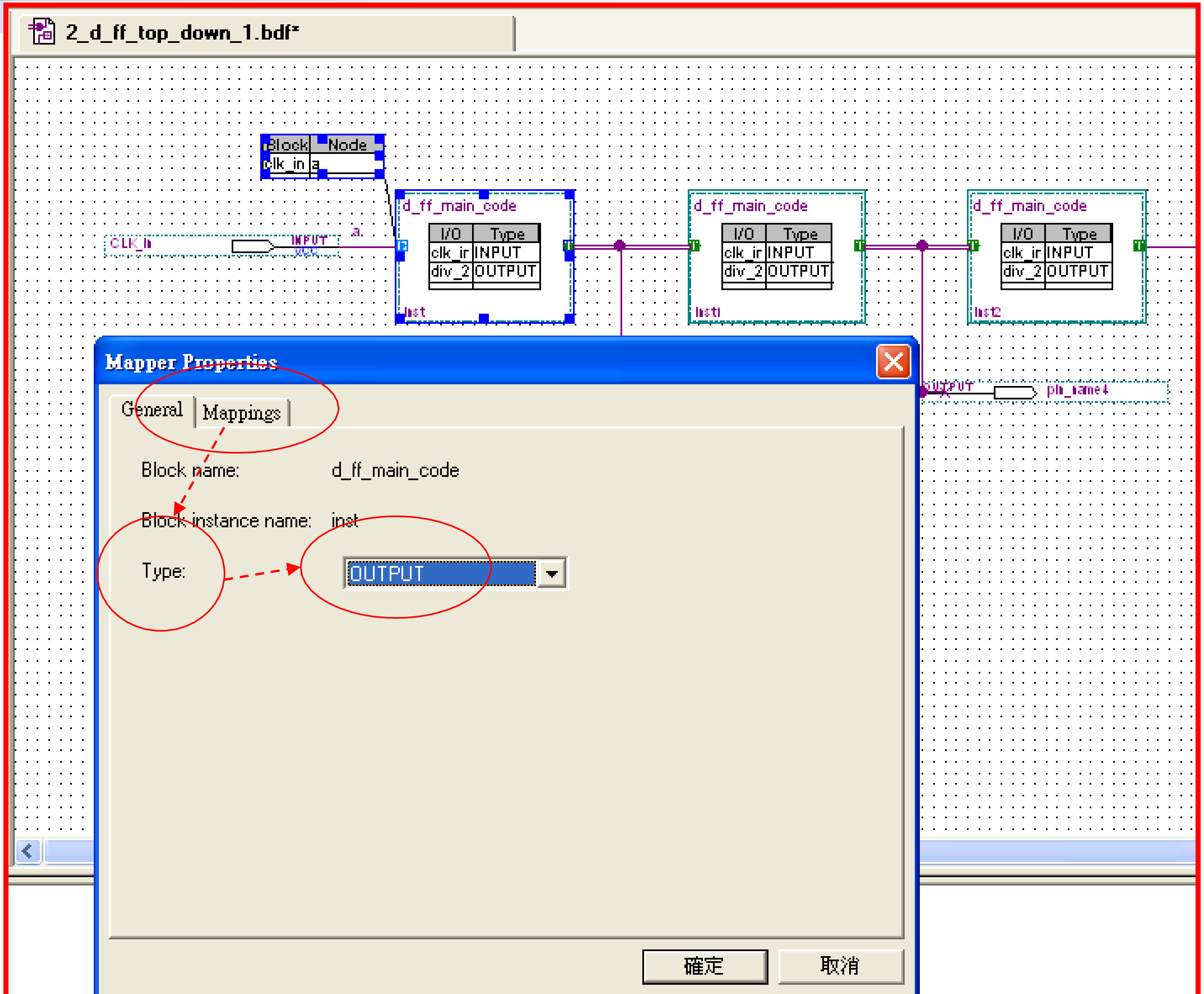
(1).修正輸入接腳為”clk_in”.
(2).於圖框輸入端出現”◀▶”雙端輸出入＝＞點二下=>General=>Type=>選”input”

Top-Down Design (20)

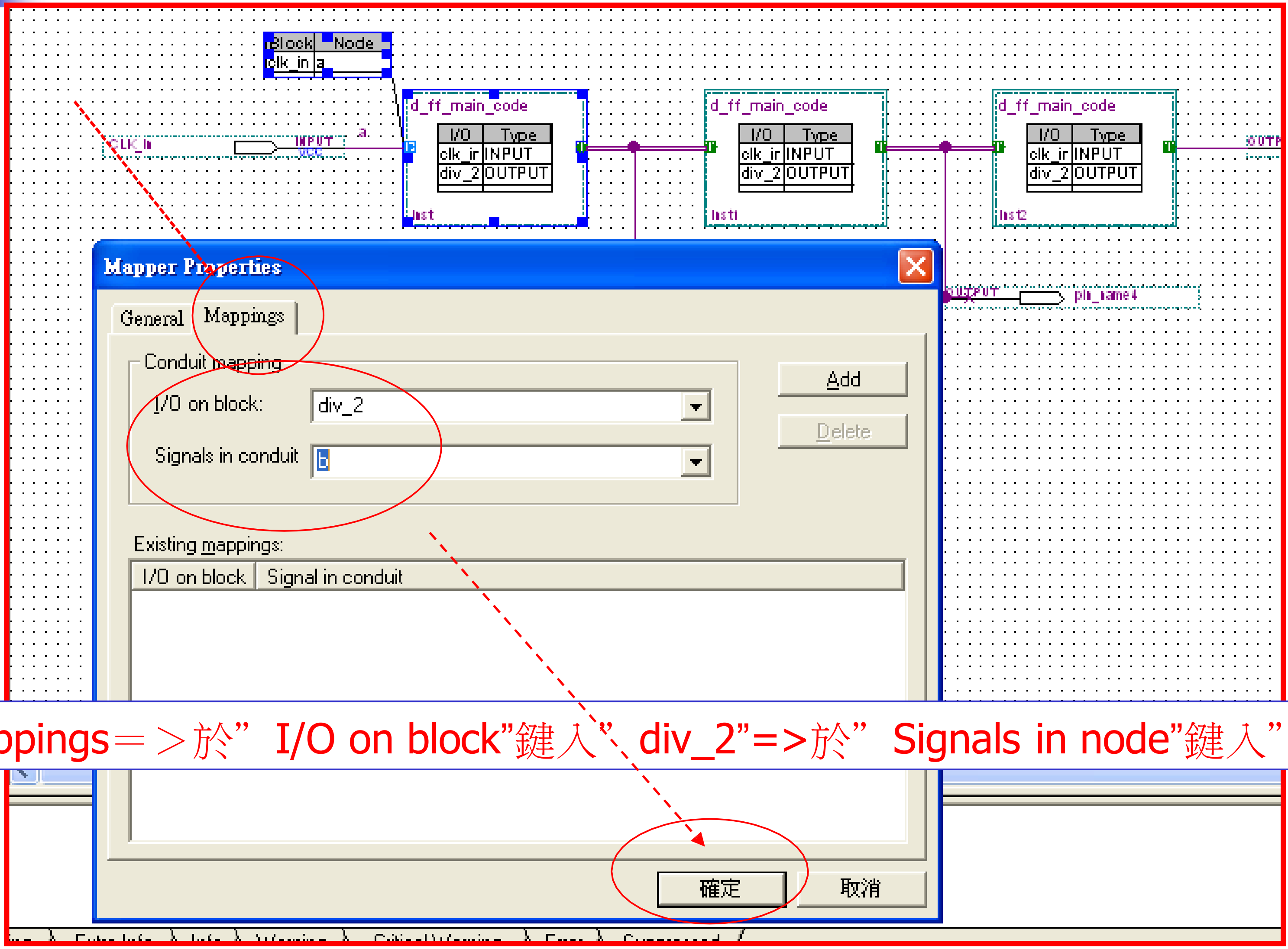


選Mappings＝＞於”I/O on block”鍵入”clk_in”=>於”Signals in node”鍵入”a”=>確定.

Top-Down Design (21)

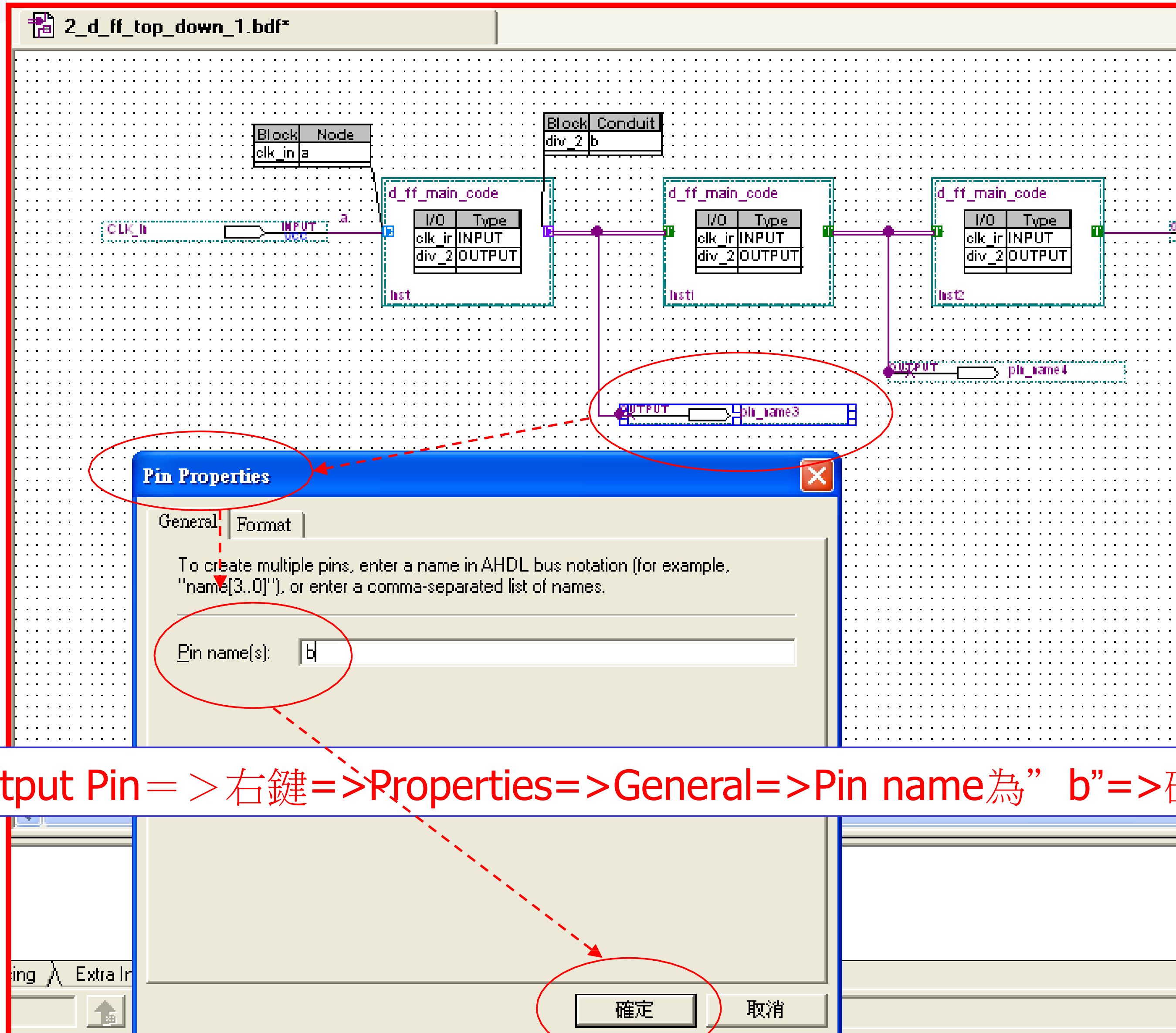


Top-Down Design (22)



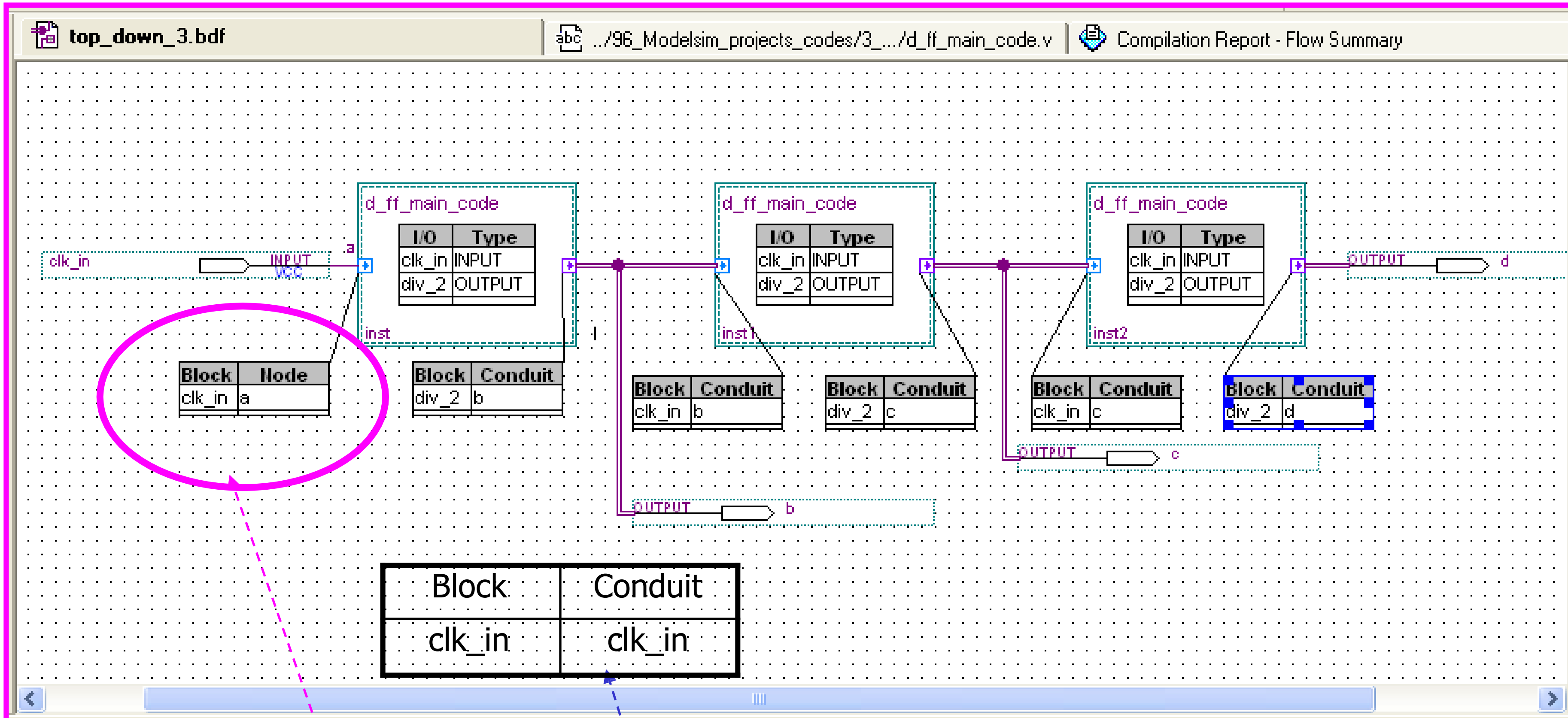
選Mappings＝＞於”I/O on block”鍵入”div_2”=>於”Signals in node”鍵入”b”=>確定

Top-Down Design (23)



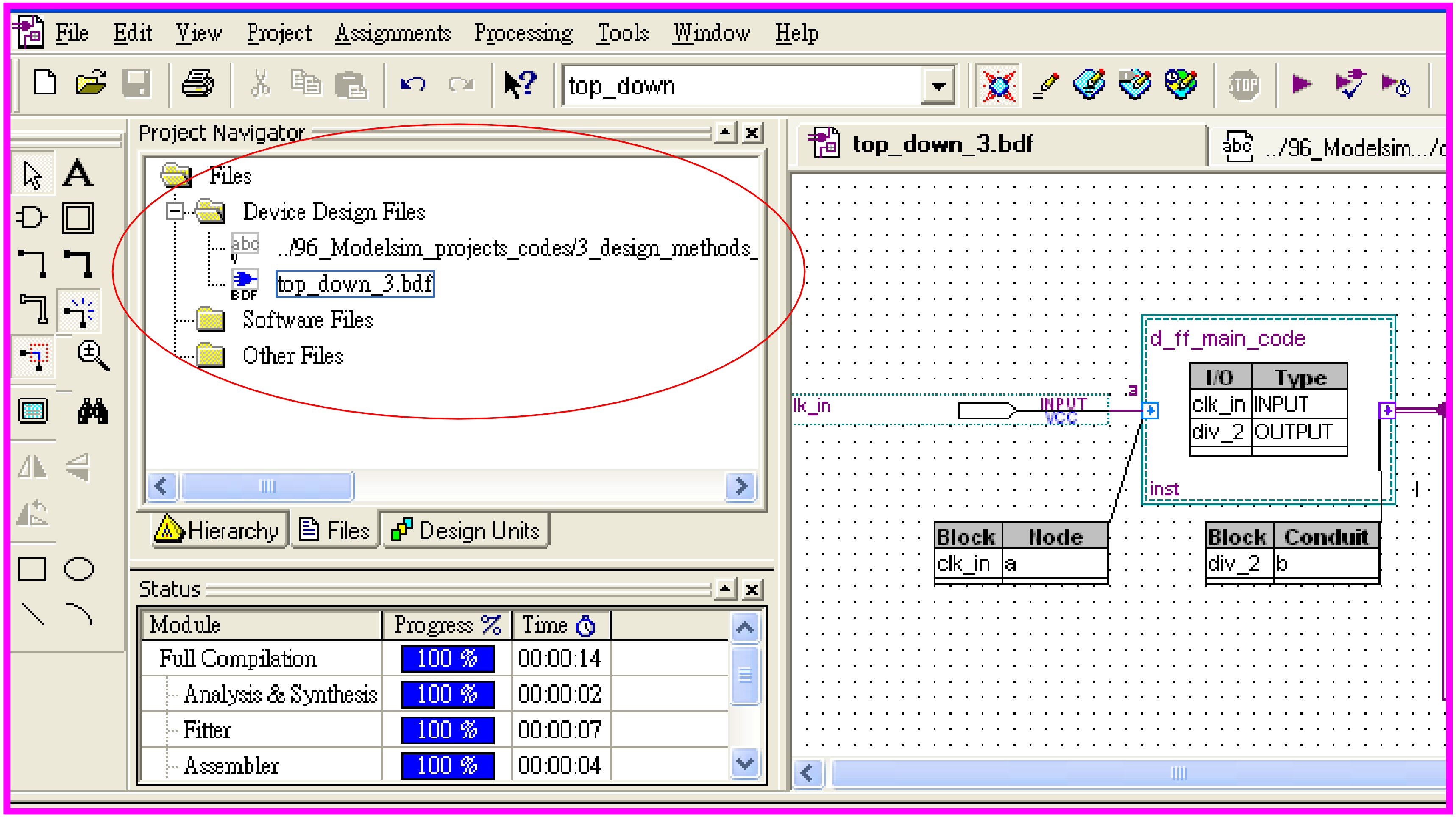
Output Pin＝＞右鍵=>Properties=>General=>Pin name為”b”=>確定.

Top-Down Design (24)

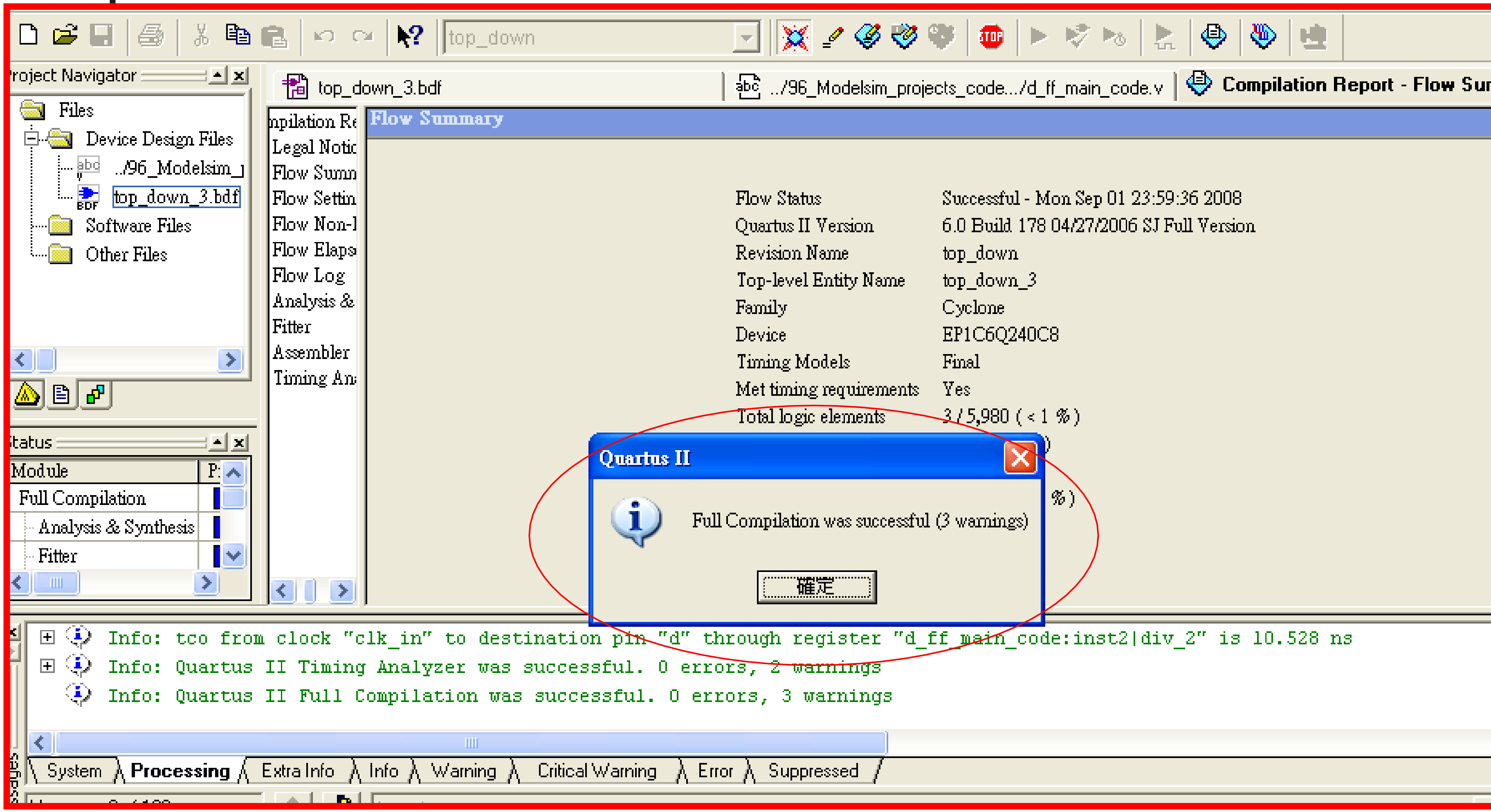


(1). 輸出入接腳的名稱若與節點名稱相同，使用”單位元接線（實線）”或”管線”均可。
(2). 若輸出入接腳的名稱若與節點名稱不同，僅能使用”管線”。
(3). 嚐試使用管線或位元連線看看結果如何?(單位元均可適用,惟多位元會有誤).

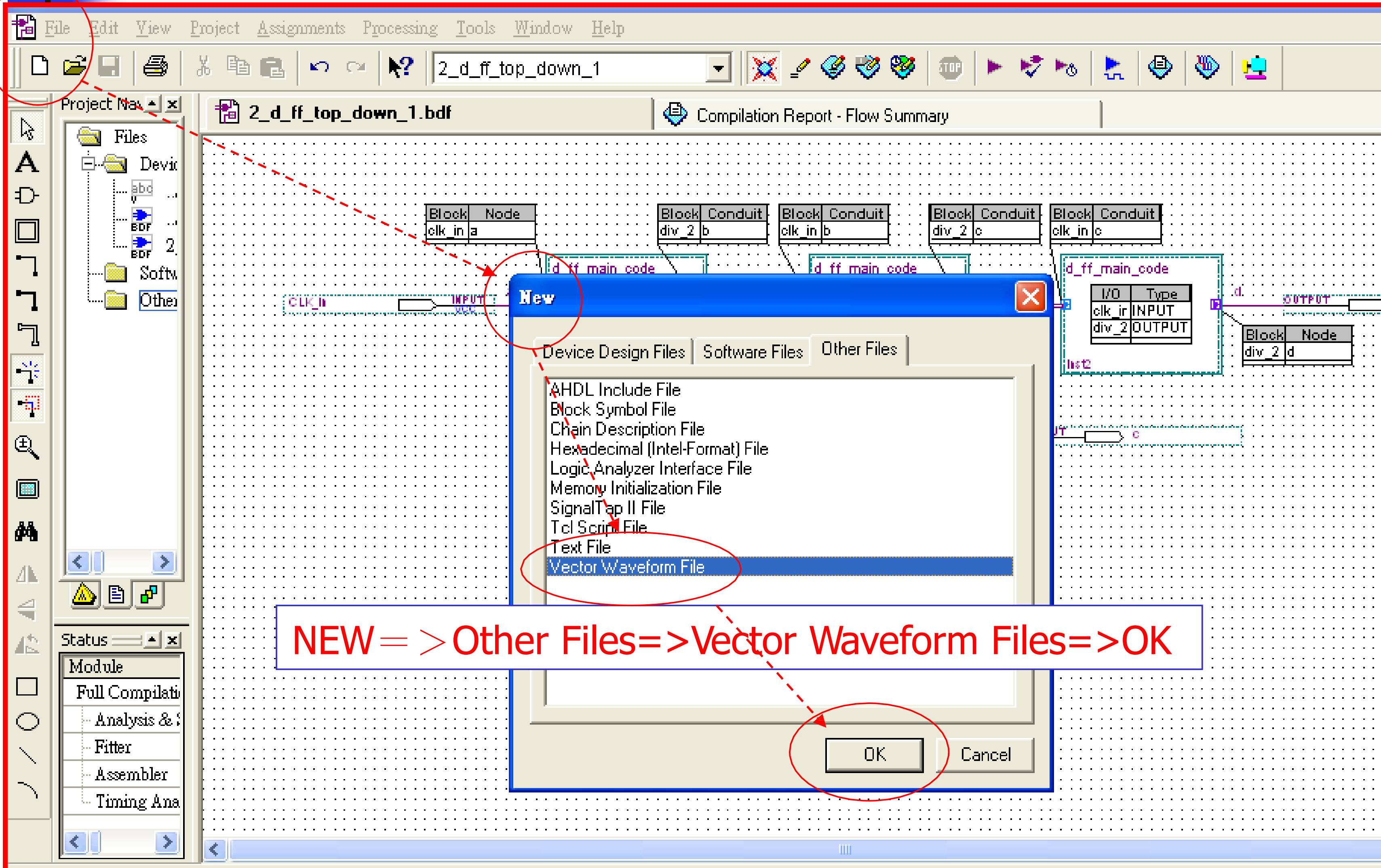
Top-Down Design (25)



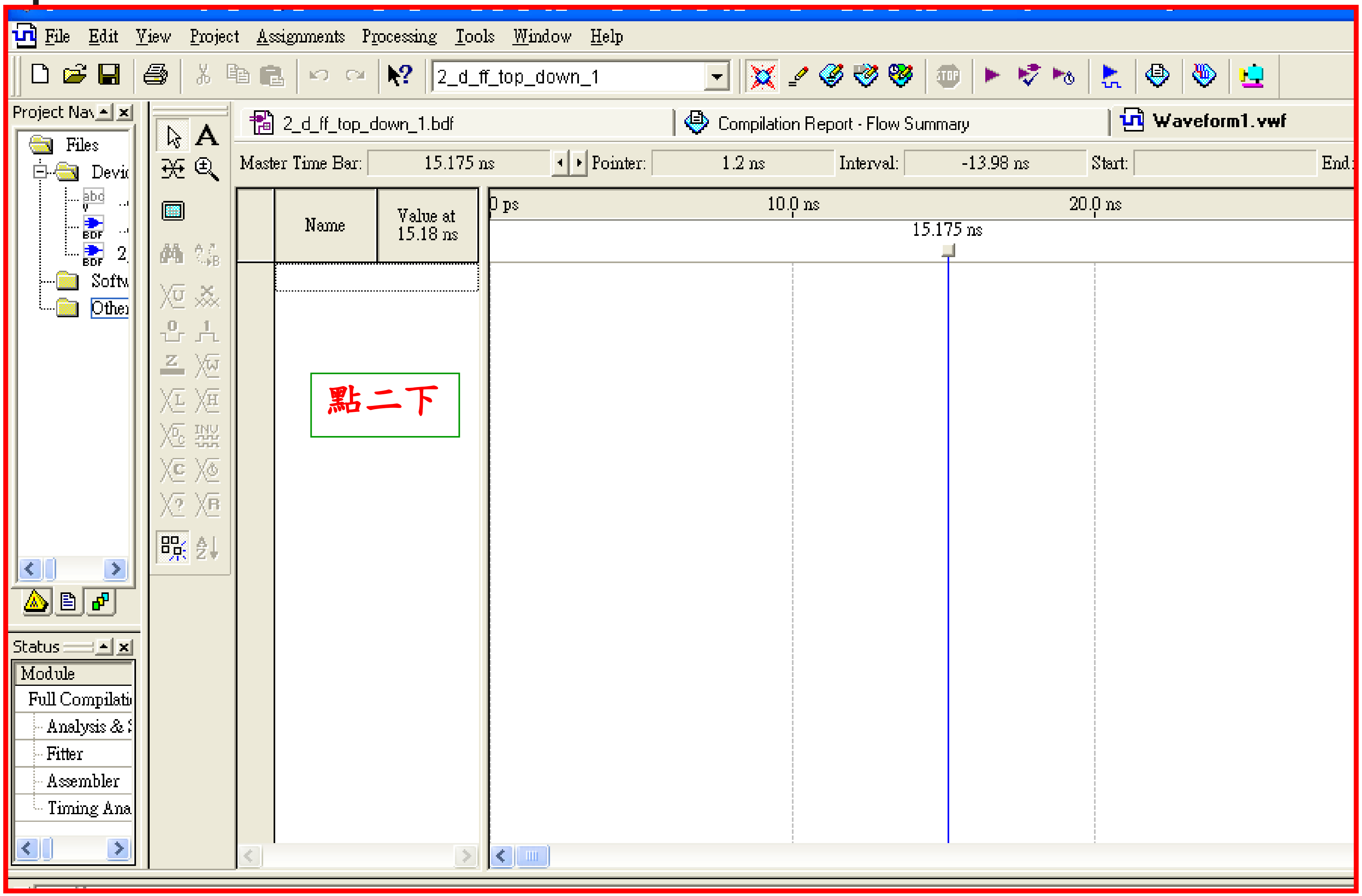
Top-Down Design (26)



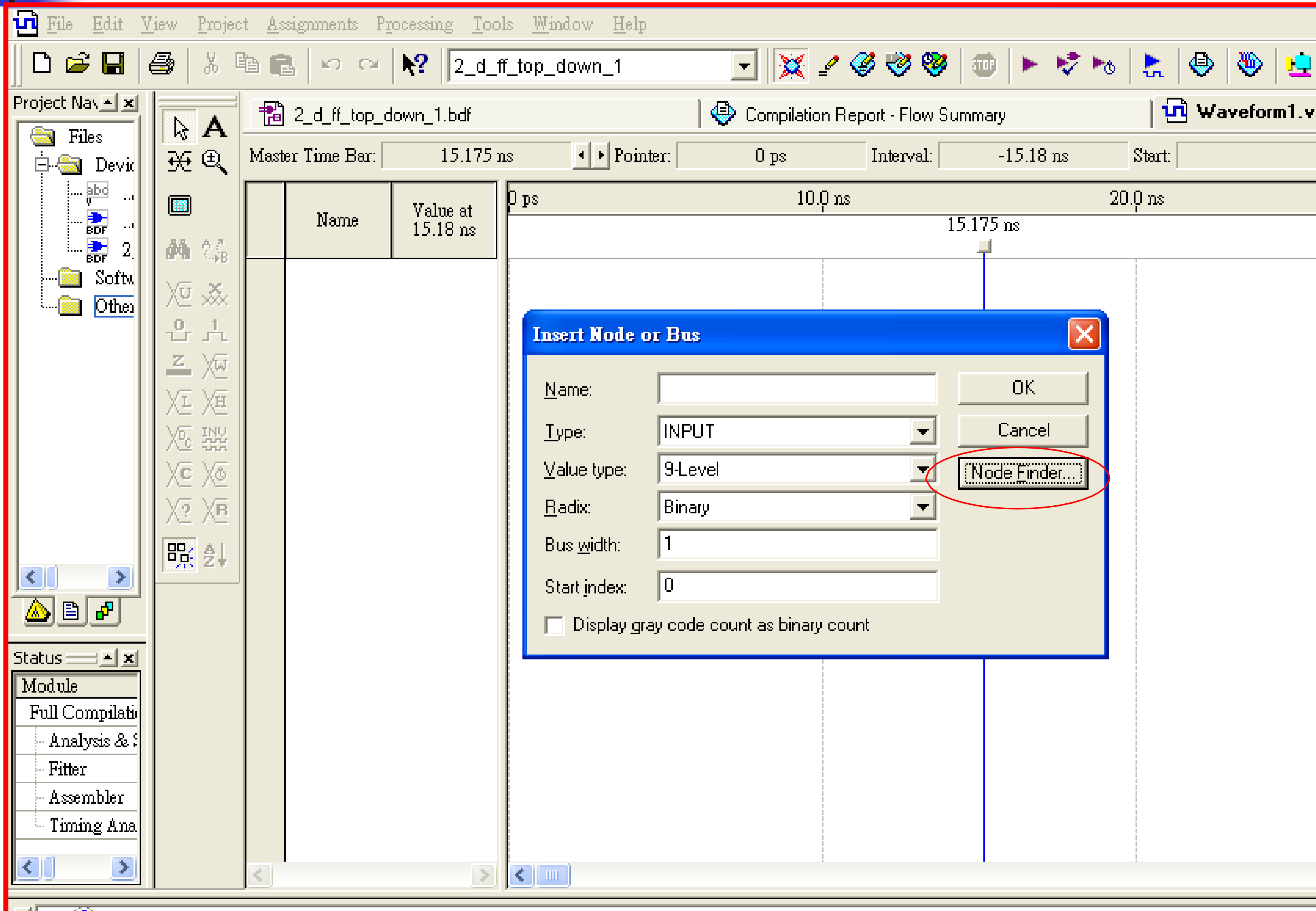
Top-Down Design (27)



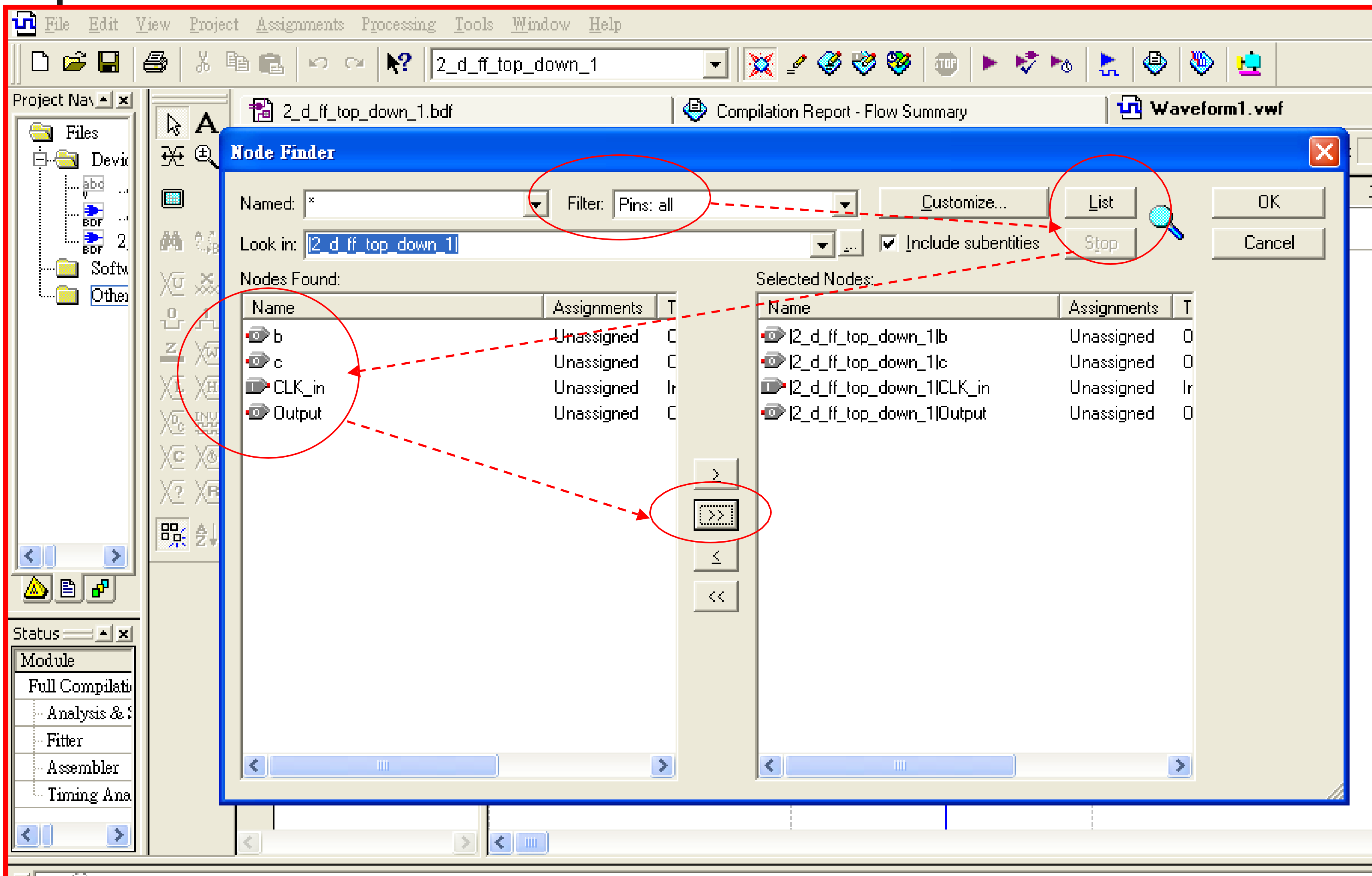
Top-Down Design (28)



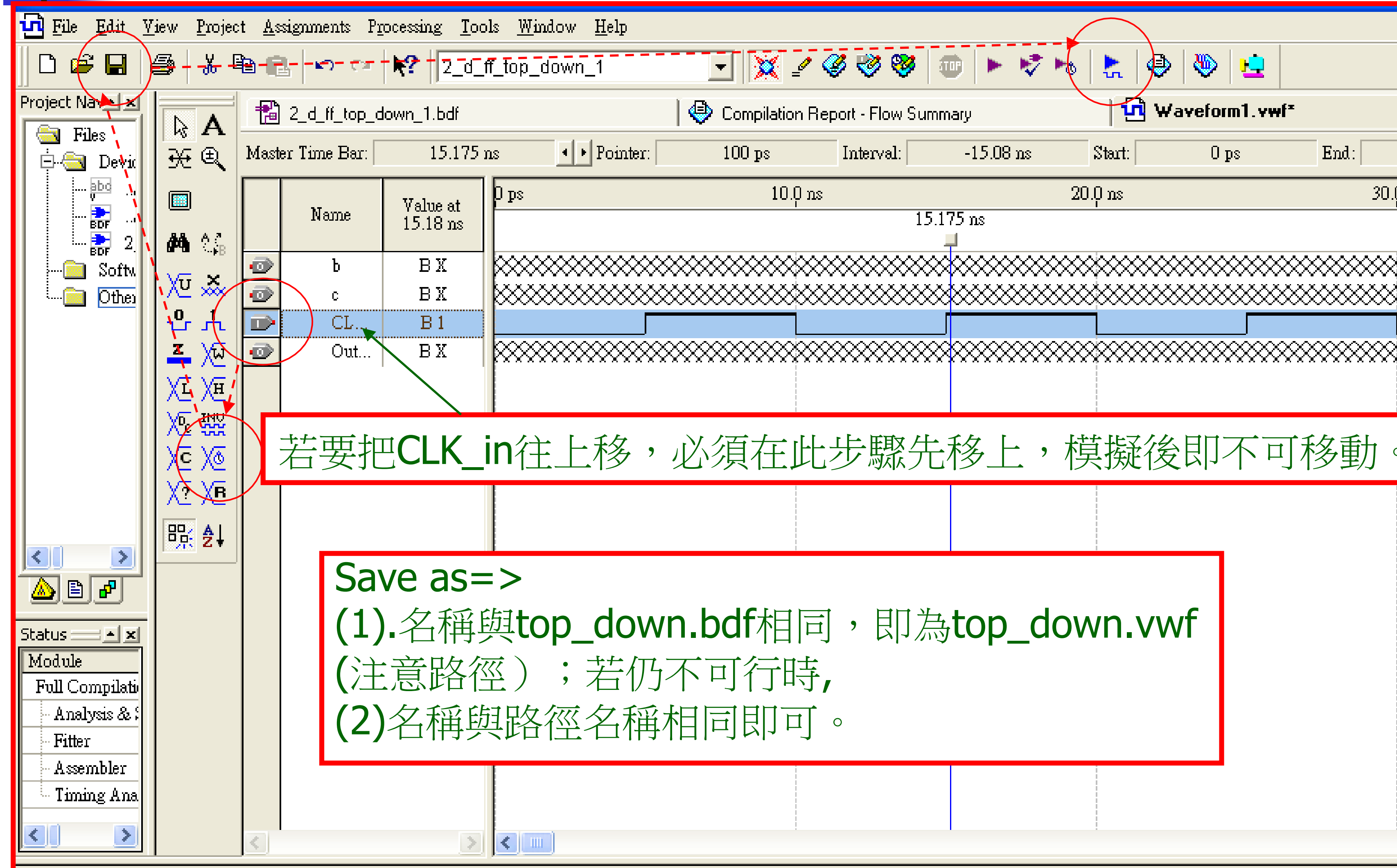
Top-Down Design (29)



Top-Down Design (30)



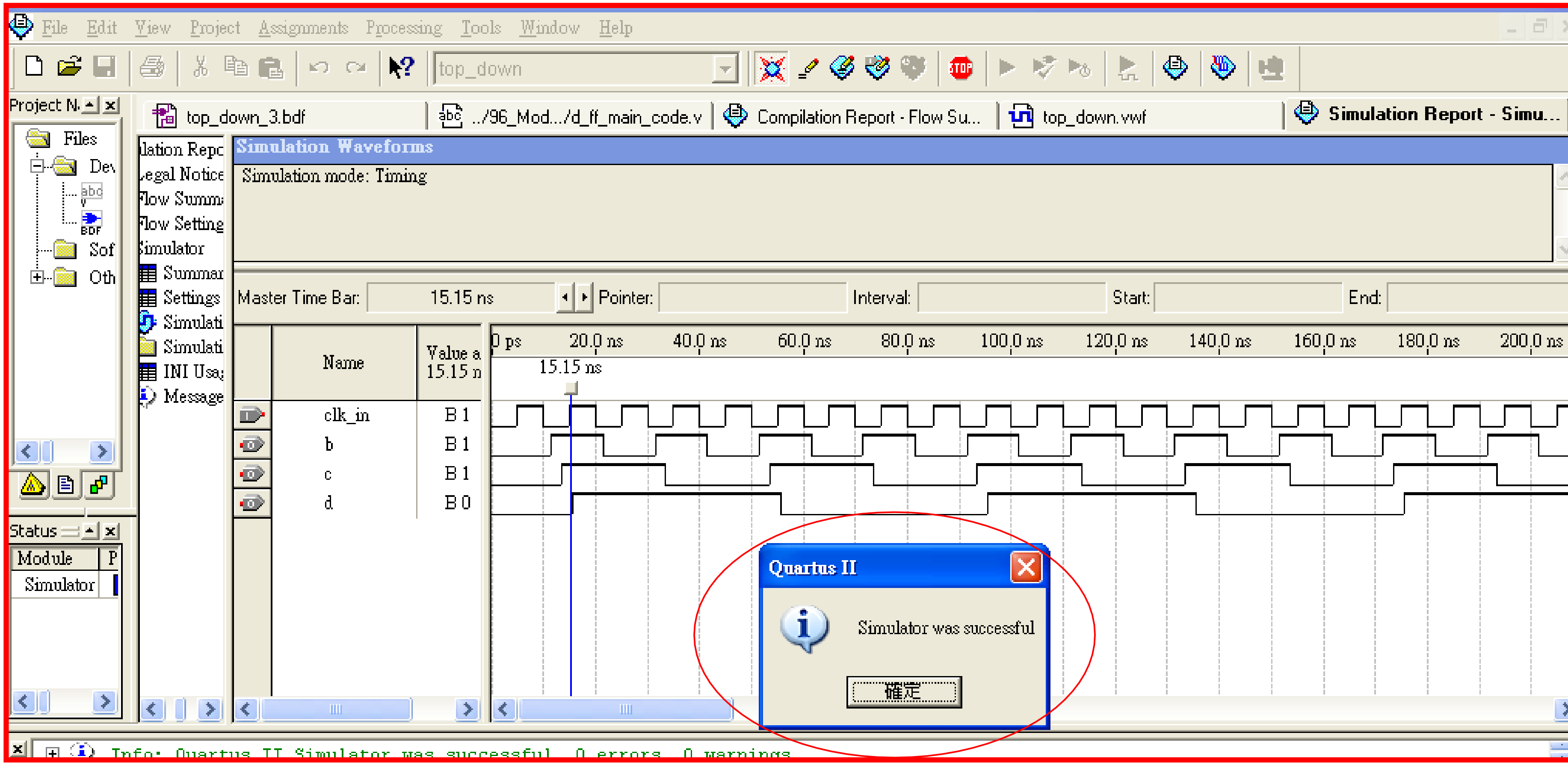
Top-Down Design (31)



若要把CLK_in往上移，必須在此步驟先移上，模擬後即不可移動。

Save as=>
(1).名稱與top_down.bdf相同，即為top_down.vwf
(注意路徑）；若仍不可行時,
(2)名稱與路徑名稱相同即可。

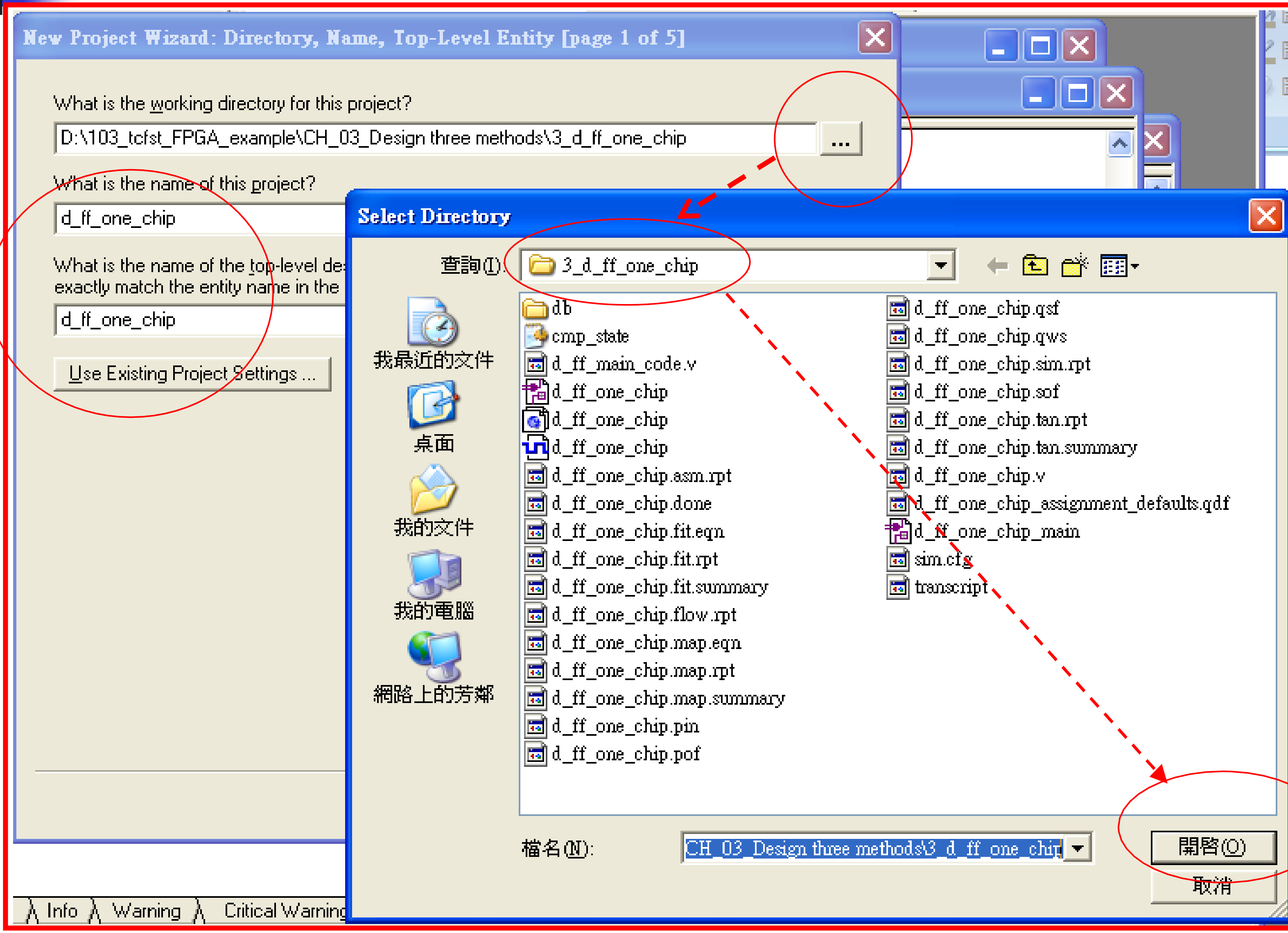
Top-Down Design (32)



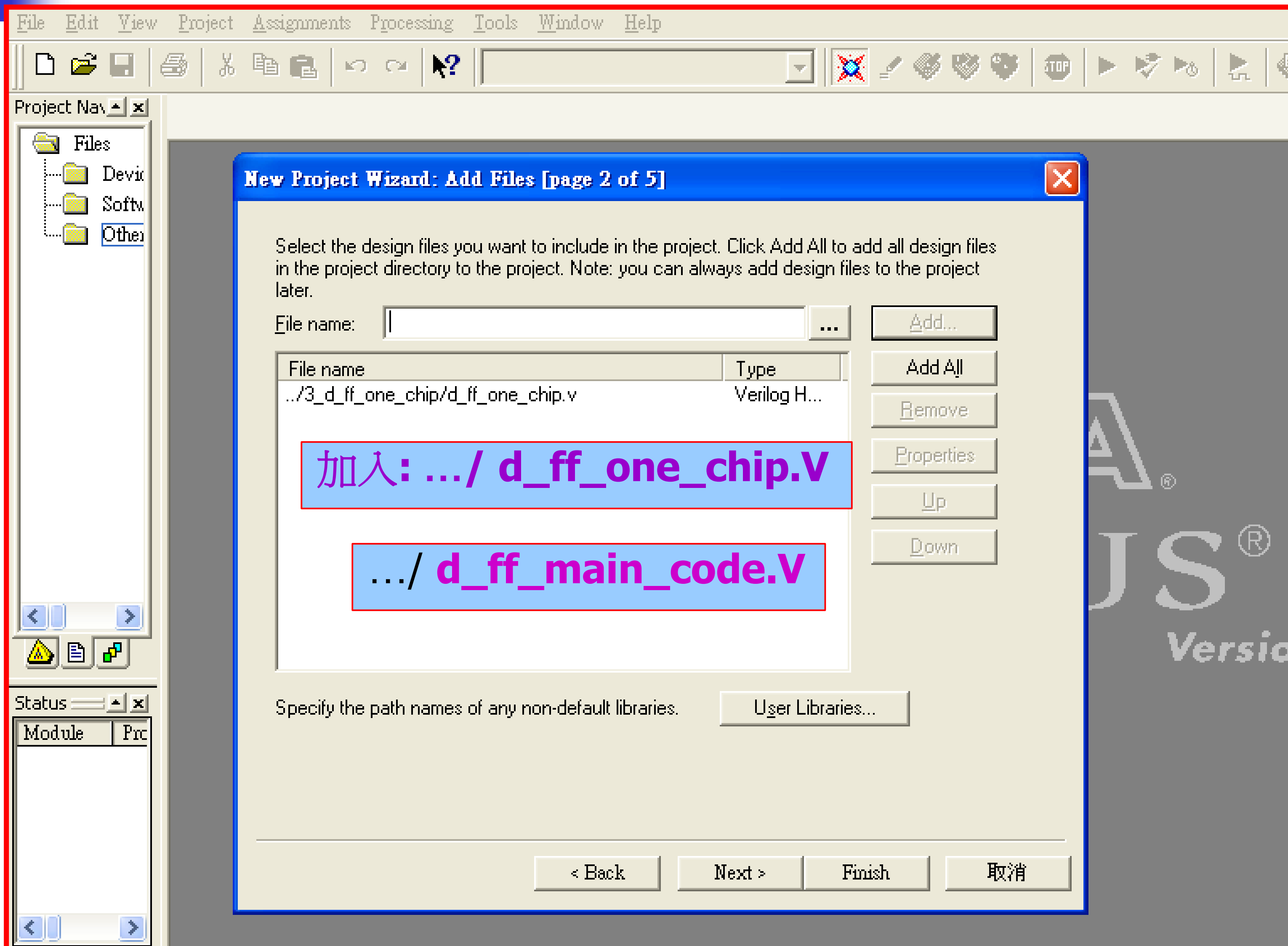
One-Chip Design

One-Chip Design Flow

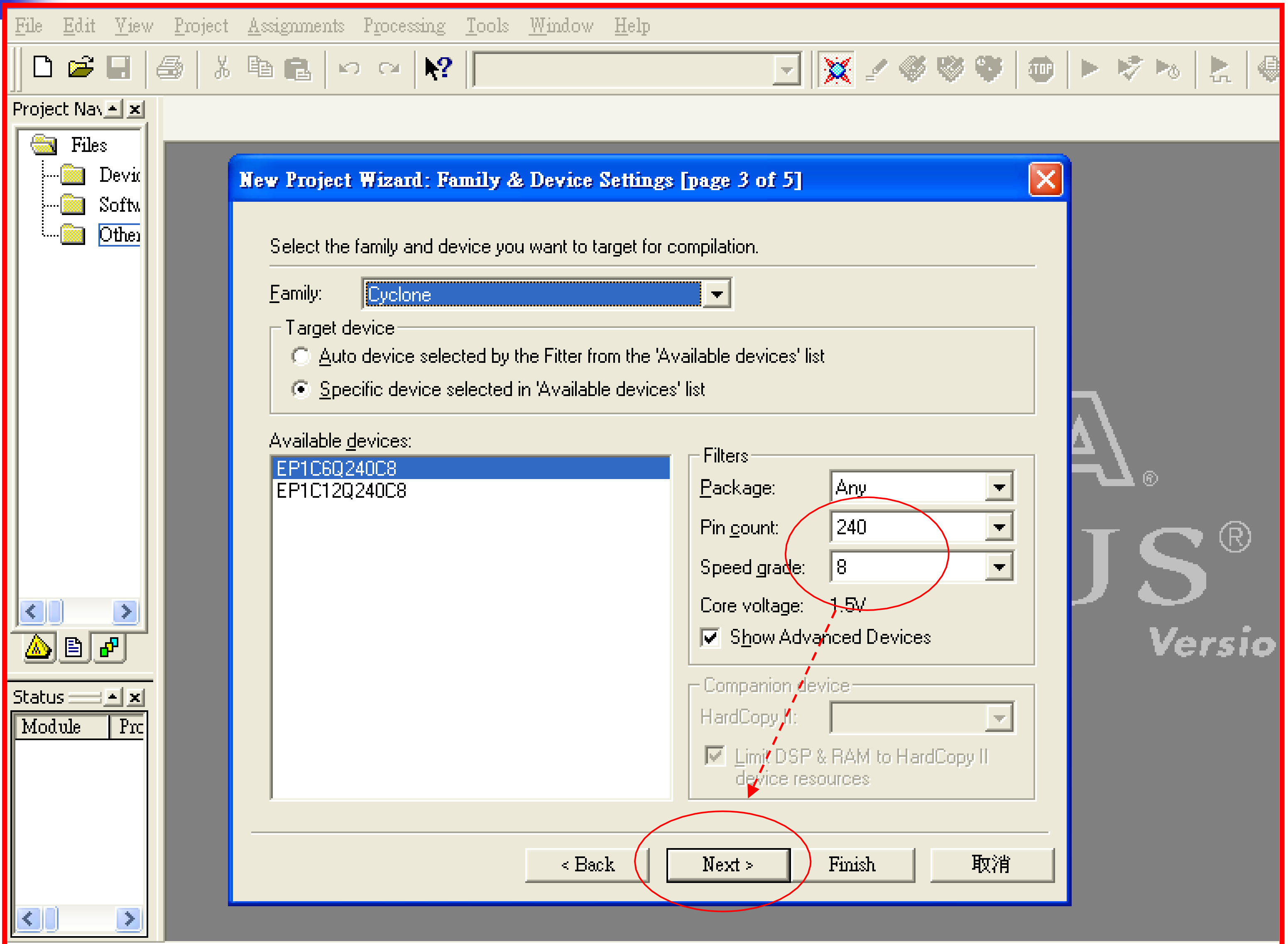
One-Chip Design (1)



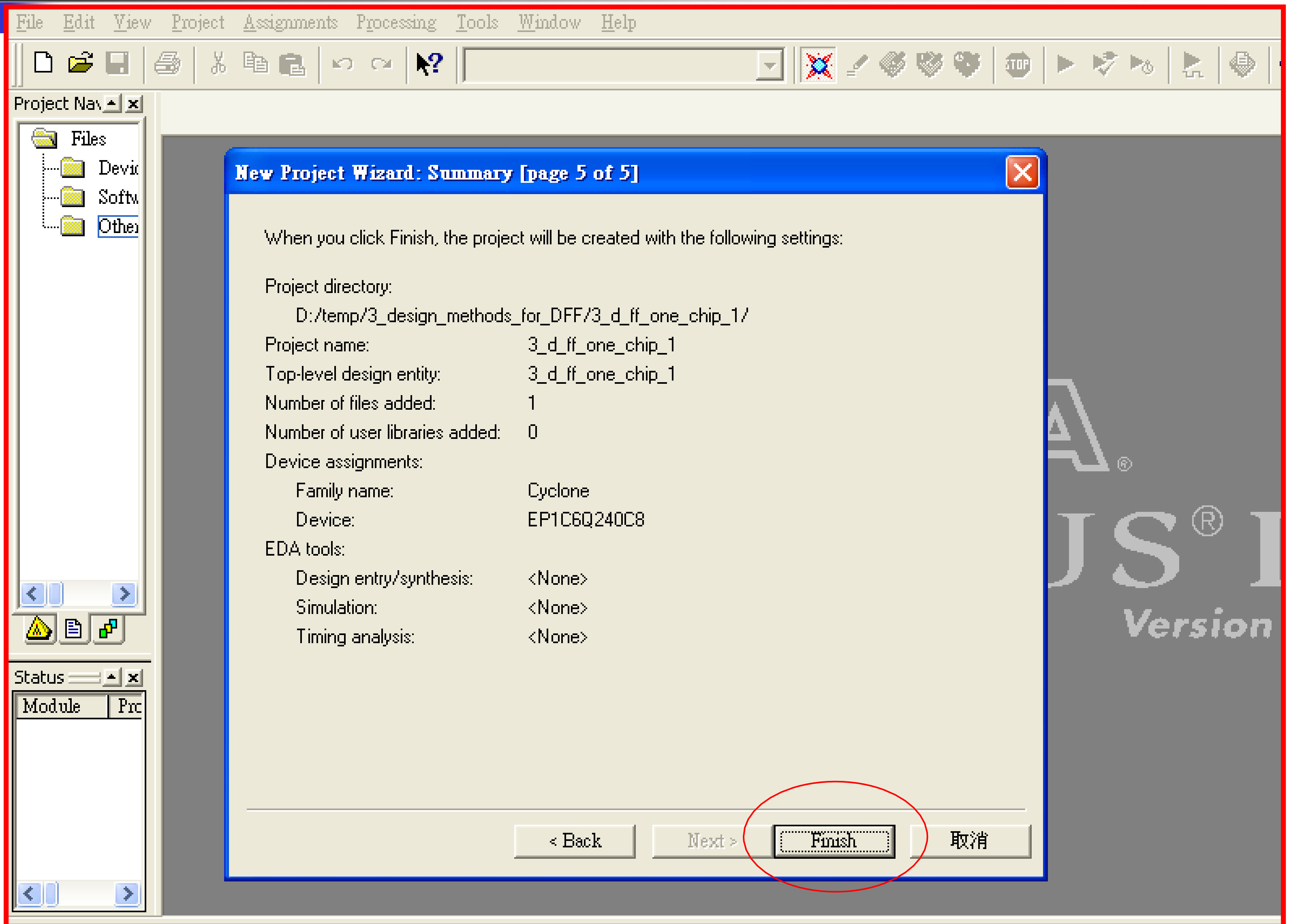
One-Chip Design (2)



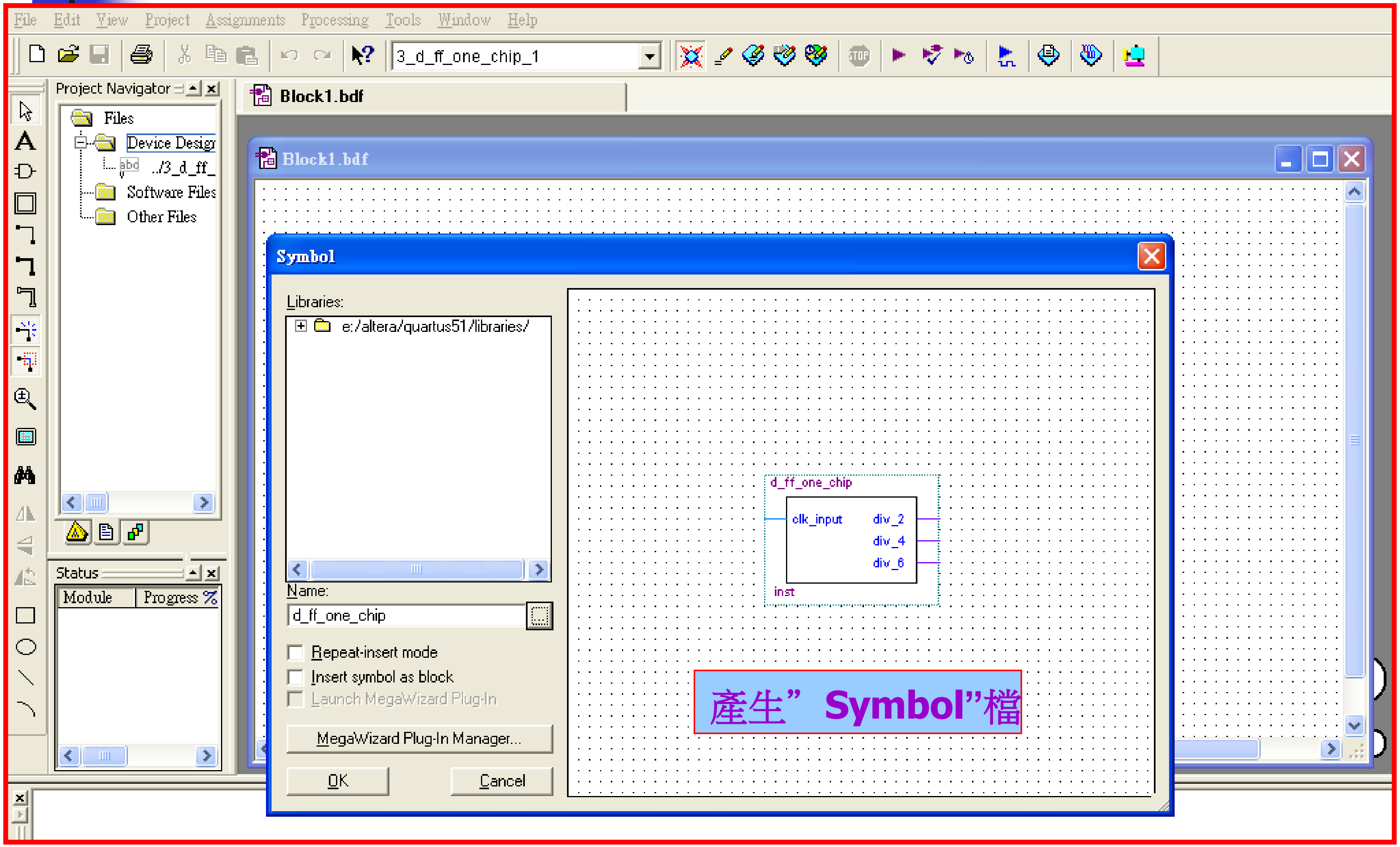
One-Chip Design (3)



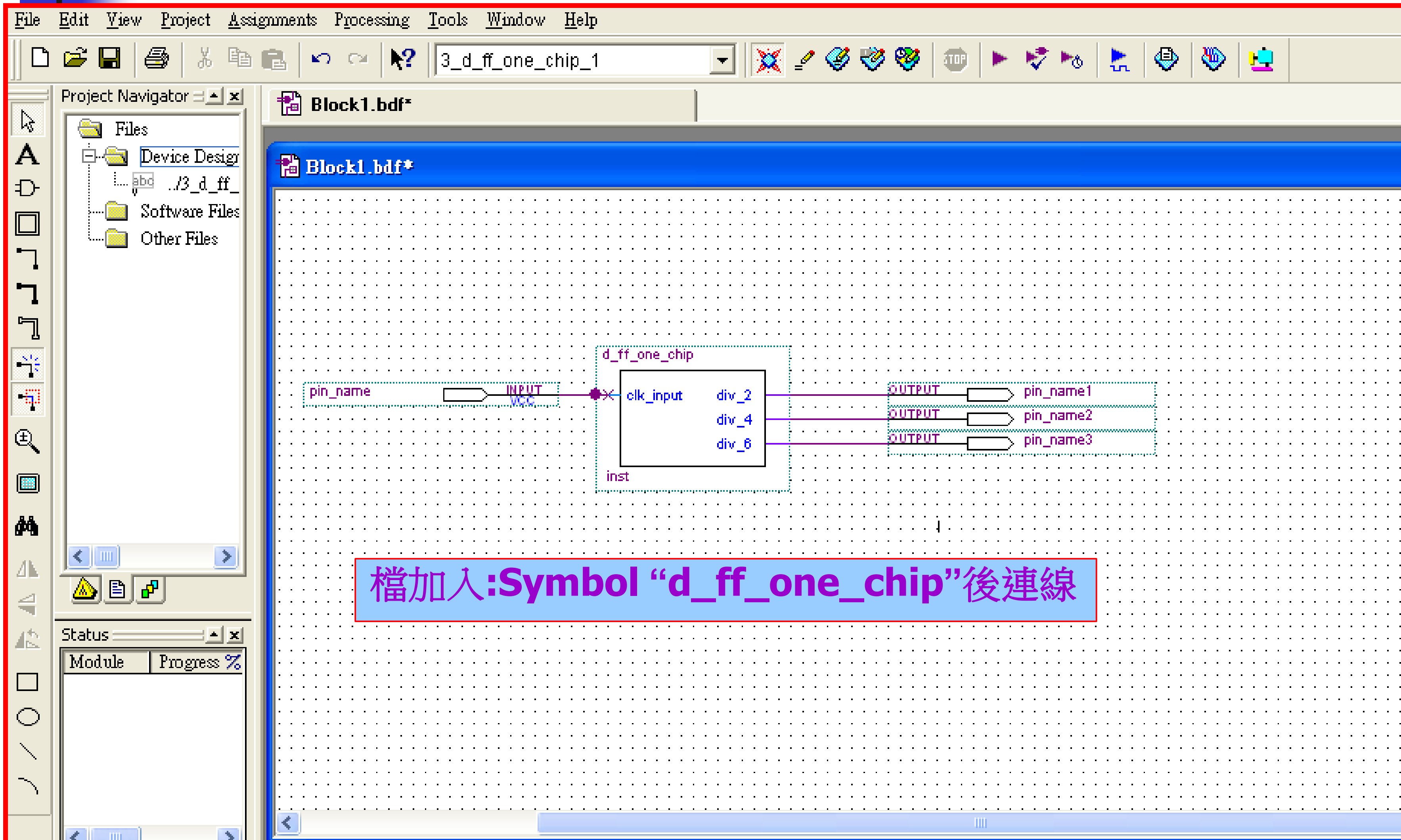
One-Chip Design (4)

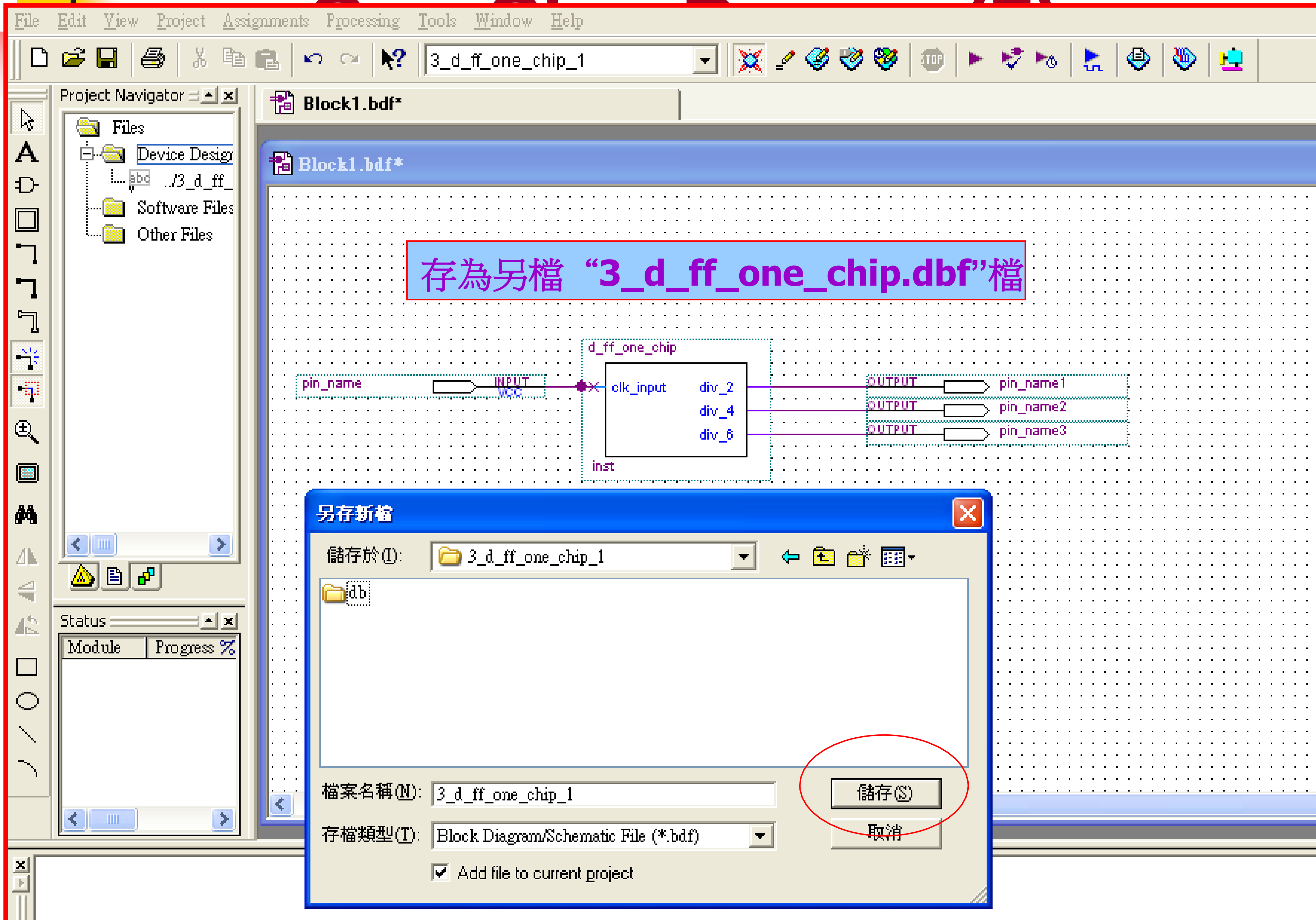


One-Chip Design (5)

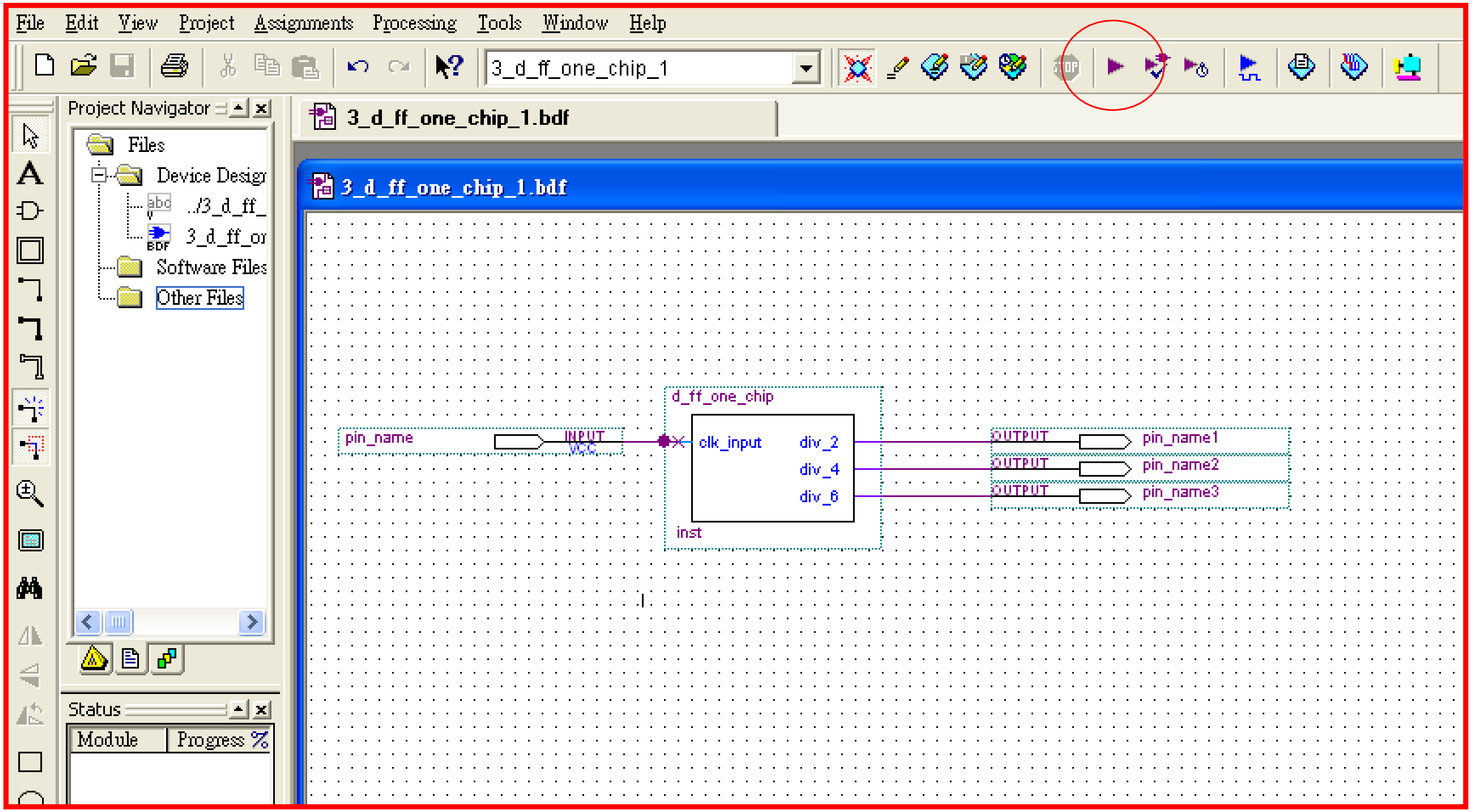


One-Chip Design (6)

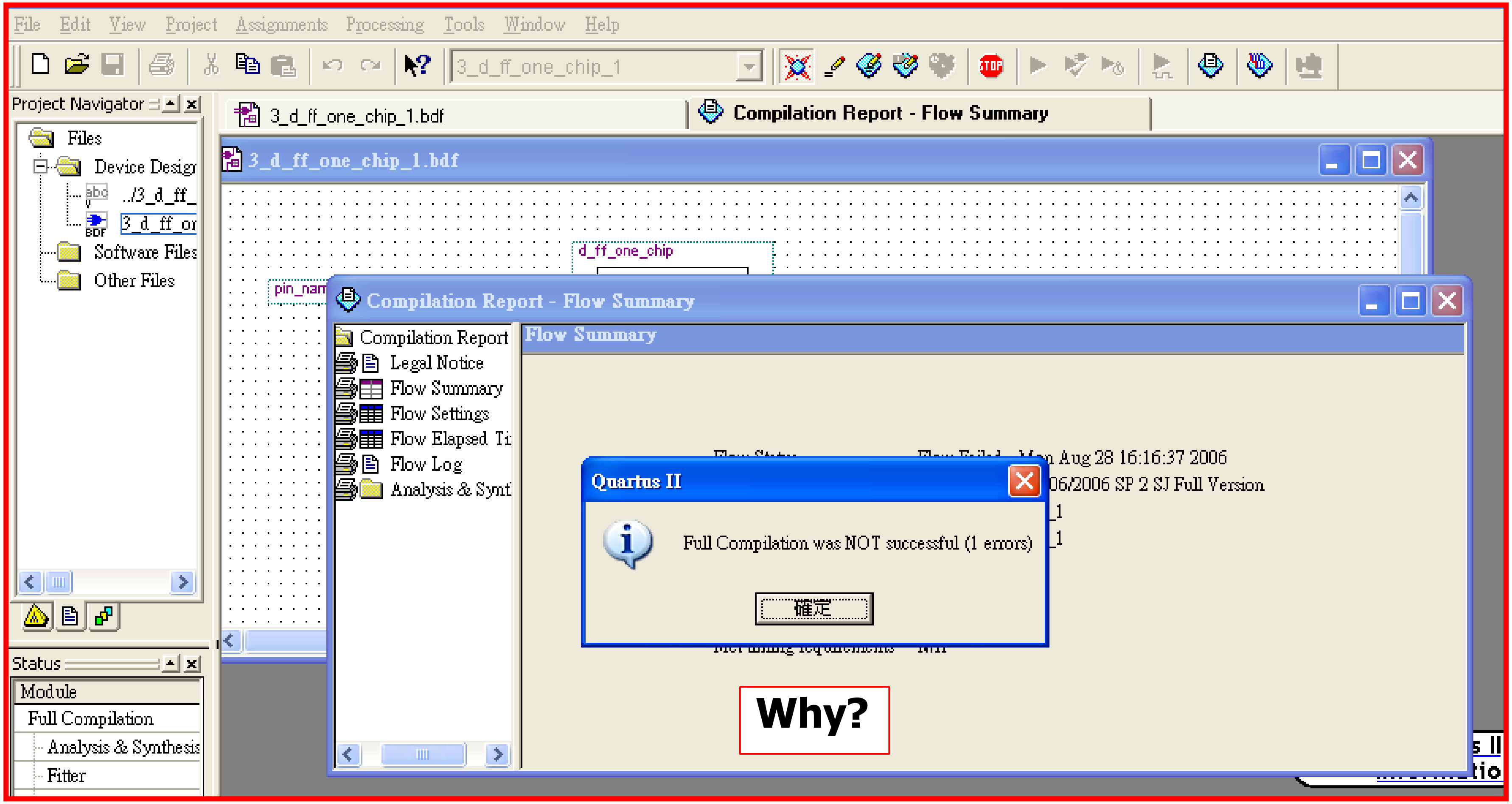



File Edit View Project Assignments Processing Tools Window Help
3_d_ff_one_chip_1
Project Navigator
Files
Device Design
../3_d_ff_
Software Files
Other Files
Block1.bdf*
存為另檔“3_d_ff_one_chip.dbf”檔
d_ff_one_chip
pin_name
INPUT
VCC
clk_input
div_2
div_4
div_6
inst
OUTPUT
pin_name1
OUTPUT
pin_name2
OUTPUT
pin_name3
Status
Module
Progress %
另存新檔
儲存於(I):
3_d_ff_one_chip_1
db
檔案名稱(N):
3_d_ff_one_chip_1
存檔類型(T):
Block Diagram/Schematic File (*.bdf)
儲存(S)
取消
Add file to current project

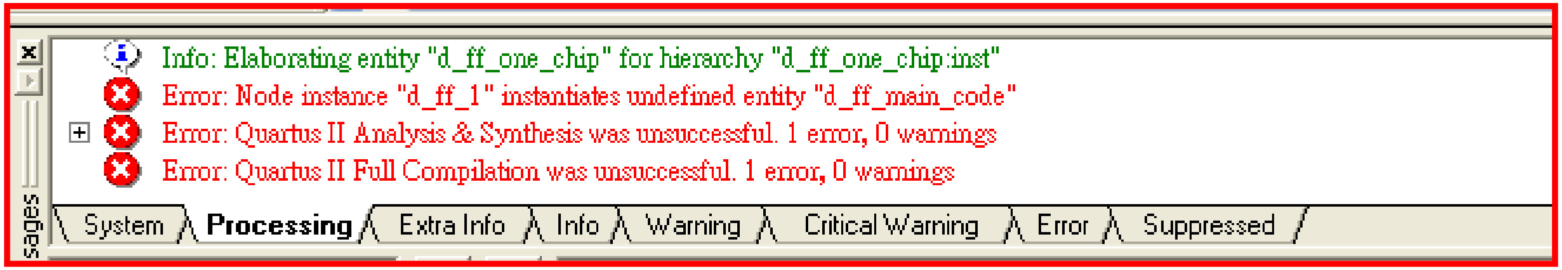
One-Chip Design (8)



One-Chip Design (9)



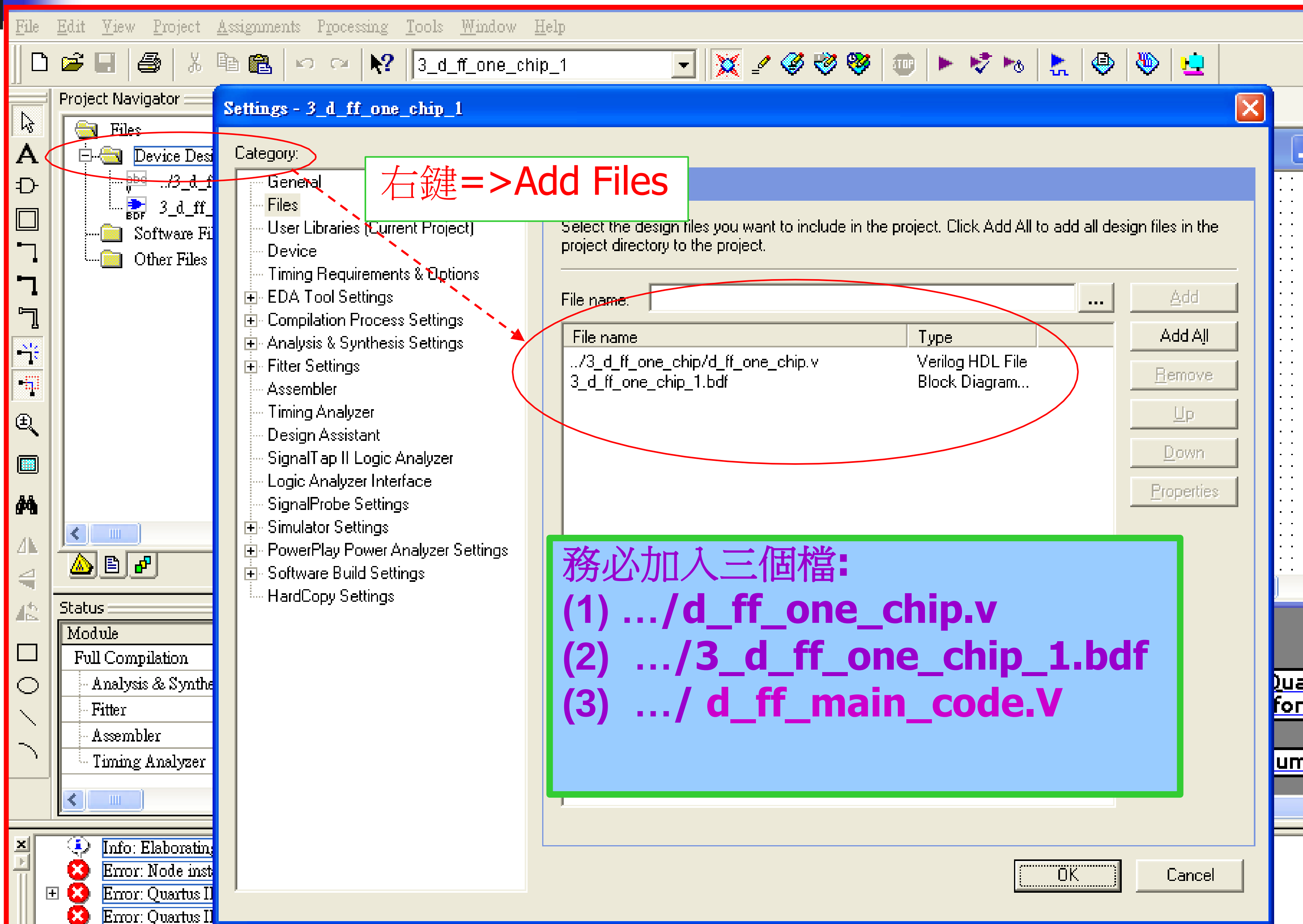
One-Chip Design (10)



Error: Node instance "d_ff_1" instantiates undefined entity "d_ff_main_code “
=> 表示別名” d_ff_1” 所對應的” d_ff_main_code” 主程式未見,必項加入.

Error: Quartus II Analysis & Synthesis was unsuccessful. 1 error, 0 warnings Error: Processing ended: Mon Aug 28 16:16:37 2006
Error: Elapsed time: 00:00:01
Error: Quartus II Full Compilation was unsuccessful. 1 error, 0 warnings

One-Chip Design (11)



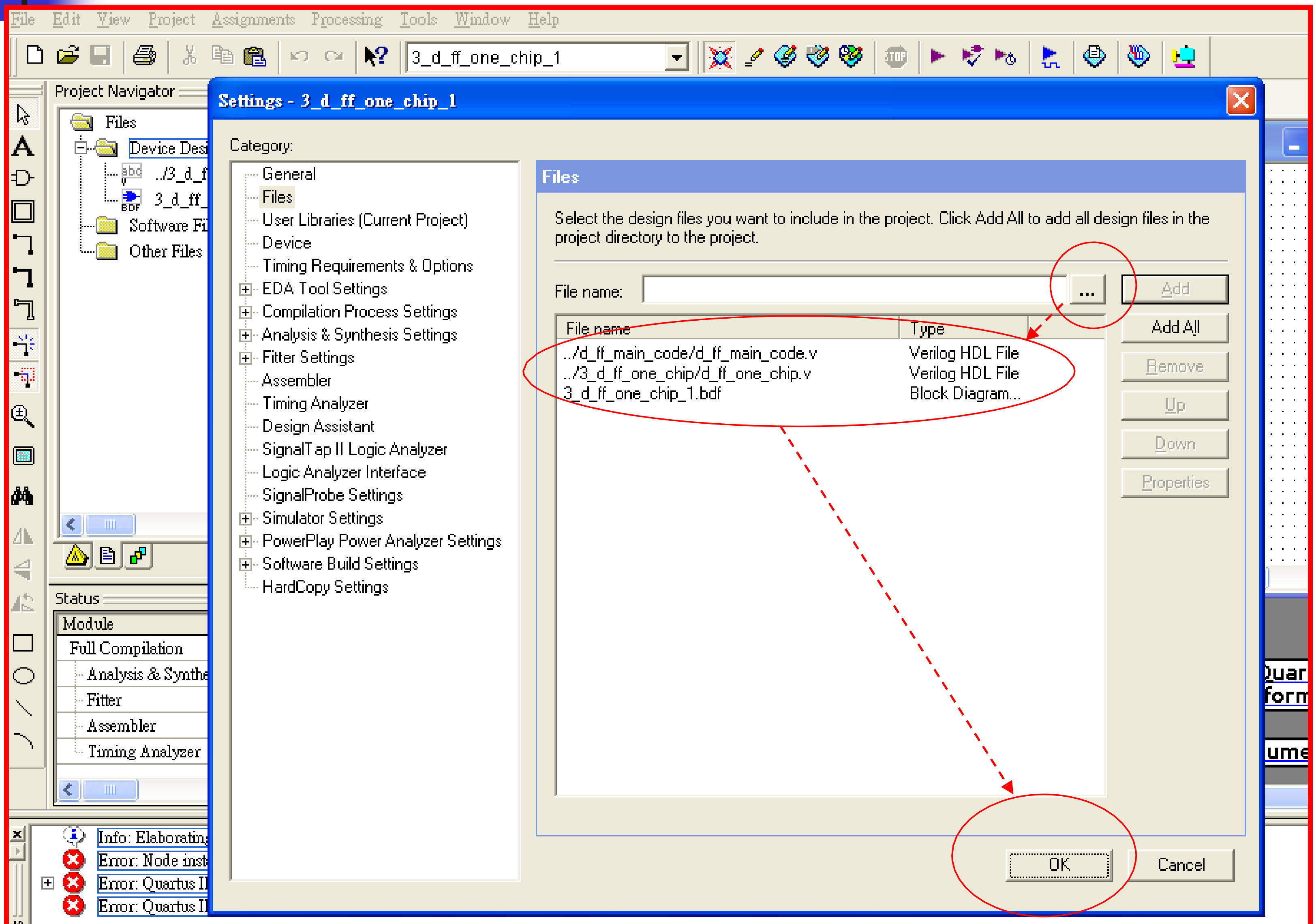
務必加入三個檔:

(1) …/d_ff_one_chip.v

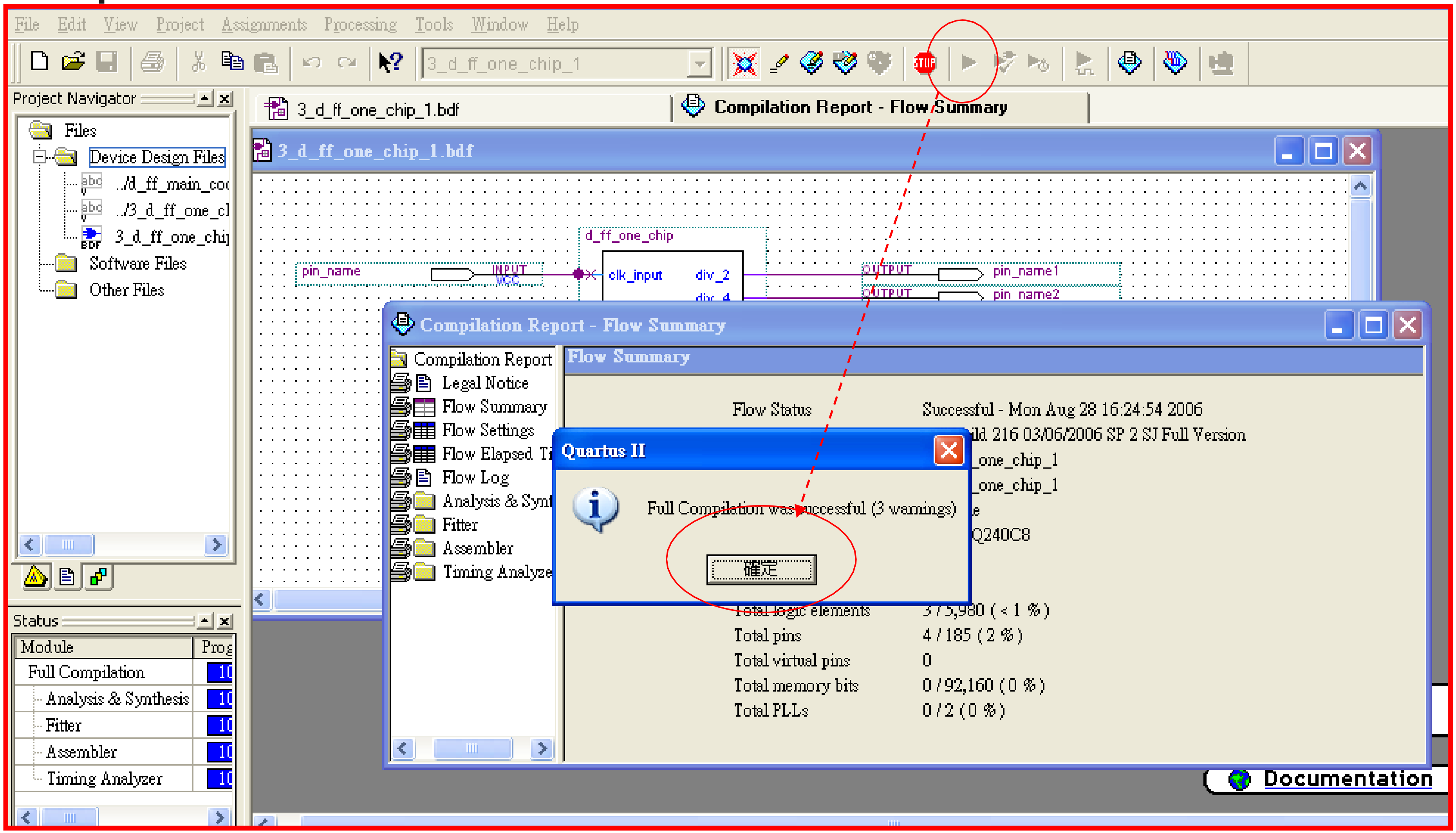
(2) …/3_d_ff_one_chip_1.bdf

(3) …/ d_ff_main_code.V

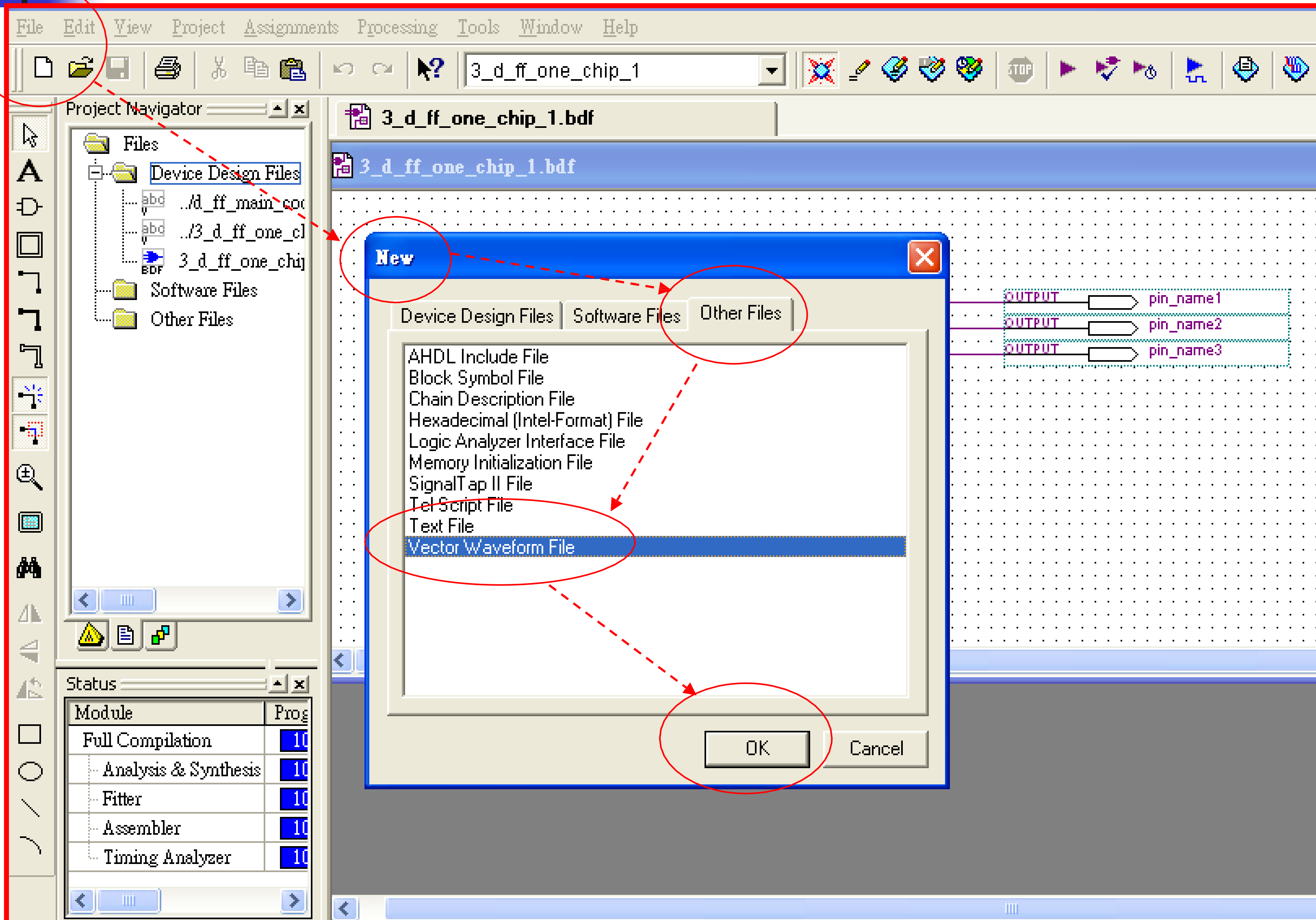
One-Chip Design (12)



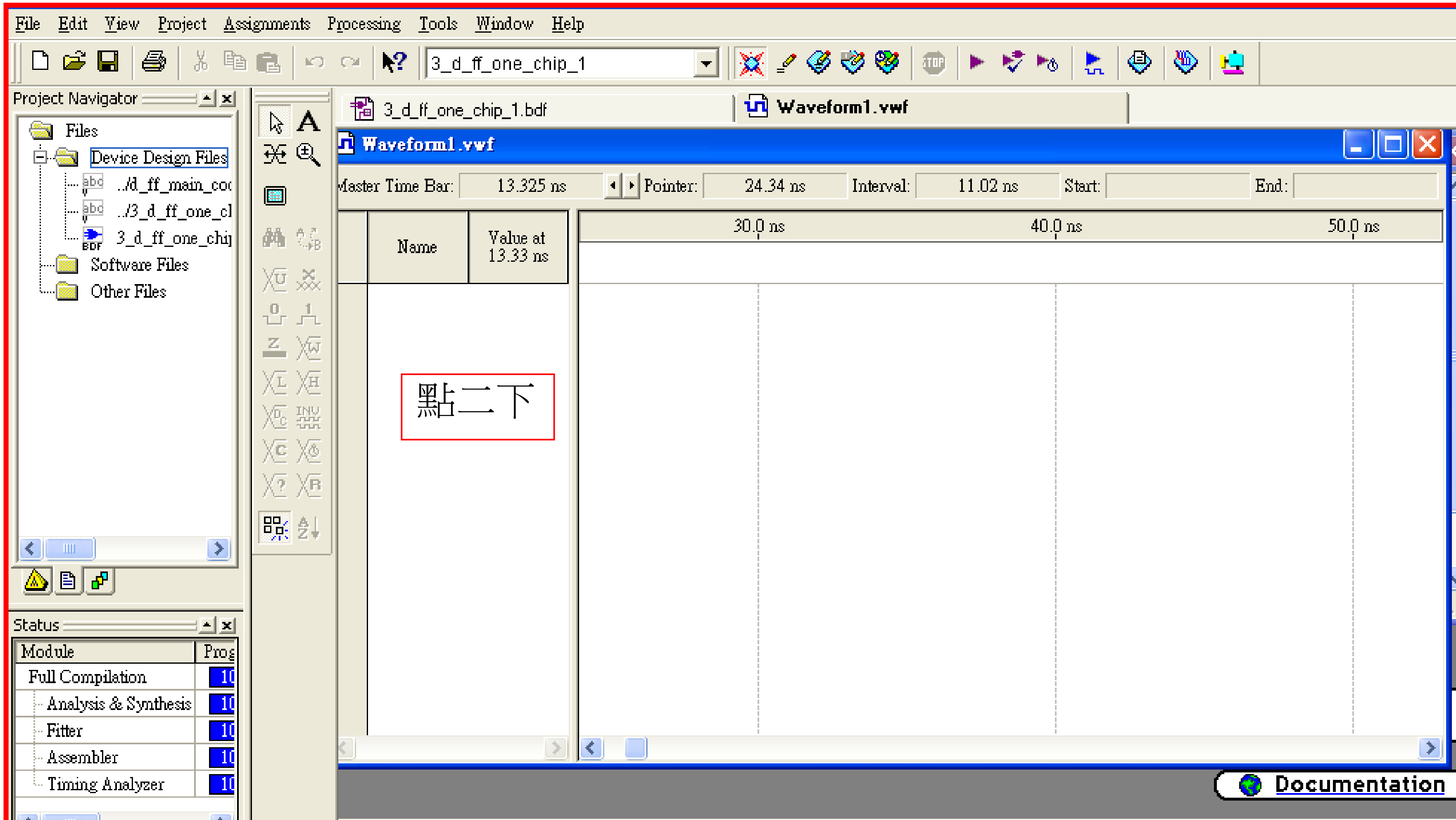
One-Chip Design (13)



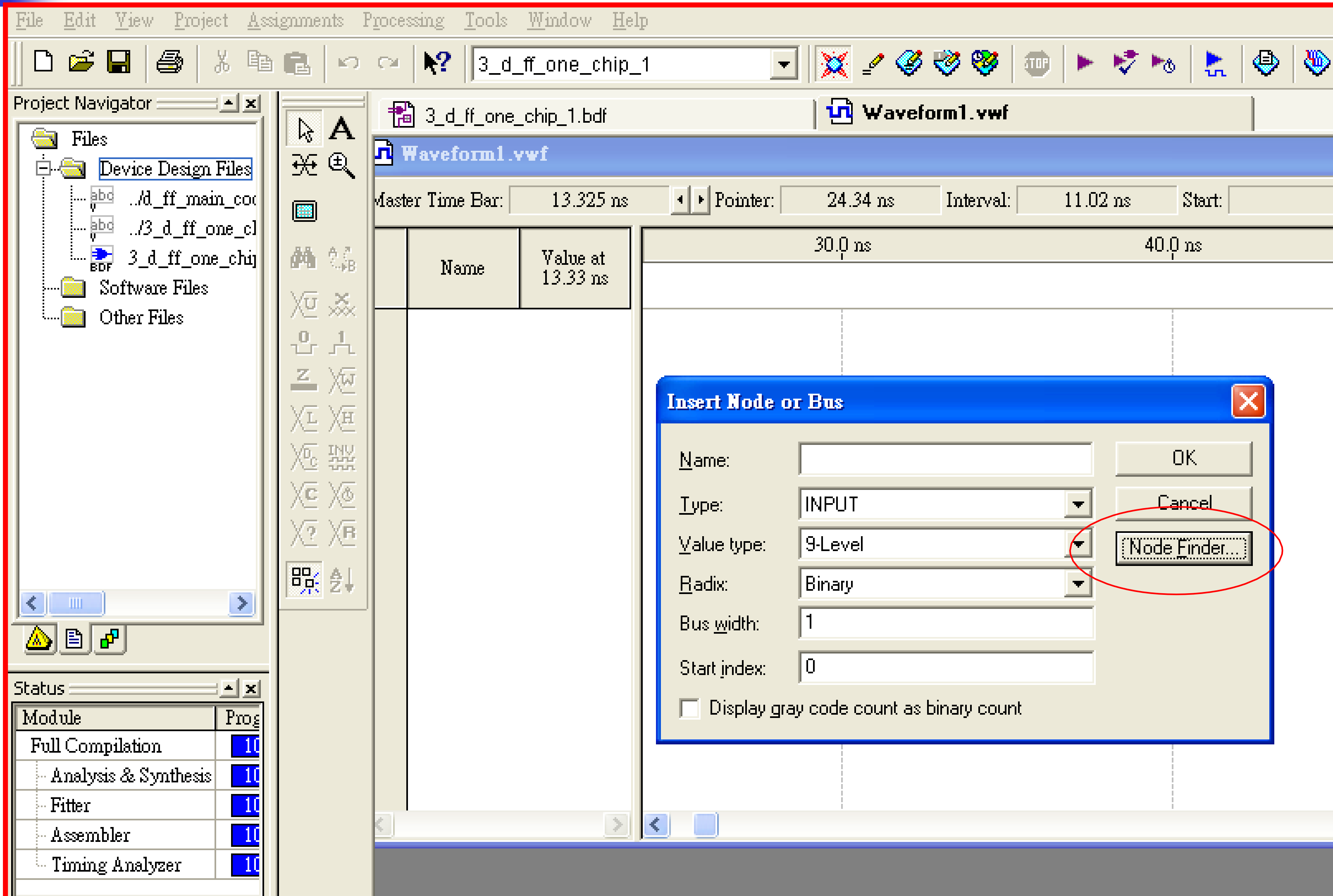
One-Chip Design (14)



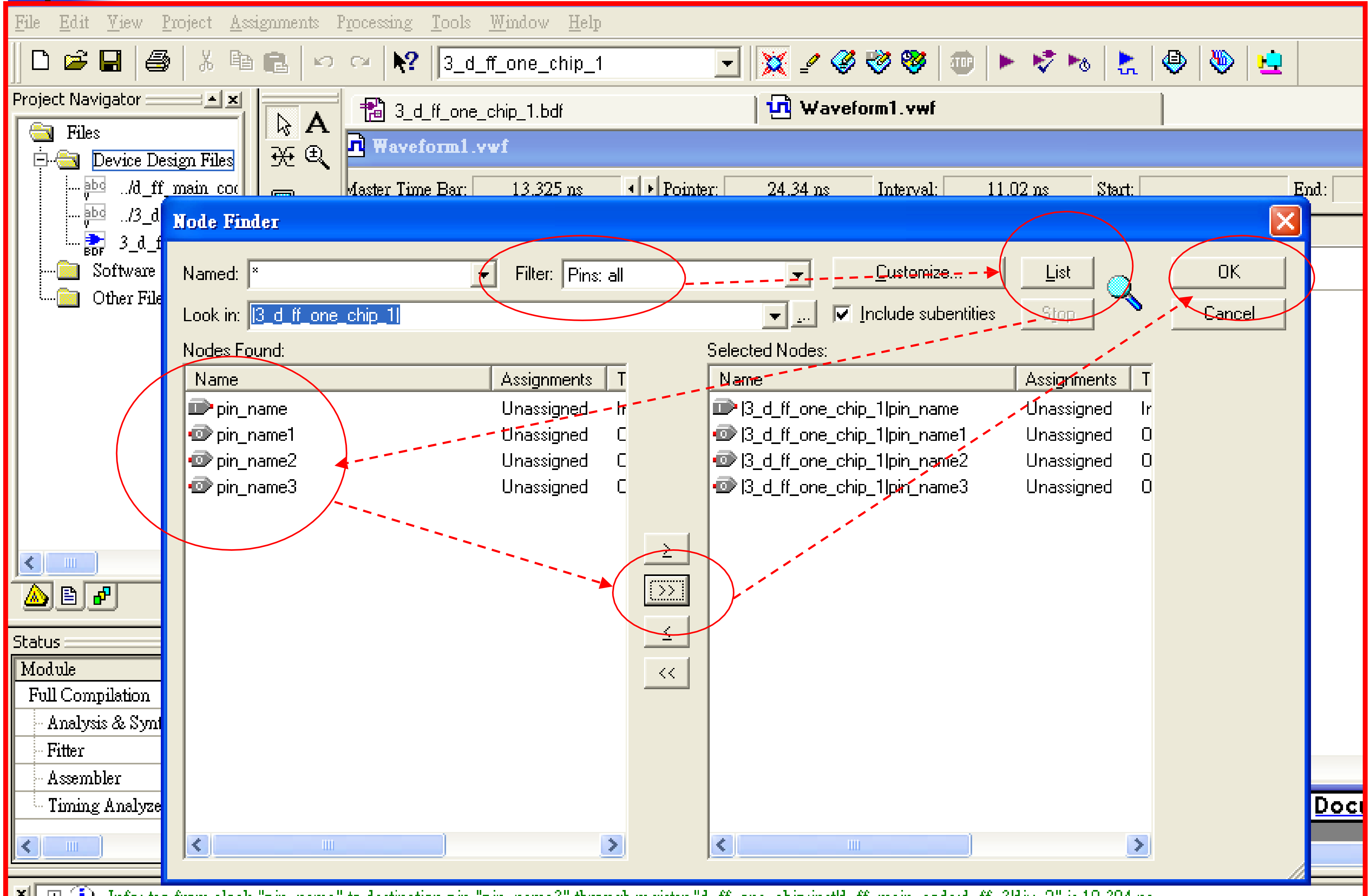
One-Chip Design (15)



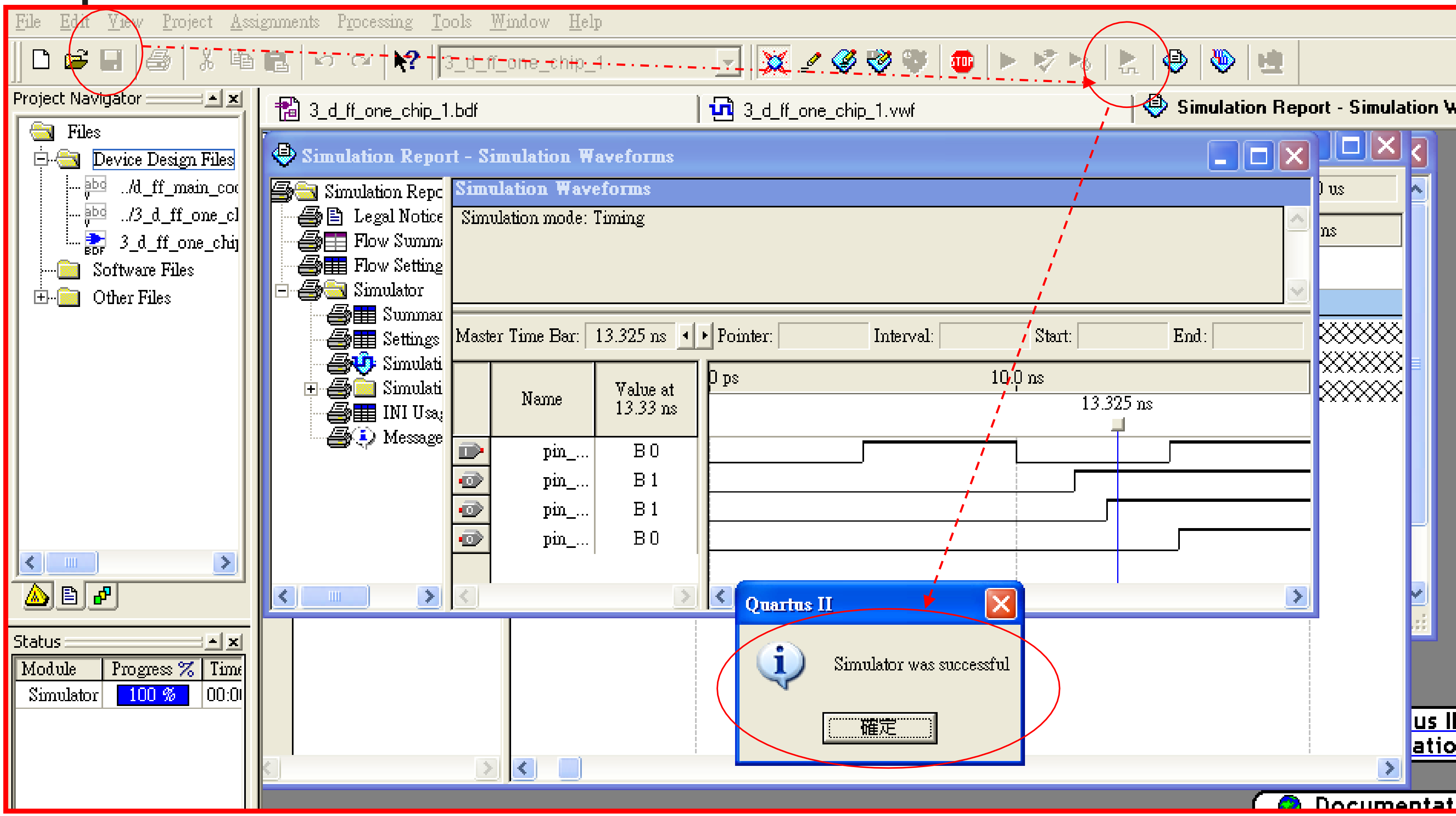
One-Chip Design (16)



One-Chip Design (17)



One-Chip Design (18)



One-Chip Design (19)

```
module d_ff_one_chip(clk_input, div_2, div_4, div_6);
input       clk_input;
output      div_2;
output      div_4;
output      div_6;
wire        a;
wire        b;

assign      div_2 = a;
assign      div_4 = b;

// by name
d_ff_main_code  d_ff_1(.clk_in(clk_input),.div_2(a));

d_ff_main_code     d_ff_2(.clk_in(a),.div_2(b));

d_ff_main_code     d_ff_3(.clk_in(b),.div_2(div_6));

endmodule
```

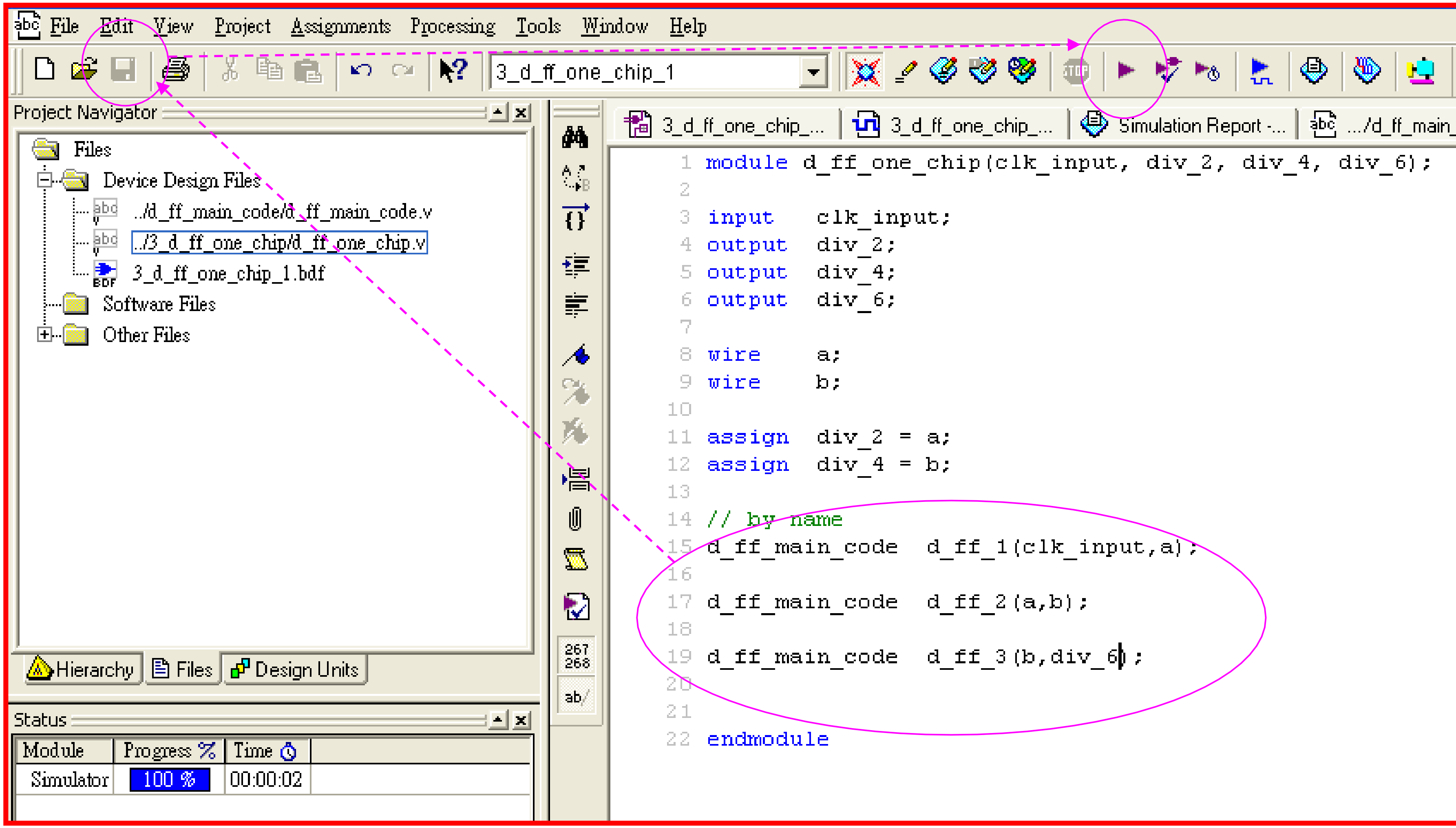
單晶片之原程式
(By Name)

One-Chip Design (20)

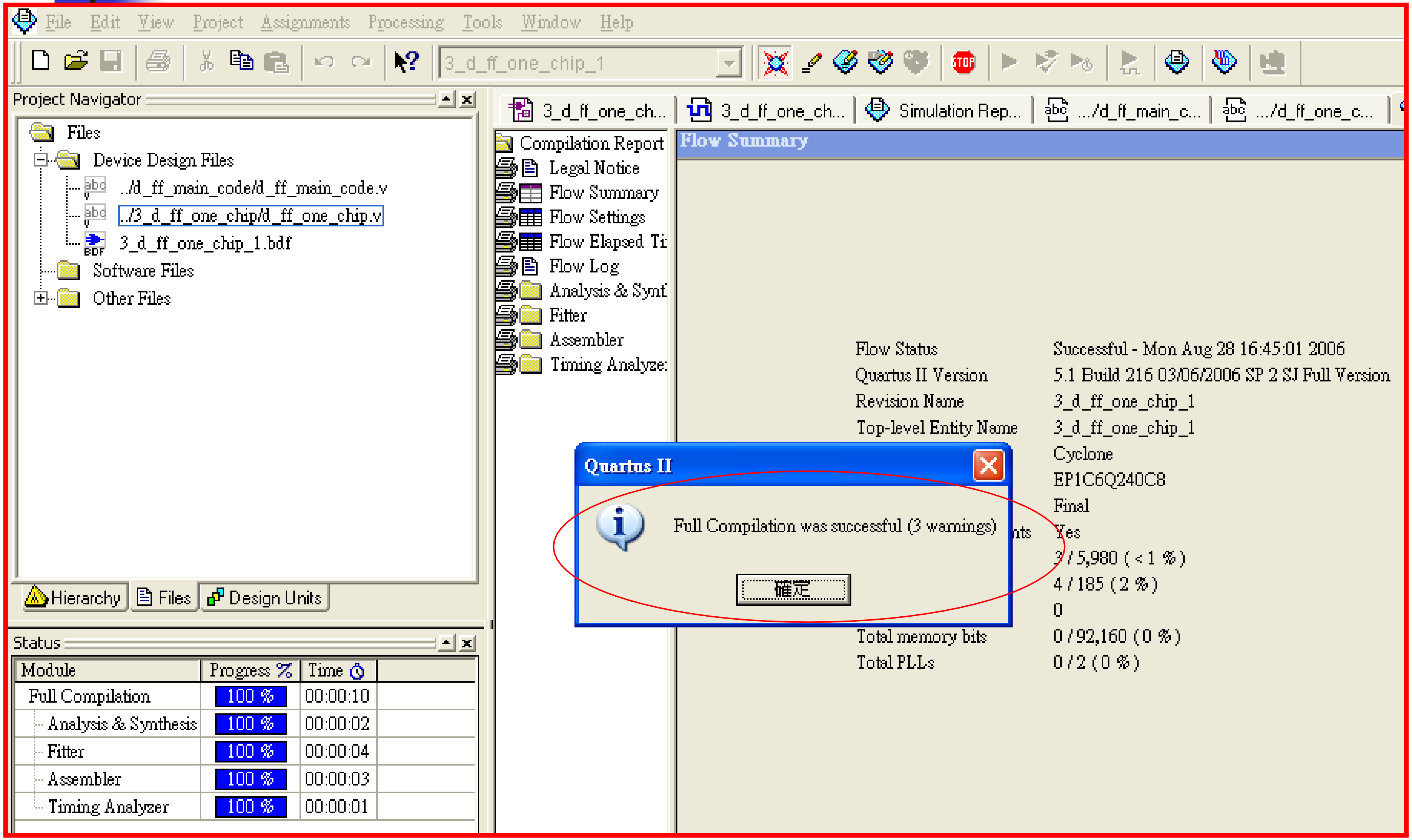
```
module d_ff_one_chip(clk_input, div_2, div_4, div_6);
input       clk_input;
output      div_2;
output      div_4;
output      div_6;
wire        a;
wire        b;

assign      div_2 = a;
assign      div_4 = b;

// by name
d_ff_main_code  d_ff_1(clk_input,a);

d_ff_main_code    d_ff_2(a,b);

d_ff_main_code    d_ff_3(b,div_6));

endmodule
```

單晶片之修正程式
(By Order)

One-Chip Design (21)



One-Chip Design (22)



One-Chip Design (23)